BC State-Space Computer

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Abstract

We propose a next-generation paradigm of state-space computing based on the arithmetic covariance structure of Bost–Connes (BC), establishing a correspondence between the composition of data, the task constraints of programs, and the physical execution of computation within a common structure. Prime-factor composition is carried by mode occupations, multiplication changes the corresponding occupations, and the logarithmic energy spectrum relates these data changes to definite energy differences. Programs select admissible states and exchange processes through programmable boundaries, allowing task conditions to participate in the formation of results. The arithmetic module performs internal processing, while the interaction module brings external conditions into a run and passes the resulting records and states to subsequent computations. On finite carriers, we establish a compilation from divisibility conditions to occupation windows, derive the target distribution, convergence time, and joint error bound for an independent-exchange model, and provide correctness guarantees for condition updates and multistage runs. These results show how a common arithmetic structure can connect data representation, task compilation, and result formation, providing a unified mathematical description of component reuse and computational continuation, and a concrete structural and dynamical basis for the BC state-space computer.

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1 BC State-Space Computer

Every computer computes through physical states and their changes. Its state space specifies which states it can carry, while operations and observations specify how those states change and yield results. A computing paradigm further specifies how a task enters this physical process: what carries data, what a program controls, and how results arise. Starting from these relations, we study a way to organize computation through the Bost–Connes (BC) state space.

1.1 Difficulties of Existing Paradigms

Classical digital computation expresses a task as symbols and rules for manipulating them, which physical devices then execute. This organization allows programs to be reused across devices and complex tasks to be decomposed into verifiable steps. When a task concerns many-body evolution, probability distributions, or a continually changing environment, many relations in the target process still have to be represented, compiled, and executed before they become actual machine behavior. Quantum computation and quantum simulation already allow superposition, entanglement, and some target dynamics to be carried directly by physical systems; realizing target relations as controllable processes remains a common problem across computing approaches [1,2].

The dynamical difficulty begins with implementing state relations and evolution rules. A dense classical representation of a general many-body quantum state grows rapidly with the number of degrees of freedom, whereas specially structured states admit compressed representations. Quantum devices can carry quantum states directly, but the evolution required by a task must still be realized through gate sequences or tunable couplings. For example, a target interaction unavailable directly through device couplings must be decomposed into executable operations, with errors controlled throughout their composition. Computational cost therefore depends both on the structure of the target and on how the machine can realize it.

Statistical tasks also require the machine to produce outcomes with specified weights. Sampling algorithms construct distributions through update rules; physical systems produce statistics through energy exchange, relaxation, and measurement. When target constraints have not entered the executable update relations, a run must additionally check candidate states, calculate weights, or reject inadmissible samples. The target distribution, actual exchange mechanism, and completion time must be connected: specifying an equilibrium distribution identifies only where evolution may tend; reaching the required statistical accuracy from a preparable initial state also depends on the specific dynamics.

Tasks continuously coupled to an environment also involve sensing and return. External conditions are sensed and processed internally to form controls, whose execution changes those conditions in turn. When sensing records, internal models, and actuation variables use different representations, conversion, calibration, and time synchronization are needed between stages. Which variables must be retained, which relations can be carried by the device structure itself, and how sensing, processing, and action are connected all affect the complexity of the overall process. In such computation, the environment both supplies inputs and participates in result formation.

These difficulties all concern the organization of computation. When the compositional, dynamical, and statistical relations of a task can be carried by the same physical structure, the machine may reduce conversions between different representations and operational layers. This motivates studying computation in terms of a state space and its controllable processes as a whole, and determining the consequences of this organization in concrete tasks.

1.2 Requirements for a New Paradigm

Organizing computation around the state space as a whole requires connecting data composition, state changes, outcome statistics, and environmental participation. This leads to five design requirements.

(1) Compositional relations are carried by the state-space structure. The basic degrees of freedom and their composition must be specified so that compositional relations in a task can correspond to those of joint states. Both the range of representable states and the range of states that can actually be prepared and controlled must have a clear physical meaning.

(2) Computation is performed by controllable dynamics. Programs must be able to select admissible states, interactions, and evolution conditions so that the state changes required by a task become physically realizable processes.

(3) Statistical and thermal processes can serve as computational resources. Measurement, exchange, and relaxation must be able to perform sampling, filtering, and distribution formation. Statistical weights and the time required to establish them must be determined jointly by the observations and dynamics employed.

(4) The environment participates through physical boundaries. External conditions must be able to enter preparation and control, and the records and final states of a run must be able to return to the environment or become inputs to the next computation.

(5) Continuous evolution has an explicit algebraic organization. Physical evolution may proceed continuously, while the relations between data and operations must remain precisely expressible, composable, and verifiable; the domain and error of their realization by finite devices must be assessable.

These five requirements connect the constitution of a machine with its operation. A new organization must supply a common structure through which they constrain one another within a single system and support reusable programs.

1.3 The BC State-Space Computer

The BC system connects the multiplicative structure of integers, phase actions, and a logarithmic energy spectrum [3]. The physical interpretation of the BC state space and the dynamics of its finite reduction provide further foundations for this construction [4,5].

This structure addresses the requirements beginning with state composition. Unique factorization of positive integers supplies a composition rule along prime directions, and integer multiplication corresponds to increasing the occupations of the relevant modes. With the BC logarithmic spectrum, the same multiplication label also determines the energy difference of that change. Compositional and dynamical relations are thus connected by common arithmetic labels, which also impose explicit algebraic constraints on continuous-time evolution.

This connection can also enter statistical processes. In exchange mechanisms satisfying the appropriate thermal equilibrium conditions, energy differences and temperature jointly constrain the ratio of forward and reverse exchanges, allowing arithmetic labels to participate in equilibrium weights. Actual exchange rates and relaxation times depend on the specific couplings, while readout statistics are supplied by quantum instruments on the carrier. External fields, thermal contacts, and system ports then allow boundaries to select admissible states, exchange relations, and observations, turning these structures into physical processes that can be arranged repeatedly.

We call computation in which the composition of a physical state space, constrained evolution, observation statistics, and return coupling jointly perform a task **state-space computing**. In this organization, states carry data, programs specify reusable boundary and operational relations, and actual runs produce results.

A **BC state-space computer** is a computing system that realizes the BC covariance relations in a restricted form through physical degrees of freedom that can be prepared, controlled, and read out within finite experimental boundaries. Its finiteness is determined jointly by addressable modes, occupation ranges, and the auxiliary degrees of freedom participating in a run. A boundary selects a run from the states and operations realizable by the machine, allowing the same carrier to perform different tasks.

On this basis, we propose a next-generation computing paradigm organized by the BC arithmetic covariance structure. Its core is to use the same arithmetic relations to connect state composition, operational energy changes, constrained statistics, and the composition of runs. Data representation, program selection, and physical processes can thereby be related within a common structure. We substantiate this organizing principle through finite carriers, compilable task families, and their operational results; specific computational capabilities and resource advantages depend on the corresponding tasks and implementation conditions.

Programmable quantum simulation, dissipative quantum computation, and measurement-based quantum computation have developed ways to organize computation through tunable couplings, dissipative processes, and measurement feedback, respectively [2,6,7]. Concrete quantum-simulation constructions also use logarithmic spectra and autocorrelation responses to realize finite Dirichlet sums and thereby approximate the Riemann–Hurwitz ζ function [8]. The correspondence between prime-factor multiplicities, bosonic occupations, and logarithmic prime energy levels has been studied explicitly [9]; logarithmic spectra have also been used in integer-factorization schemes that seek unknown factors through controlled excitation and energy measurement [10,11].

Building on these connections and the BC covariance structure, this paper organizes reusable finite boundary programs. We focus on compiling divisibility conditions into occupation windows, proving target distributions and finite-time guarantees for specified exchange processes, and passing condition updates, records, and states to subsequent computations. The window tasks analyzed in full below take already supplied prime exponents as inputs and perform conditional sampling and updating. The role of the common structure is demonstrated by this concrete compilation-and-execution chain; physical evolution, dissipation, and feedback themselves are shared by different computing approaches.

The construction begins with a finite state space and then establishes a complete run through preparation, evolution, observation, and return coupling.

2 State Space and Its Operation

The BC state-space computer carries states through finite physical degrees of freedom and uses boundaries to select the processes those states can undergo. A complete computation includes state preparation, evolution, and observation, as well as how external conditions enter the machine and how results leave it or continue to participate in its operation.

2.1 Finite State Space

The constitution and operations of a finite machine arise from a restricted realization of the BC structure. We first state the relations on the global carrier, then restrict the modes and occupation ranges addressable by an actual device. Prime modes provide the basic composition: each mode occupation records its excitation number, and joint occupations record the composition of these excitations. For a positive integer label n, let $v_p(n)$ be the exponent of the prime p in its unique factorization. Then

$$n = \prod_p p^{v_p(n)}, \qquad |n\rangle = \bigotimes_p |v_p(n)\rangle_p.$$

The corresponding global Hilbert carrier is

$$\mathcal{H}_{\mathrm{BC}} = \ell^2(\mathbb{N}^\times) \simeq \bigotimes_p{}' \, \ell^2(\mathbb{Z}_{\geq 0}).$$

The tensor product uses the vacuum of each mode as its reference. Every basis label has only finitely many nonzero occupations, while Hilbert completion still permits square-summable superpositions of infinitely many basis vectors.

We use the BC logarithmic spectrum. Expressing energies in a common energy unit and using the corresponding dimensionless time, let N_p denote the occupation-number operator of mode p. Then

$$H = \sum_p (\log p) N_p, \qquad H|n\rangle = (\log n)|n\rangle.$$

In this BC spectrum, the energy per occupation quantum of each mode is $\log p$, and multiplicative composition of integers corresponds to additive composition of energies [9].

The global multiplication action μ_m sends $|n\rangle$ to $|mn\rangle$. Writing $\sigma_t(A) = e^{itH} A e^{-itH}$ for the time evolution of an operator A gives

$$\mu_m \mu_n = \mu_{mn}, \qquad \sigma_t(\mu_m) = m^{it} \mu_m.$$

In the standard representation, a phase element $e(r)$ is specified by a rational phase label $r \in \mathbb{Q}/\mathbb{Z}$ and acts as $e(r)|n\rangle = e^{2\pi i n r}|n\rangle$. Phase and multiplication satisfy

$$\mu_m e(r) \mu_m^\dagger = \frac{1}{m} \sum_{ms=r} e(s),$$

where the sum runs over phase labels in $\mathbb{Q}/\mathbb{Z}$ satisfying $ms = r$. These relations between phase, multiplication, and time evolution jointly give the BC covariance structure [3]. The same integer label thus participates in state composition, occupation changes, and energy changes.

A finite device addresses only finitely many modes and operates within finite occupation ranges. For a finite prime set Λ, let L_p be the occupation cutoff of mode p. Its occupation carrier is

$$\mathcal{H}_{\text{occ}} = \bigotimes_{p \in \Lambda} \text{span}\{|0\rangle_p, \ldots, |L_p\rangle_p\}.$$

Multiplication corresponds to occupation increments when both input and output lie within this range. A finite device uses the compression of this action; the complete operation reports out-of-range branches as failures or continues them on an enlarged carrier.

We write $\mathcal{H}_\Lambda$ for the finite carrier actually used by the device. Besides occupation carriers, a device may use finite residue and phase carriers; conversion between carriers requires a specified encoding and physical implementation. A change of basis within one carrier and an encoding conversion between different carriers have distinct roles.

A state is described by a density operator ρ on $\mathcal{H}_\Lambda$, with $\rho \geq 0$ and $\text{Tr}\,\rho = 1$. In the occupation basis, diagonal entries give occupation probabilities and off-diagonal entries describe coherence between occupation states; the joint state also contains correlations between modes. Occupation, phase, and correlation are different properties of the same state. Let $\mathfrak{D}_\Lambda$ denote the reachable state class under the device's declared preparation and control capabilities. The information that can be read from it is determined by actual observational capabilities. Auxiliary degrees of freedom involved in operations and measurements must remain in the joint state until they are released.

2.2 The Complete Run

The unit through which the machine performs a task is a **run**: external conditions enter through a boundary, an available state is prepared and undergoes constrained evolution, terminal readout produces a record and a post-measurement state, and return coupling passes the results to the environment or the next run.

Let s_{phys} denote external conditions, ρ^- the available state before the run, and b the resulting boundary conditions. Write ρ_0 for the prepared initial state, ρ_* for the pre-readout state along the actual monitoring history, and r and ρ^+ for the terminal record and post-measurement state. Figure 1 places these objects in a complete run and identifies control within a run and return between runs.

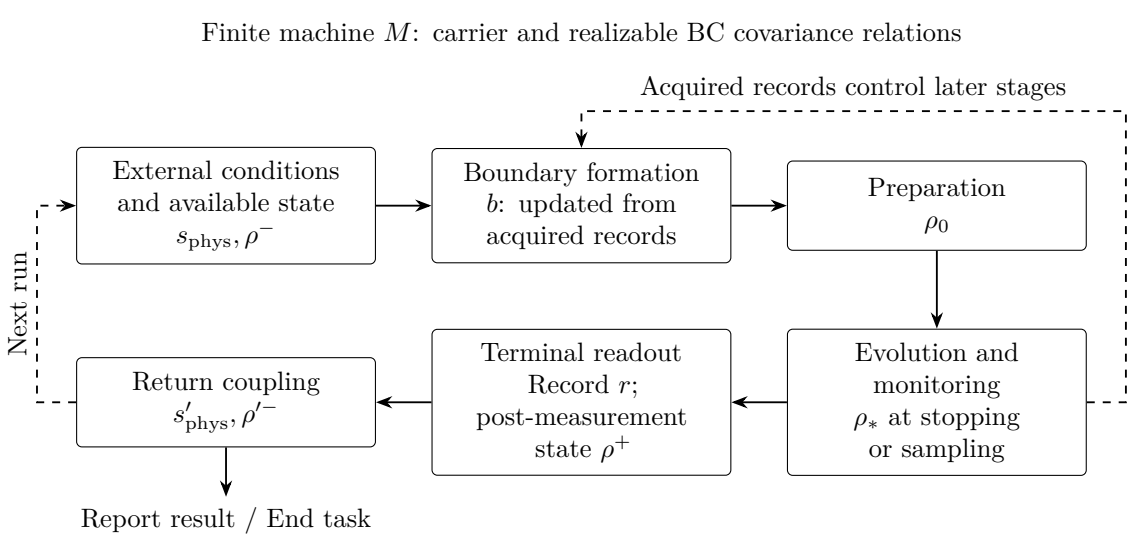


Figure 1: A complete run on the same finite machine. Solid arrows connect the physical stages of the current run. Dashed arrows show how acquired records control later boundaries and how the current output connects to the next run. Outcome branches and their probabilities are retained through the instrument description in the text.

Solid arrows connect the stages in the order of the physical process. Records already read through monitoring can control the boundary at later times; the returned external conditions s'_{phys} and available state ρ'^- can initiate the next run. These links organize feedback within a run and composition between runs, respectively. Evolution and readout can produce different records, and the probability of every branch belongs to the complete run.

This chain distinguishes the machine, boundary, and results. The machine supplies realizable states and operations; the boundary selects the conditions of this run; actual preparation, evolution, and observation jointly produce its results. The boundary itself does not prescribe trajectories or readings that have yet to occur.

2.3 Boundary Formation and Run Conditions

External light, sound, force, temperature, or port states of other systems enter a run through the device's transduction coupling. Writing $\mathcal{T}_\Lambda$ for the correspondence established by this coupling gives

$$\mathcal{T}_\Lambda : s_{\text{phys}}(t) \longmapsto b(t).$$

Boundary parameters can include pumping, external fields, temperature, chemical potentials, selection rules, and connections to other systems. They determine which degrees of freedom are excited, which exchanges are allowed, and which quantities can be observed. External variables absent from the transduction coupling do not participate in this run.

The boundary selects an admissible state set from the reachable states, denoted by

$$\mathcal{S}_b = \{\rho \in \mathfrak{D}_\Lambda : \rho \text{ satisfies boundary } b\text{'s support, conservation, and selection conditions}\}.$$

The same boundary also specifies admissible couplings, evolution schedules, monitoring, terminal readout, and stopping criteria. Together, these conditions determine how a run can proceed. If the boundary includes feedback, subsequent parameters can depend only on records already produced; b then represents a causal control policy, whose actual parameters unfold with the run history.

2.4 Preparation and Constrained Evolution

Once run conditions have been formed, the device must bring the currently available state into the required initial state. For deterministic preparation, let $\mathcal{P}_b$ denote the preparation channel. Then

$$\rho_0 = \mathcal{P}_b(\rho^-) \in \mathcal{S}_b.$$

The initial state may be formed by cooling and reset, by writing occupations and phases through particle or energy exchange, or by inheriting the preceding post-measurement state. If preparation selects a measurement outcome, a preparation instrument describes all branches; ρ_0 is the normalized conditional state of the selected branch, whose probability is retained separately.

The system then evolves under its intrinsic spectrum, available couplings, and boundary conditions. Let $\Phi^b_{t,0}$ denote the unconditional evolution from the end of preparation to time t. Then

$$\rho(t) = \Phi^b_{t,0}(\rho_0).$$

Evolution may be unitary in a closed system; when the system exchanges with auxiliaries or the environment, it is given by the corresponding open-system process. The unconditional state here aggregates all monitoring branches; once a particular record is obtained, the state must also be updated conditionally on it.

Eliminating the environment and describing evolution by a quantum channel on the system requires specifying how the environment is prepared and related to the system. This reduced description applies when the environment is independently prepared in a given state. An environment that retains correlations and will interact again must remain in the joint run state. Evolution therefore always acts on a state sufficient to describe the physical process, with its specific generator determined by the actual couplings and exchange mechanisms.

2.5 Monitoring and Stopping

The time at which evolution enters terminal readout can be prescribed in advance or determined by records obtained during the run. Scheduled sampling reads the state at a given time; an event-triggered run requires an actual monitoring process to supply the stopping information.

Let $Y_b[0,t]$ denote the monitoring record obtained up to time t, and let h_b be a stopping criterion taking values in $\{0,1\}$. The event-triggered stopping time is

$$\tau_b = \inf\{t \geq 0 : h_b(Y_b[0,t]) = 1\}.$$

The criterion uses only records available up to the current time, and measurement back-action is included in the evolution.

A stopping event means that the machine has entered the specified readout stage. Whether this also means task completion depends on the task's acceptance conditions. If a run does not stop within the specified time limit, that outcome and its probability must be retained, so that completion time, success probability, and output statistics belong to the same run description.

2.6 Terminal Readout and Statistics

Terminal readout produces both a record that can be retained and a state available for subsequent use. The pre-readout state depends on the evolution and monitoring actually experienced by the system. We must therefore identify the history reaching the terminal stage before calculating terminal-outcome probabilities.

Starting from a given prepared initial state ρ_0 and considering discrete records, let h be a monitoring history leading to stopping, and let $\mathcal{J}^b_h$ be the instrument component incorporating

evolution, monitoring, and stopping selection. It is a completely positive, trace-nonincreasing map, whose output trace gives the probability of that history:

$$p(h) = \operatorname{Tr} \mathcal{J}_h^b(\rho_0), \qquad \rho_*^h = \frac{\mathcal{J}_h^b(\rho_0)}{p(h)}, \quad p(h) > 0.$$

For scheduled sampling without selection of monitoring outcomes, the pre-readout state is given by the unconditional channel at the corresponding time. For event-triggered stopping, ρ_*^h also includes conditional information from the records already obtained and must be determined by the history instrument.

Let $\{\mathcal{I}_{b(h),r}\}_r$ be the terminal quantum instrument selected according to the available history. Each component is completely positive and trace-nonincreasing, and the sum over all terminal outcomes is trace-preserving. The conditional probability of outcome r and the post-measurement state are, respectively,

$$p(r \mid h) = \operatorname{Tr} \mathcal{I}_{b(h),r}(\rho_*^h),$$

$$\rho_{h,r}^+ = \frac{\mathcal{I}_{b(h),r}(\rho_*^h)}{p(r \mid h)}, \qquad p(r \mid h) > 0.$$

Connecting the history to terminal readout gives the joint probability

$$\boxed{p(h,r) = \operatorname{Tr}\Big[\mathcal{I}_{b(h),r} \circ \mathcal{J}_h^b(\rho_0)\Big].}$$

Summing over unreported histories yields the probability of the retained record. For continuous records, $\mathcal{J}_h^b$ and $p(h)$ are instrument and probability densities relative to a chosen reference measure on records; sums become integrals, and conditional states are defined almost everywhere.

If ρ_0 comes from a selected preparation branch of probability q, the formula above gives the joint probability conditional on that branch; its contribution to the complete run probability is $q\,p(h,r)$. Other preparation branches, as well as non-stopping and out-of-range outcomes, also carry probability and are included in the complete statistics.

Individual records, post-measurement states, and distributions obtained from repeated runs provide different forms of output. Records can directly answer discrete questions, post-measurement states can participate in subsequent computation, and repeated runs estimate probabilities and correlations. We use the quantum-instrument statistics above; the conditions concerning finite joint states and records are discussed in [5]. The thermal KMS condition constrains equilibrium states and the corresponding exchange relations; observation statistics are supplied by the adopted instruments and statistical conditions.

2.7 Return Coupling and the Run Law

Readout results enter subsequent physical processes through return coupling. Records may control external fields or actuators, post-measurement states may pass to another system, and both may jointly determine preparation for the next run. Denoting return coupling by $\mathcal{W}_b$ gives

$$\mathcal{W}_b : (r, \rho^+) \rightsquigarrow (s'_{\text{phys}}, \rho'^-).$$

Here r includes monitoring records and terminal outcomes actually retained for feedback; the arrow allows the return process to be stochastic. A return ending in display or recording gives a standalone run. A return changing the environment, subsequent boundary, or available state gives sequential runs or a sensing–actuation loop.

Return is implemented by record-controlled physical channels or quantum instruments. The controller selects operations from records already read, while the post-measurement state continues to participate as a physical input. A correlated environment participating across runs remains in the

joint state; fresh auxiliaries prepared independently conditional on known history enter the reduced channel as in Section 2.4.

Let the machine be M. With external initial conditions and interface preparation specified, the boundary policy b and pre-run state ρ^- determine, through the physical processes above, a joint probability law for records, post-measurement states, and return outcomes:

$$\mathsf{Run}_M(b, \rho^-) = \mathrm{Law}(r, \rho^+, s'_{\mathrm{phys}}, \rho'^- \mid b, \rho^-).$$

When environmental memory must be retained, the pre-run state is the corresponding joint state. With feedback, b is a prescribed causal policy, and the actual controls along each history are generated by that policy.

This run law places records, post-measurement states, and environmental responses within one physical description. The boundary selects a run, repetitions with the same preparation produce output statistics, and retained final states and return actions continue the subsequent process. Programmability is thus expressed through the selection and composition of these processes.

For a run to perform a specific computation, we must also specify the inputs to be processed, the target to be reached, and the interpretation of the results. Tasks and programs connect these requirements to the physical process already established.

3 Computational Semantics

Section 2 established the physical run from boundary formation to return of results. A task further specifies which states carry its inputs, which available actions realize its target relations, and how records or delivered states are interpreted as answers. A program organizes these arrangements into a single run, repeated sampling, or continuous feedback, allowing actual outputs to be tested against the task requirements.

3.1 Tasks and Boundary Programs

A task first specifies the relation to be established. For example, an arithmetic task may require samples satisfying divisibility conditions, a state-preparation task may require delivery of a state in a specified class, and a sensing–actuation task may require reports or responses determined by external conditions. Each includes admissible inputs, target results, and a way to assess those results. Denoting the input and result classes by X and Y, respectively, we write the task specification as

$$\mathscr{T} = (\mathsf{X}, \mathsf{Y}, \Gamma_{\mathscr{T}}, \varepsilon).$$

where $\Gamma_{\mathscr{T}}(x)$ gives the target relation for input $x \in \mathsf{X}$ and ε is the allowed error. The target may be a numerical value, probability distribution, physical state, or environmental response; its acceptance criterion is specified together with the chosen object.

Inputs enter through actual states or boundary variables. On an arithmetic carrier, joint occupations of prime modes carry integer labels, while finite phase and residue coordinates can carry corresponding arithmetic data. External conditions change preparation, fields, or exchange conditions through transduction coupling; a quantum state of another system participates through a physical port. For an unknown quantum input, the relevant coupling performs the operation without requiring the full numerical density matrix in advance.

A **boundary program** π specifies state constraints, preparation and couplings, observation schedules, causal control, and result interpretation for this input class. These relations are the reusable content of the program; only after a particular input is supplied do they become the boundary of a concrete run. Let M denote the machine, e the environmental conditions entering the run, and u the controls actually applied. We write boundary formation as

$$b_{\pi,x,e,u}(t) = \mathcal{B}_M\big(\pi, x, e(t), u(t)\big).$$

With feedback, the program prescribes control rules in advance, and current controls use only readable inputs and records already produced. Thus, even with the program and control policy fixed, different records may lead to different subsequent processes. The program, input, and environment supply the reusable arrangement, conditions to be processed, and actual external actions, respectively; together they determine the current boundary.

On the same occupation carrier, changing the admissible occupation range changes the arithmetic conditions to be satisfied, while changing couplings and observations selects different processing and readout methods. The machine supplies available states and actions, from which the program organizes a process corresponding to the task. Compilation converts the task into this physical arrangement.

3.2 Physical Compilation of Tasks

Compilation turns task relations into conditions that the machine can implement. It determines how inputs are written or connected, which admissible states and couplings perform the required changes, and which observations or state-delivery procedures produce the result. For a machine M, **physical compilation** is a partial map

$$\mathfrak{C}_M : \mathscr{T} \longmapsto \pi.$$

The states, couplings, and instruments used by compilation must lie within the device's realizable range; the implementation also specifies the required occupation, coherence, control, and readout conditions. The compiled program arranges preparation, evolution, and observation together. Admissible states and actual dynamics jointly determine the output.

Multiplicative conditional sampling directly illustrates this connection. Suppose the machine has declared a finite prime set Λ and occupation cutoffs L_p, restricting integer labels to that occupation carrier. The input supplies lower and upper exponents ℓ_p, u_p for each mode, satisfying $0 \le \ell_p \le u_p \le L_p$, and thereby determines

$$n_- = \prod_{p\in\Lambda} p^{\ell_p}, \qquad n_+ = \prod_{p\in\Lambda} p^{u_p}.$$

The task requires producing, among labels satisfying $n_- \mid n \mid n_+$, the distribution at a given inverse temperature $\beta > 0$

$$\varpi_{n_-,n_+;\beta}(n) = \frac{n^{-\beta}}{Z_{n_-,n_+;\beta}} \mathbf{1}_{\{n_-|n|n_+\}},$$

where $Z_{n_-,n_+;\beta}$ is the sum of weights over admissible labels. The input here is already in prime-exponent representation; obtaining these exponents from another representation is part of input conversion.

By unique factorization, the required divisibility relation is equivalent, in this occupation domain, to the modewise conditions

$$\boxed{n_- \mid n \mid n_+ \quad \Longleftrightarrow \quad \ell_p \le v_p(n) \le u_p \quad (p \in \Lambda).}$$

The program can therefore compile the input into admissible occupation windows for the individual modes. Their state support is

$$\mathcal{H}_{[\ell,u]} = \bigotimes_{p\in\Lambda} \operatorname{span}\{|\ell_p\rangle_p, \dots, |u_p\rangle_p\}.$$

The device preserves this support by selecting admissible transitions and disabling those crossing the boundary, while preparation places the initial state inside the window. The divisibility conditions thereby become actual restrictions on states and actions.

The window specifies admissible samples; their output weights must be formed by the corresponding exchange process. The BC spectrum $H = \sum_{p\in\Lambda}(\log p)N_p$ gives an energy difference $\log p$ for a single occupation increment in mode p. The program uses the fixed nearest-neighbor exchange law defined in Section 5.1: occupation distributions evolve autonomously, modes are independent, the mode clock rate is $\kappa_p > 0$, and the forward-to-reverse rate ratio is $p^{-\beta}$; transitions beyond the boundary are disabled. The specific rates and stationary distribution are derived in the operational analysis of that task.

The actual program prepares the lower-corner state $|n_-\rangle$, evolves under this exchange law, and reads each mode occupation. Section 5.1 derives the target stationary distribution and a finite-time bound on joint-distribution error from this generator, thereby determining a readout time meeting the tolerance. Compilation thus specifies support, exchange mechanism, preparation, and readout together. When the same program processes different exponent inputs, the mode boundaries change, while the spectrum and the exchange law inside each window retain their prescribed form.

Phase processing and environmental interaction likewise require correspondences between task relations, physical arrangements, and outputs; their available operations are supplied by the specific modules.

3.3 Results and Completion

A physical run produces records, post-measurement states, and environmental responses; the task determines which of them constitute the computational result. For a classical report, the program specifies an interpretation map

$$\mathcal{D}_\pi : r_{\mathrm{all}} \longmapsto y \in \mathsf{Y},$$

where r_{all} comprises the monitoring, terminal, sensing, and verification records actually retained for interpreting the output. Quantum outputs are delivered through declared physical channels; when a task uses both records and states, their joint output is retained. An environmental task is tested against the specified response or actuation outcome.

In multiplicative conditional sampling, occupation readings k_p give the sample $n = \prod_p p^{k_p}$. These readings determine whether one output belongs to the divisibility window; whether the output has the required distribution depends on the probability law of the complete sampling process. A task estimating probabilities or correlations must also reach the specified accuracy through repeated runs and the corresponding estimation rule.

The program's run law builds on the physical run of Section 2. For a single-run program, one substitutes the boundary policy, preparation conditions, and available state determined by the program and input into Run_M, then extracts the required output. For repeated sampling or feedback across runs, runs are connected by the prescribed causal rules while retaining the records, states, and environment that will still participate. Let λ collect the given environmental preparation, causal control policy, and available pre-run state. The joint run law for input x is

$$[\![\pi]\!]^{\lambda}_{M}(x) = \mathrm{Law}(r_{\mathrm{all}}, \rho_{\mathrm{out}}, s_{\mathrm{out}} \mid \pi, x, \lambda).$$

where ρ_{out} denotes the conditional states delivered on the record branches and s_{out} the retained environmental outputs. The report distribution, state, or response required by the task is extracted from this law and denoted collectively by $O^{\lambda}_{M,\pi}(x)$. For a classical report, this is the probability law after the interpretation map; for states and their correlations, the required joint physical object is retained.

Correctness requires the actual output to meet the task's target. Let $d_{\mathscr{T}}$ be a distance for the chosen output object and $\mathfrak{A}_\pi(x)$ the predeclared class of admissible run conditions. Then

$$\boxed{d_{\mathscr{T}}\Big(O^{\lambda}_{M,\pi}(x), \Gamma_{\mathscr{T}}(x)\Big) \le \varepsilon, \qquad \lambda \in \mathfrak{A}_\pi(x).}$$

This requirement applies to every admissible input specified by the task. Probability distributions may use total variation distance, states the corresponding trace distance, and numerical values

or environmental responses a task-specific error. A set-valued target uses distance to that set or an explicit acceptance test. Comparisons always concern the same type of output object and its corresponding acceptance criterion.

Task completion is determined by this requirement. Scheduled readout, monitoring events, and program stopping specify the physical timing; tasks requiring verification determine delivery through an executable verification procedure, while tasks requiring statistical accuracy use evolution and sampling schedules with established error guarantees. In the sampling example, the program delivers a sample after running for the proved time bound. Correctness states that the output distribution meets the tolerance; it does not require reconstructing the entire distribution from one sample.

Continuous sensing and feedback tasks may report or actuate throughout a specified time window. Cutoffs, timeouts, preparation failures, and other incomplete outcomes also belong to the program output; if retries are allowed, continuation follows rules fixed in advance. The required output, the process reaching it, and its completion criterion jointly define a computation.

Program meaning is determined by the input-to-result correspondence: the task specifies the target, compilation selects realizable states, actions, and observations, and the run produces outputs tested against the task (Figure 2). Window sampling implements divisibility through occupation boundaries and illustrates this correspondence. The components performing preparation, exchange, and readout make the compiled arrangement executable.

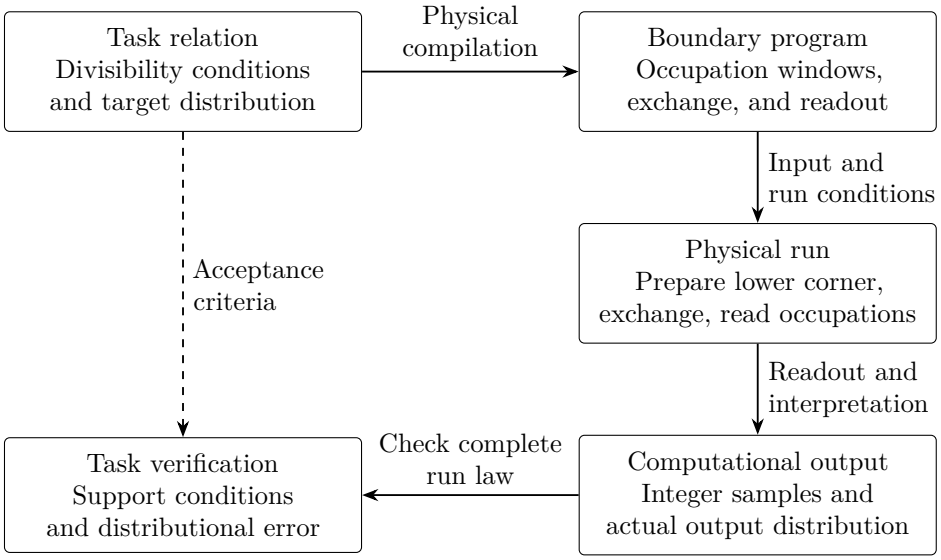


Figure 2: How a task enters BC computation. Compilation turns divisibility conditions into local occupation boundaries; the specified exchange process forms output statistics, and occupation readouts are interpreted as integer samples. Support conditions and distributional error jointly implement the task requirements. A single reading supplies only that run's sample.

4 Arithmetic and Interaction Modules

The BC state-space computer consists jointly of arithmetic and interaction modules. The arithmetic module carries data in occupations and phases and processes them through controlled operations, exchange, and observation. The interaction module brings external conditions into these processes and converts run results into reports, actuation, or actions on other systems. Both modules share the machine's joint state space, with preparation, coupling, and readout selected by boundary programs.

These components implement the boundary relations compiled in Section 3. Figure 3 shows how the two modules connect within the complete machine. Starting with data entry, we now establish the available actions for internal processing and external interaction.

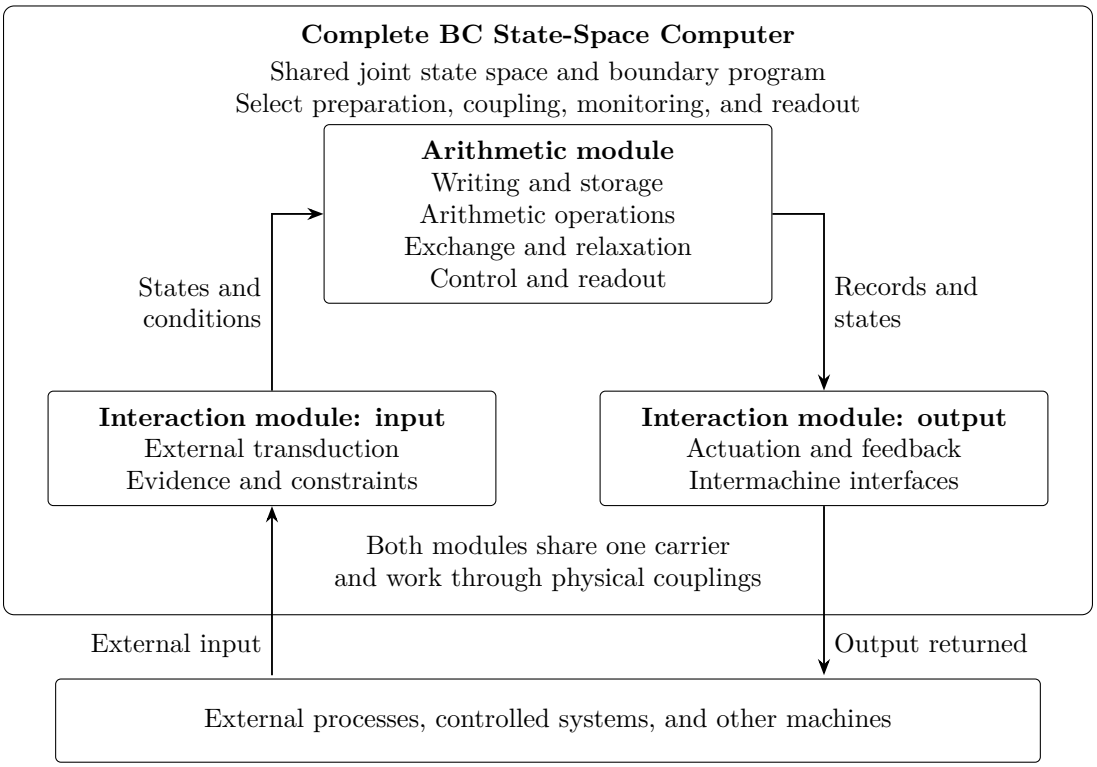


Figure 3: The two module types together constitute the complete machine. The interaction module is shown through its input and output functions, displaying how external conditions enter computation and results return to the environment. The arithmetic module lists composable capabilities rather than a fixed sequence. The program selects coherent operations and exchange as required.

4.1 Arithmetic Module

The arithmetic module organizes data through occupations, phases, and correlations on the same carrier. For each task, the program selects states to prepare, operations to apply, and quantities to observe. Coherent operations and exchange processes may be used separately, in successive stages, or alternately. The module's construction begins with data entry.

4.1.1 Writing and Storing Data

Writing occupations and phases. A supplied input must first be placed in a state the machine can process. Let Λ be the finite prime set and L_p the occupation cutoffs. If the prime support and occupations of an integer n lie within this range, and the input already supplies its prime exponents, one can prepare mode by mode

$$|n\rangle = \bigotimes_{p\in\Lambda} |v_p(n)\rangle_p.$$

If the input is an integer supplied in positional notation, obtaining $v_p(n)$ remains a conversion that must be performed. A positive rational a/b can be carried by separate nonnegative occupations for numerator and denominator; signed valuation coordinates require a carrier or auxiliary record for the negative direction.

Phase inputs use another declared finite carrier. For an odd prime q, take the position basis $|a\rangle$ in $\mathbb{C}[\mathbb{Z}/q\mathbb{Z}]$ and define

$$F_q|a\rangle = \frac{1}{\sqrt{q}} \sum_{x \bmod q} e^{2\pi i a x/q}|x\rangle.$$

Setting this phase slope on a prepared equal-amplitude superposition writes the value a into $F_q|a\rangle$. Here F_q changes basis within the same finite carrier; conversion between occupation and residue representations requires a separate encoding and actual coupling.

For pairwise coprime moduli $q_1, \ldots, q_s$, if their product Q covers the chosen integer domain so that distinct integers have distinct residue tuples, one can define an isometric encoding

$$J|n\rangle = \bigotimes_{i=1}^{s} |n \bmod q_i\rangle.$$

The encoded state lies in the image of J, and operations and the energy spectrum must transform with the encoding. This image need not be the entire residue tensor space, and neither $JHJ^\dagger$ nor encoded multiplication necessarily acts separately on residue channels. Figure 4 distinguishes input encoding from phase transformations within a carrier.

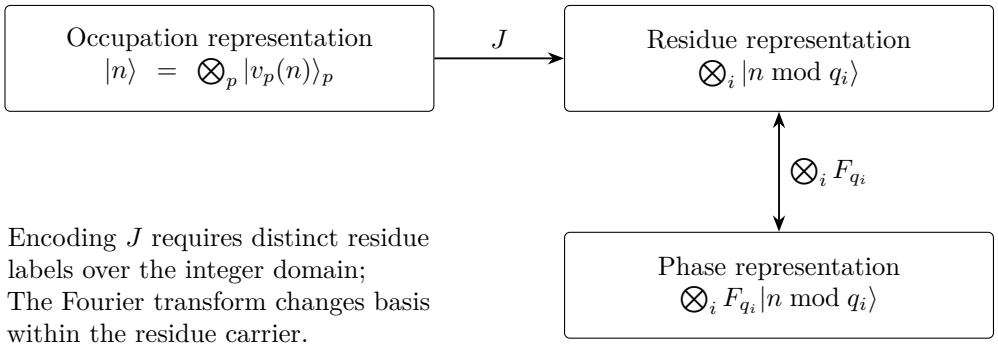


Figure 4: Occupation encoding, residue encoding, and phase representation. The map J changes both the tensor decomposition and the representation of operations. Correlations and operational locality must therefore be assessed on the encoded carrier. The transform F_q alone performs neither integer factorization nor occupation-to-residue conversion.

Retaining and reading data. Written data must remain available while used. Let Φ_t be the evolution channel during that time window and Φ_t^* its trace adjoint. If an observable X satisfies $\Phi_t^*(X) = X$, then

$$\mathrm{Tr}[X\Phi_t(\rho)] = \mathrm{Tr}[\Phi_t^*(X)\rho] = \mathrm{Tr}(X\rho).$$

For occupation storage, the program temporarily opens a coupling to write a selected mode and then disables actions changing its occupation during retention. The diagonal Hamiltonian and the occupation-diagonal character action U_χ defined in Section 4.1.2 satisfy $[N_p, H] = [N_p, U_\chi] = 0$. Let P_k be the spectral projections of N_p. If the remaining actions also preserve their sectors, then $\Phi_t^*(P_k) = P_k$. Following ideal readout by these projections, the conditional state is $\rho_k = P_k\rho P_k/\,\mathrm{Tr}(P_k\rho)$, and the probability of reading the same value again is

$$\mathrm{Tr}[P_k\Phi_t(\rho_k)] = \mathrm{Tr}(P_k\rho_k) = 1.$$

This gives quantum nondemolition (QND) readout under the retention conditions above [12]. Preserving an occupation eigenvalue and preserving coherence between different occupations impose different requirements; decay from residual couplings is calculated from the corresponding generator. These relations specify how occupation labels are retained and repeatedly read while in use.

Preparing coherent probes. We next choose weighted pure-state probes so that amplitudes of different labels jointly participate in subsequent evolution and readout. Factorization along prime

directions determines whether these amplitudes admit local preparation, while normalization determines whether a limit exists as truncations grow. The mean response of diagonal evolution can also be obtained from a mixed state with the same occupation distribution; we first establish pure-state preparation and give specific readout methods later. To describe factorization and normalization, consider an unnormalized vector $|\psi_c\rangle = \sum_n c(n)|n\rangle$ on the global occupation carrier, take $c(1) = 1$, and write $s_p = \sum_{k\geq 1} |c(p^k)|^2$. If c is multiplicative on coprime arguments, unique factorization gives

$$|\psi_c\rangle = \bigotimes_p \left(|0\rangle_p + \sum_{k\geq 1} c(p^k)|k\rangle_p \right), \qquad \||\psi_c\rangle\|^2 = \prod_p (1 + s_p).$$

Expanding the tensor product over successive finite prime sets gives coefficient $\prod_p c(p^{v_p(n)}) = c(n)$ for each integer label, while squared norms multiply. The squared norm of the difference between finite-product vectors is the sum over newly added occupation terms. Convergence in the vacuum-reference tensor product therefore requires exactly $\prod_p(1+s_p) < \infty$, equivalently $\sum_p s_p < \infty$. Conversely, if a product state has global vacuum coefficient 1, every local vacuum coefficient is nonzero and can be normalized to 1; expansion then gives the same multiplicative relation. Thus

$c(1) = 1$: $|\psi_c\rangle$ is a normalizable product state

$\Longleftrightarrow$

c is multiplicative on coprime arguments and $\displaystyle\sum_p \sum_{k\geq 1} |c(p^k)|^2 < \infty$.

If each prime direction additionally satisfies $c(p^{j+k}) = c(p^j)c(p^k)$, then $c(p^k) = c(p)^k$, giving complete multiplicativity. Normalizability then reduces to $|c(p)| < 1$ for every mode and $\sum_p |c(p)|^2 < \infty$, with

$$\||\psi_c\rangle\|^2 = \prod_p \big(1 - |c(p)|^2\big)^{-1}.$$

Product states with zero vacuum coefficient are not covered by this equivalence, for example $|11\rangle$. For a nonzero product state in this vacuum-reference carrier, only finitely many local factors can have zero vacuum component. Taking their lowest nonzero occupations as references reduces the state, by finite occupation shifts, to one with nonzero local vacuum coefficients.

The vector with amplitudes $c(n) = n^{-\sigma}$ has squared norm $\zeta(2\sigma)$ for $\sigma > 1/2$. Its normalized pure state factorizes along prime directions; a finite device prepares mode and occupation truncations of it. The number of local preparation calls is proportional to the retained mode count; precision, truncation, and physical cost per call are accounted for separately. This pure state and the mixed Gibbs state below describe coherent amplitudes and thermal occupation probabilities, respectively.

Superposition may also require correlations between modes. Take

$$|\phi\rangle = c_1|n\rangle + c_2|m\rangle, \qquad n \neq m, \qquad |c_1|^2 + |c_2|^2 = 1,$$

and partition the prime modes into $A \sqcup B$. Write $|n\rangle = |n_A\rangle|n_B\rangle$ and $|m\rangle = |m_A\rangle|m_B\rangle$. If occupations on either side coincide, that factor can be extracted and the state remains a product. If occupations differ on both sides, both pairs of local basis vectors are orthogonal, so absorbing coefficient phases into the local basis vectors gives a Schmidt decomposition. Hence

$$E_A(\phi) = -|c_1|^2 \log |c_1|^2 - |c_2|^2 \log |c_2|^2.$$

For a single-prime partition $A = \{r\}$, differing on both sides is equivalent to $v_r(n) \neq v_r(m)$ and $n/m \notin r^{\mathbb{Z}}$. For example, $(|143\rangle + |13\rangle)/\sqrt{2}$ changes only the 11 mode and is a product state. The same superposition expression has different correlation structures under different encodings or tensor partitions; preparation must match the actual carrier.

Occupation labels, phase labels, and coherent probes thus have distinct preparation and retention conditions. The program selects the appropriate carrier and applies subsequent operations within the interval during which the data remain available.

4.1.2 Arithmetic Operations

Arithmetic operations change written occupations and phases. Occupation multiplication implements multiplicative label changes and divisibility selection; phase operations implement addition and probe transformations on residue carriers.

Occupation multiplication and constraint selection. For a finite occupation projection P, write restricted multiplication as $K_m = P\mu_m P$, where m is a positive integer. It sends $|n\rangle$ to $|mn\rangle$ when both n and mn lie in the admissible domain; out-of-range inputs require corresponding outcomes from the complete instrument. Since P commutes with the diagonal Hamiltonian, the admissible action satisfies

$$[H_P, K_m] = (\log m)K_m, \qquad H_P = PHP.$$

The same label m therefore specifies both occupation increments and the energy difference. For the complete rectangular occupation box used here, denote its integer-label set by $\mathcal{N}_{\Lambda,L}$ and define the valid domain $D_m = \{n \in \mathcal{N}_{\Lambda,L} : mn \in \mathcal{N}_{\Lambda,L}\}$. Distinct inputs still have distinct outputs, so evaluation on basis vectors gives

$$K_m^\dagger K_m = P_{D_m}, \qquad e^{itH_P} K_m e^{-itH_P} = m^{it} K_m.$$

Multiplication increases each prime occupation by a nonnegative amount. Thus, whenever the final occupation remains inside the box, the intermediate occupation does as well. Consequently $K_m K_s = K_{ms}$ holds on a complete occupation box; domains with internal holes require checking intermediate occupations. The output projection $Q_m = K_m K_m^\dagger$ selects representable multiples of m in the box, giving

$$Q_m|n\rangle = \mathbf{1}_{m|n}|n\rangle, \qquad \boxed{Q_m Q_s = Q_{\mathrm{lcm}(m,s)\cdot}}$$

This turns divisibility constraints into composable sector selections. The projection is zero when no multiple is representable; reverse multiplication reduces occupations on divisible states that remain in the admissible domain.

Phase probing can use a multiplicative character χ modulo q, acting diagonally as $U_\chi|n\rangle = \chi(n)|n\rangle$. This action is unitary on the sector coprime to q. On the full occupation domain, the character vanishes on noncoprime labels and must be understood as a compressed operation embedded in a complete quantum instrument. Evaluating admissible multiplication on basis vectors gives

$$U_\chi K_m|n\rangle = \chi(mn)|mn\rangle = \chi(m)\chi(n)|mn\rangle, \qquad U_\chi K_m = \chi(m)K_m U_\chi.$$

For $(m, q) = 1$, this is a nonzero phase covariance relation; the multiplier m and character modulus q have distinct roles.

Phase addition and encoding conversion. On a finite phase carrier, relative phases and the Fourier transform implement addition. Defining $V_c|x\rangle = e^{2\pi icx/q}|x\rangle$ gives

$$V_c F_q|a\rangle = \frac{1}{\sqrt{q}} \sum_{x \bmod q} e^{2\pi i(a+c)x/q}|x\rangle = F_q|a+c\rangle.$$

Translation in position encoding is therefore $S^c = F_q^{-1} V_c F_q$, with outputs interpreted modulo q. Multiplicative scaling $M_g|a\rangle = |ga\rangle$ is unitary for $g \in \mathbb{F}_q^\times$, and

$$M_g S M_g^{-1}|a\rangle = |a+g\rangle, \qquad M_g S M_g^{-1} = S^g.$$

When g generates $\mathbb{F}_q^\times$, translation and scaling generate the affine group $\mathbb{F}_q \rtimes \mathbb{F}_q^\times$. Constant vectors form the trivial subspace, while the zero-sum subspace has dimension $q-1$. In the Fourier basis, translations distinguish nonzero frequencies and scalings act transitively on them, so the zero-sum subspace is irreducible. All irreducible representations consist of $q-1$ one-dimensional representations of the multiplicative quotient and this $q-1$-dimensional representation; their squared dimensions sum to $q(q-1)$.

The same arithmetic data can also be processed in parallel modulo several coprime moduli. After applying the appropriate $S_{q_i}^c$ to each residue channel, the result is reconstructed in the declared integer range. The encoding must cover the input and output domains being processed; outside that range, outputs are congruence classes modulo Q and cannot be read directly as untruncated integer addition.

Conversion can also change correlations between channels. For example, $(|1\rangle + |11\rangle)/\sqrt{2}$ is a single-mode superposition in occupation representation. After encoding into residue carriers modulo 13 and 17, both channels distinguish the two terms, producing an equal-amplitude state of Schmidt rank 2. Addition of a fixed constant, $\bigotimes_i S_{q_i}^c$, is a local unitary and does not change entanglement across this residue partition; correlations are established by the actual encoding or joint operation.

Transforming character probes. Coherent evaluation also requires control of probe amplitudes in different representations. The transformation between multiplicative characters and phase vectors can be calculated directly. Extend a nontrivial χ by zero at 0 and take

$$e_a(x) = q^{-1/2} e^{2\pi i a x/q}, \qquad v_\chi(x) = (q-1)^{-1/2}\chi(x), \qquad \tau(\chi) = \sum_{x \bmod q} \chi(x) e^{2\pi i x/q}.$$

For $b \neq 0$, substitution gives $\sum_x \chi(x) e^{2\pi i b x/q} = \bar{\chi}(b)\tau(\chi)$. Character orthogonality determines the magnitude of the Gauss sum:

$$|\tau(\chi)|^2 = \sum_{t\neq 0} \chi(t) \sum_{y\neq 0} e^{2\pi i (t-1)y/q} = (q-1) - \sum_{t\neq 1} \chi(t) = q.$$

Thus, for $a \neq 0$,

$$\langle e_a, v_\chi\rangle = \frac{\bar{\chi}(-a)\tau(\chi)}{\sqrt{q(q-1)}}, \qquad |\langle e_a, v_\chi\rangle| = \frac{1}{\sqrt{q-1}},$$

whereas the inner product vanishes for $a = 0$. There are $q-2$ nontrivial multiplicative characters, spanning the subspace of functions vanishing at 0 and having zero sum. Two complete mutually unbiased bases of the entire $\mathbb{C}^q$ are instead the position and Fourier bases, whose overlap magnitudes are all $q^{-1/2}$.

The same calculation for $F_q v_\chi$ gives

$$(F_q v_\chi)(a) = \frac{\tau(\chi)\bar{\chi}(a)}{\sqrt{q(q-1)}}, \qquad \boxed{F_q v_\chi = \frac{\tau(\chi)}{\sqrt{q}} v_{\bar{\chi}}.}$$

The Fourier transform thus maps this character family to itself. Among nontrivial multiplicative characters, the quadratic character χ_2 is the unique real character satisfying $\bar{\chi}_2 = \chi_2$ and hence gives an eigendirection. Its phase follows from exact evaluation of the quadratic Gauss sum: every nonzero square has two square roots, so $\tau(\chi_2) = \sum_{x \bmod q} e^{2\pi i x^2/q}$. Taking parameters $(2, q)$ in the finite Gauss reciprocity formula [13] and complex-conjugating gives

$$\frac{\tau(\chi_2)}{\sqrt{q}} = \frac{e^{\pi i/4}}{\sqrt{2}}\left(1 + e^{-\pi i q/2}\right) = \begin{cases} 1, & q \equiv 1 \pmod 4, \\ i, & q \equiv 3 \pmod 4. \end{cases}$$

These are the two possible eigenvalues of the quadratic-character vector under the positive-sign Fourier convention, depending on the modulus.

Translations, phase modulation, and Fourier transforms also combine with quadratic phases to form finite Heisenberg and Weil actions [14]. Appendix A gives their generating relations and closure as a projective representation at odd prime-power modulus; general position-dependent phase operations belong to a larger operation class. Occupation arithmetic, residue arithmetic, and phase processing are thereby connected through encodings and operations on their respective carriers.

4.1.3 Exchange and Relaxation

Exchange moves states between admissible occupations, forms statistical distributions, and affects retention of phase and label information. Component design must determine separately the target distribution, occupation relaxation rate, and availability time of written information.

Formation of equilibrium statistics. Suppose a restricted-multiplication edge has energy difference $\log m$, and its forward and reverse weak-coupling transitions arise from the same paired matrix element and an equilibrium environment at inverse temperature β. The environmental KMS relation [15] then gives

$$\frac{\Gamma(n \longrightarrow mn)}{\Gamma(mn \longrightarrow n)} = e^{-\beta \log m} = m^{-\beta}.$$

This constrains a pairwise rate ratio. Actual couplings determine absolute rates and admissible edges; combining distinct channels also requires accounting for channel multiplicities. The window sampling of Section 3 specifies a concrete boundary and rate law on this basis.

For a finite occupation carrier, the equilibrium target can be determined by maximizing entropy at fixed mean energy [16]. Variation with respect to the density operator yields $-\log \rho - I - \alpha I - \beta H = 0$, and normalization gives $\rho_\beta = e^{-\beta H}/Z$. Strict concavity of entropy gives the unique maximizer for an interior constraint value. For an additive spectrum,

$$\boxed{Z_{\Lambda,L}(\beta) = \prod_{p\in\Lambda} \sum_{k=0}^{L_p} p^{-\beta k}, \qquad P_p(k) = \frac{p^{-\beta k}}{\sum_{j=0}^{L_p} p^{-\beta j}}.}$$

These finite sums are normalizable for every finite real β, giving the uniform distribution at $\beta = 0$. For a nonconstant spectrum, $\mathrm{d}\langle H\rangle_\beta/\mathrm{d}\beta = -\mathrm{Var}_\beta(H) < 0$, so a mean energy between spectral endpoints determines β uniquely; endpoints are attained in the corresponding limits. The product formula applies to an occupation box with an additive spectrum. With cross-mode constraints or evidence energies, the partition function must be recalculated from the actual total energy.

This factorization also gives composition relations for independent mode groups. If two occupation boxes have disjoint prime supports, unique factorization makes $W_\times(|n\rangle \otimes |s\rangle) = |ns\rangle$ a unitary isomorphism onto the combined box. Since $\log(ns) = \log n + \log s$,

$$W_\times(H_1 \otimes I + I \otimes H_2)W_\times^\dagger = H_{12}, \qquad Z_{12} = Z_1 Z_2,$$

and $\rho_{12,\beta} = W_\times(\rho_{1,\beta} \otimes \rho_{2,\beta})W_\times^\dagger$. If primes are shared, $|p\rangle|1\rangle$ and $|1\rangle|p\rangle$ yield the same product, so such multiplication must retain inputs or auxiliary records to preserve reversibility.

The ideal limit of all primes and unbounded occupations requires additional convergence conditions. The entropy of the single-mode geometric distribution $P(k) = (1-z)z^k$ is

$$S(z) = -\log(1-z) - \frac{z}{1-z}\log z, \qquad \frac{\mathrm{d}S}{\mathrm{d}z} = -\frac{\log z}{(1-z)^2}.$$

Under the energy constraint $\sum_p (\log p) z_p/(1-z_p) = E$, using $\mathrm{d}[z/(1-z)]/\mathrm{d}z = (1-z)^{-2}$ makes the stationarity condition $-\log z_p = \beta \log p$, hence

$$z_p = p^{-\beta}, \qquad Z = \prod_p (1-p^{-\beta})^{-1} = \zeta(\beta), \qquad \beta > 1.$$

In this range, $E = -\zeta'(\beta)/\zeta(\beta)$ decreases strictly with β. Maximum entropy determines equilibrium weights; the time to reach them must be calculated from a specified generator.

Relaxation of occupation distributions. Once stationary weights are fixed, the required waiting time depends on exchange rates along admissible edges. Nearest-level exchange on a tree gives a calculable construction [17]. After choosing a direction on a $(q+1)$-regular tree, each vertex has one upward edge and q downward edges. Equal edge weights give levelwise upward and downward rates $a = \kappa/(q+1)$ and $b = q\kappa/(q+1)$, with ratio $1/q$. This level-model ratio equals the thermal ratio for gap $\log q$ at $\beta = 1$.

For finite occupation levels, take $H_S = (\log q)N$ and couple raising and lowering actions to an equilibrium reservoir. If reservoir correlation functions are integrable and satisfy the conditions for the weak-coupling limit, the Davies construction gives a GKSL generator [18,19]. Forward and reverse coefficients obey the KMS ratio; specific jump channels, actions within degenerate spaces, and absolute rates are determined by the coupling.

For a birth–death chain with $n \geq 2$ levels and reflecting endpoints, let the constant upward and downward rates be $a, b > 0$. A diagonal similarity transform makes the generator real symmetric and tridiagonal, with off-diagonal entries $\sqrt{ab}$. Reflecting boundary conditions give angular parameters $m\pi/n$ for the nonzero modes, hence

$$\gamma_m = a + b - 2\sqrt{ab}\cos\frac{m\pi}{n}, \qquad m = 1, \ldots, n-1.$$

Substituting the equal-edge level rates gives

$$\boxed{\gamma_m^{(n)} = \kappa\left[1 - \frac{2\sqrt{q}}{q+1}\cos\frac{m\pi}{n}\right].}$$

As n increases, the finite-level gap approaches $\kappa[1 - 2\sqrt{q}/(q+1)]$. A general thermal ratio is realized by $a = \kappa/(1+q^\beta)$ and $b = \kappa q^\beta/(1+q^\beta)$, which can be substituted into the same spectral formula. The full finite-chain derivation agrees with the window generator in Section 5. Figure 5 shows the finite-level results and their half-line limits.

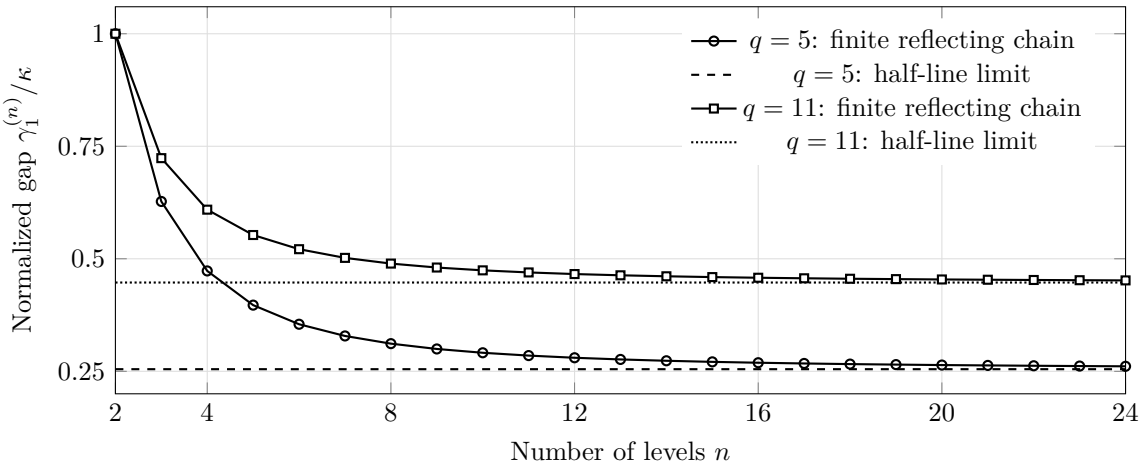


Figure 5: Finite-level gaps and half-line limits for the reflecting radial chain. Curves use the exact formula with $\beta = 1$, $a = \kappa/(q+1)$, and $b = q\kappa/(q+1)$. They describe relaxation only for this generator, not full-ring label mixing or general task completion times.

Retention of phases and labels. Phase correlations or label information may continue to decay after occupations have become stationary. Their availability time depends on exchange on the relevant carrier. For an odd prime q and $K \geq 1$, describe this exchange on the finite unit group $G = (\mathbb{Z}/q^K\mathbb{Z})^\times$. Given a nonnegative step measure ν, its generator on observable functions is

$$(Lf)(x) = \sum_{g \in G} \nu(g)\,[f(xg) - f(x)], \qquad \nu(1) = 0.$$

Multiplicative characters diagonalize this action. Writing $\widehat{\nu}(\chi) = \sum_g \nu(g)\chi(g)$ gives $L\chi = [\widehat{\nu}(\chi) - \widehat{\nu}(1)]\chi$. If the response on the entire dual group is specified and compatible with a nonnegative step measure, Fourier inversion gives

$$\nu(g) = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \widehat{\nu}(\chi)\overline{\chi(g)}.$$

The complete response and the self-jump convention at $\nu(1) = 0$ jointly determine this measure. If $\nu(g) = \nu(g^{-1})$, the generator is reversible with respect to the uniform measure.

A localized retention model uses a step-weight parameter $\beta > 1$. Writing $\Lambda_{\mathrm{VM}}(n)$ for the von Mangoldt function, specify, for $g \neq 1$,

$$\nu_s(g) = \frac{1}{2} \sum_{n \equiv g \bmod q^K} \Lambda_{\mathrm{VM}}(n) n^{-\beta} + \frac{1}{2} \sum_{n \equiv g^{-1} \bmod q^K} \Lambda_{\mathrm{VM}}(n) n^{-\beta}, \qquad \nu_s(1) = 0.$$

Absolute convergence makes this a finite measure, and symmetrization ensures reversibility. Let a_χ denote the corresponding character multiplication operator. Substitution into the generator gives

$$La_\chi = -d_\chi(\beta) a_\chi, \qquad d_\chi(\beta) = \sum_{(n,q)=1} \Lambda_{\mathrm{VM}}(n)\big(1 - \operatorname{Re}\chi(n)\big) n^{-\beta}.$$

A nontrivial character differs from 1 on some positive integer n representing a unit of the residue ring. Factoring n into primes yields at least one prime factor p satisfying $\chi(p) \neq 1$; its positive Mangoldt weight gives $d_\chi(\beta) > 0$. The corresponding expectation decays as $e^{-t d_\chi(\beta)}$; multiplying all transition rates by one constant multiplies $d_\chi(\beta)$ by the same constant. The Dirichlet series for the logarithmic derivative gives

$$d_\chi(\beta) = -\frac{\zeta'}{\zeta}(\beta) - \frac{\log q}{q^\beta - 1} + \operatorname{Re}\frac{L'}{L}(\beta, \bar{\chi}).$$

This gives the decay time of character expectations in the unit-group exchange model. Here β is a step-weight parameter; a thermal-bath interpretation requires a corresponding energy-exchange construction. Exchange along the valuation direction additionally requires retaining that direction in the joint carrier.

Another label-retention problem arises on a finite full ring. For an odd prime q and $K \geq 2$, take $\{x, qx \bmod q^K\}$ as nontrivial edges on $\mathbb{Z}/q^K\mathbb{Z}$, assigning symmetric rate $\kappa/(q+1)$ to every undirected edge. The pointwise stationary measure is uniform, while level probabilities include level-size degeneracies. This symmetric exchange between points is defined separately from the occupation thermal chain assigning $H_S = (\log q)N$ above.

In this full-ring model, primitive character components have damping rate $\kappa/(q+1)$, whereas the slowest rate of nonzero-label exchange through the 0 sector is $\Theta(q^{-K})\kappa$. Appendix B reduces the slowest decay in a nontrivial Teichmüller-label sector to an absorbing chain of depth K.

Using κ as the rate unit, its mean absorption time is

$$h_j = \frac{q+1}{(q-1)^2}\big(q^{K+1} - q^{j+1}\big) - \frac{q+1}{q-1}(K-j), \qquad 0 \leq j \leq K.$$

The positive lowest eigenfunction and variational estimates using absorption times bound the lowest rate by

$$\frac{1}{h_0} \leq \frac{\gamma_{\mathrm{tag}}}{\kappa} \leq \frac{1}{\mathbb{E}_\pi[h]}, \qquad \frac{\gamma_{\mathrm{tag}}}{\kappa} = \frac{(q-1)^2}{(q+1)q^{K+1}}\big(1 + O(Kq^{-K})\big),$$

The labels here are Teichmüller labels of nonzero elements: unit classes modulo q retained after stripping powers of q. Nonabsorbed levels are $j = 0, \ldots, K-1$; $j = K$ is absorbing with $h_K = 0$. The level weights $\pi_j = q^{-j} / \sum_{r=0}^{K-1} q^{-r}$ are normalized reversible weights on the nonabsorbed levels.

For $q \geq 5$ and sufficiently deep towers, the slow label rate, primitive-character rate, and radial band provide distinct representative scales. Low-conductor characters generally contain several decay modes, and complete mixing requires the corresponding full generator. Each rate must therefore be used together with its carrier, boundary, and generator.

Approximation of coupled responses. Nonlinearly coupled components can also be analyzed by mean-field approximation within a specified model and time window. Let $A(t)$ be the exact first-moment vector and $A^{\mathrm{MF}}(t)$ the mean-field solution with the same initial condition; set $\mathrm{err}(t) = \|A(t) - A^{\mathrm{MF}}(t)\|$ and use a fixed Euclidean norm. Suppose the connected-correlation quantity $C(t)$ and the first-moment error initially vanish. On a time window where the actual model satisfies the growth bounds

$$C' \leq 2M\|f\|gC + c_1 M^3 \|f\|g,$$
$$\mathrm{err}' \leq 2M\|f\|g\,\mathrm{err} + \|f\|gC,$$

with all quantities nonnegative and $M \geq 1$, two integrations give

$$\mathrm{err}(t) \leq \frac{c_1}{2} M^3 \|f\|^2 (gt)^2 e^{2M\|f\|gt}.$$

Here g is the nonlinear coupling strength and f a fixed coupling tensor; the quantities M, c_1 are uniformly controlled on the time window used. The two growth bounds must be established from moment and correlation estimates for the actual generator, including dissipative and correlation-source contributions. Appendix C gives the subsequent integration in full. The approximation accuracy is thereby tied to the actual component dynamics.

4.1.4 Control and Readout

Control arranges preparation, arithmetic operations, and exchange into the process required by the program. Boundaries may select coupling switches, temperature schedules, observation times, and feedback branches; later choices depend only on records already obtained. Coherent, exchange, monitored, and feedback processes can be connected as needed within one program. Control may therefore also act at intermediate stages of the process.

At completion or within a specified sampling window, instruments acquire occupation, phase, or response data. The history instruments of Section 2 give record probabilities and conditional states together; the interpretation relations of Section 3 turn those records into task outputs. An occupation control register may use the decoupling and spectral-projection readout of Section 4.1.1 to retain records during subsequent control. Unknown quantum states participate through admissible channels without the controller first obtaining their complete classical description.

Readout capabilities must be specified relative to the actual carrier. For example, on the finite unit group G, character multiplication operators generate the commutative algebra

$$\mathcal{A} = \mathrm{span}_{\mathbb{C}}\{a_\chi : \chi \in \widehat{G}\}, \qquad (a_\chi f)(x) = \chi(x) f(x).$$

Define unit-group inversion $Rf(x) = f(x^{-1})$ and $\sigma(A) = RAR^\dagger$. Inversion here always means $x \mapsto x^{-1}$ on the unit group. The operator R is a linear unitary involution satisfying $\sigma(a_\chi) = a_{\bar\chi}$. If direct measurement cannot distinguish this inversion pair, its reportable self-adjoint quantities belong to

$$\boxed{\mathcal{A}_{\mathrm{sa}}^{\sigma} = \mathrm{span}_{\mathbb{R}}\{a_\chi + a_{\bar\chi} : \chi \in \widehat{G}\}.}$$

For $\chi \neq \bar\chi$, inversion invariance of $c_1 a_\chi + c_2 a_{\bar\chi}$ requires $c_1 = c_2$, while self-adjointness requires $c_2 = \bar{c}_1$; the coefficients are therefore equal and real. Real characters are themselves self-adjoint invariants, and $a_\chi a_{\bar\chi} = I$. The inversion-odd quantity $i(a_\chi - a_{\bar\chi})$ is not reported by this direct readout, but subsequent couplings may convert its differences into other readable quantities.

This inversion symmetry corresponds only to character conjugation. It is weaker than the full Galois action and differs from antiunitary complex conjugation of state amplitudes. If an added

occupation algebra commutes with $\mathcal{A}$ and inversion acts as the identity on the occupation factor, the two form a joint commutative readout algebra. If inversion acts on several phase carriers simultaneously, products of local odd quantities may be globally invariant; if all operators are allowed, the fixed part is the commutant of R and is generally noncommutative. The measurement scope must follow the joint carrier and instruments used.

Readable records may be displayed, used for statistical estimation, or used to select subsequent preparation and couplings. If an external electronic controller makes these selections, its state, timing, and communication belong to the actual control process. The arithmetic module thus supplies a complete set of actions from data entry and retention through processing to acquisition of results.

4.2 Interaction Module

The interaction module connects state processing in the arithmetic module to external processes. Input transduction allows fields or other systems to affect occupations, phases, and boundaries; output actions send records or physical states outward. Both modules use the same joint machine carrier: incoming external information becomes the states and conditions being processed, and outgoing results become inputs to other physical processes.

4.2.1 Transduction of External Conditions

Transduction uses actual coupling to convert distinguishable external features into quantities available to the machine. Light, sound, force, temperature, or chemical actions may change field parameters or produce occupation increments, phase changes, or constraint energies. Writing s_{phys} for external conditions, transduction gives

$$\mathcal{T}_\Lambda : s_{\mathrm{phys}}(t) \longmapsto b(t).$$

where $b(t)$ collects actual conditions entering preparation, operations, and observation. For a given sensor, response range, noise, and bandwidth determine which external differences can be transmitted. Weak signals may be read through counting or phase accumulation over time; continuous tracking requires both transduction and subsequent processing within the task's time window.

The correspondence between features and internal degrees of freedom is determined jointly by sensing coupling, task definition, and calibration. Visual phase delays, frequency bands, or motion features, and acoustic harmonics, directions, or source features, all require appropriate devices to form usable data. Transduction first supplies physical input; an explicit statistical model then specifies its evidential role in the task.

4.2.2 Formation of Evidence and Constraints

An inference task expresses the relation between input readings and candidate states as a likelihood. Let $\boldsymbol{v}$ denote a candidate state carried by the machine, with observed data given and likelihood $L_i(\boldsymbol{v}) > 0$ for channel i. Define the dimensionless evidence potential

$$\Phi_i(\boldsymbol{v}) = -\log L_i(\boldsymbol{v}).$$

Choose a reference inverse temperature $\beta_0 > 0$ and implement the energy increment $\Delta E_i = \Phi_i/\beta_0$, giving $e^{-\beta_0 \Delta E_i} = L_i$. In units where $\beta_0 = 1$, this becomes $E_i = -\log L_i$. A common shift of the energy zero leaves normalized results unchanged.

When the data channels are conditionally independent given $\boldsymbol{v}$,

$$L(\boldsymbol{v}) = \prod_i L_i(\boldsymbol{v}), \qquad \Phi(\boldsymbol{v}) = \sum_i \Phi_i(\boldsymbol{v}).$$

If the original energy and boundary produce prior weights $\pi_0(\boldsymbol{v}) \propto e^{-\beta_0 E_0(\boldsymbol{v})}$ at the reference inverse temperature, adding these realizable diagonal terms gives total energy $E_{\text{tot}} = E_0 + \sum_i \Phi_i/\beta_0$. At that same reference inverse temperature, the equilibrium weights are

$$\pi(\boldsymbol{v} \mid \text{data}) \propto e^{-\beta_0 E_0(\boldsymbol{v})} \prod_i L_i(\boldsymbol{v}) \propto \pi_0(\boldsymbol{v}) L(\boldsymbol{v}).$$

Here E_0 includes the BC base energy that remains active. A separately specified prior requires implementing its corresponding total energy; the base-energy weight cannot be omitted. Keeping E_{tot} fixed and operating at inverse temperature β instead gives $\pi_\beta \propto [\pi_0 \prod_i L_i]^{\beta/\beta_0}$, a tempered posterior.

Evidence thus participates in the established statistical process through boundary energies. Realizing this relation requires a device able to write the energy terms and an exchange mechanism reaching the declared error. A zero likelihood corresponds to excluded support or an appropriate finite-penalty approximation. Conditional independence permits likelihood multiplication, while the locality of factors determines the required couplings. Several factors may depend on the same candidate variable, but correlated observations require their joint likelihood and must not be multiplied repeatedly as independent evidence.

Both predefined constraints and evidence updated after observation may select admissible states or alter their weights. Data, models, and calibration determine the update; actual couplings write it into the machine, and the resulting dynamics produce the computational output.

4.2.3 Actuation and Feedback

Actuation converts computational results into external actions. A program extracts controls from retained records and drives actuators to change fields, device positions, or inputs to other systems; the new external conditions enter the machine again through transduction. Let $u_{\text{act}} = C_\pi(r_{\text{all}})$ denote the actuator control rule. One loop can be written as

$$s_{\text{phys}} \longrightarrow b \longrightarrow \text{preparation, processing, and observation} \longrightarrow (r_{\text{all}}, \rho^+) \longrightarrow s'_{\text{phys}}.$$

Records control actuators, while post-measurement states may continue to participate through physical coupling. For occupation registers, retention and readout conditions keep control labels available over the relevant time window; other feedback schemes use their actual instruments and transmission channels. Actuation latency, resolution, and noise, together with transduction response, determine the attainable time and accuracy ranges of the loop.

Feedback always uses records already produced. If external conditions, internal states, and controllers retain correlations, the next run continues from the corresponding joint state. Fresh auxiliaries are prepared conditional on available records. Continuous computation thereby follows one physical process, with the world state after actuation providing actual conditions for subsequent computation.

4.2.4 Intermachine Coupling and External Interfaces

Another machine may itself be an input or output physical system. Couplings between compatible modes can transmit occupation and phase relations, records can select the other machine's boundary and preparation, and unknown quantum states pass through admissible physical channels. A connection must specify transmitted state ranges, records, conserved-charge flows, and timing so that the preceding output belongs to the receiving input class.

Different encodings require conversion within the connection. When connecting to conventional networks, displays, or archives, occupation, phase, or response readings can be converted into numbers, images, audio, and files; actual measurement and encoding perform these conversions. For connections retaining quantum states, readout records and state transmission are specified

separately. A mathematical description of a quantum state cannot be treated as classical data already obtained.

Intermachine coupling thus extends a single machine's operational scope, allowing records and delivered states from one stage to participate directly in the next task.

The arithmetic and interaction modules jointly constitute the complete BC state-space computer: internal data preparation, retention, transformation, and statistical formation connect to the outside through transduction and return, while resulting states and records remain available for reuse. A concrete task must select and compose these actions to determine the actual delivered result and its completion time.

5 Task Families and Example Runs

The components of Section 4 must be organized by tasks to deliver values, samples, or actions meeting specified requirements. Section 3 compiled multiplicative conditional sampling into occupation windows, an exchange mechanism, and a readout schedule. Following that program, we can calculate the target distribution and the running time required to reach a prescribed error. Other arithmetic, detection, and control tasks combine the same preparation, operation, and interaction capabilities, with their targets determining execution and delivery.

5.1 Multiplicative Conditional Sampling

5.1.1 Sample Distributions from Occupation Windows

Figure 6 gives a concrete example of window compilation. The input requires $11 \mid n \mid 11^2 13^2$ and sampling with relative weights n^{-1} at $\beta = 1$. The 11 mode therefore admits occupations $1, 2$, and the 13 mode admits $0, 1, 2$; together they give the six integers shown.

The machine first sets these windows and prepares the lower corner $|11\rangle$. It then enables nearest-neighbor exchange within the windows, bringing the occupation distribution toward the required weights. Reading both occupations at the prescribed time gives an integer sample. Independent preparation and execution repeated over several runs estimate distribution statistics. Double circles mark states retained after adding $13 \mid n$: the new condition changes the window and preparation corner, while exchange continues along the retained nearest-neighbor edges.

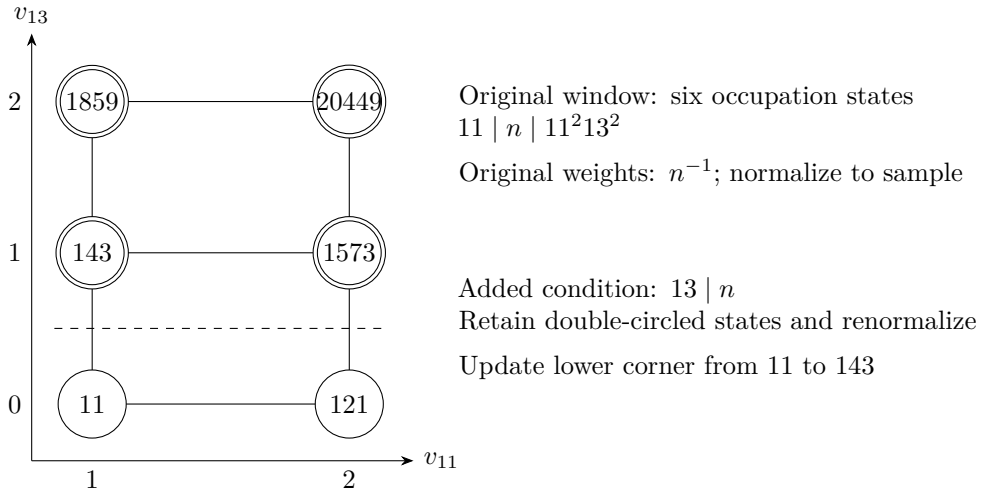


Figure 6: Compilation and updating of a divisibility window. Edges connect adjacent occupations; transitions outside the window are closed. The added condition removes the lowest row while retaining the same internal exchange law. Integers label states; circle sizes do not represent probabilities.

In general, the input supplies $n_- \mid n_+$ in prime-exponent form and an inverse temperature $\beta > 0$. Within the declared finite mode set Λ and occupation cutoffs L_p, write

$$n_- = \prod_{p\in\Lambda} p^{\ell_p}, \qquad n_+ = \prod_{p\in\Lambda} p^{u_p}, \qquad 0 \le \ell_p \le u_p \le L_p.$$

These exponents directly specify local windows. Let the mode count be $N_{\mathrm{mode}} = |\Lambda|$, the window lengths $d_p = u_p - \ell_p + 1$, and write $z_p = p^{-\beta}$. The required samples follow

$$\boxed{\varpi_{n_-,n_+;\beta}(n) = \frac{n^{-\beta}}{Z_{n_-,n_+;\beta}} \mathbf{1}_{\{n_-\mid n\mid n_+\}}.}$$

This divisibility window is specified jointly by prime occupations and differs from a numerical interval $n_- \le n \le n_+$. Substituting $v_p(n) = \ell_p + j_p$ into the weights, unique factorization separates the finite sum mode by mode:

$$Z_{n_-,n_+;\beta} = n_-^{-\beta} \prod_{p\in\Lambda} \frac{1 - z_p^{d_p}}{1 - z_p}, \qquad \varpi_p(\ell_p + j) = \frac{(1 - z_p) z_p^j}{1 - z_p^{d_p}}, \quad 0 \le j < d_p.$$

The program disables transitions beyond the window, retains the spectrum $H = \sum_p (\log p) N_p$ inside it, and uses the fixed nearest-neighbor exchange law

$$\Gamma_{p,+} = \frac{\kappa_p z_p}{1 + z_p}, \qquad \Gamma_{p,-} = \frac{\kappa_p}{1 + z_p}, \qquad \kappa_p > 0.$$

where κ_p sets the mode's time scale. Occupation distributions evolve autonomously, and exchanges are independent across modes. For example, separate jump operators $\sqrt{\Gamma_{p,+}}|k+1\rangle\langle k|$ and $\sqrt{\Gamma_{p,-}}|k\rangle\langle k+1|$ for each $k = \ell_p, \ldots, u_p - 1$, together with a diagonal Hamiltonian, yield this occupation generator. Its adjoint preserves the occupation-diagonal algebra. The forward-to-reverse ratio agrees with the thermal weight of the energy gap, while mode clock rates specify absolute rates. Section 8.2.2 supplies a finite-collision construction realizing this occupation evolution.

Every internal edge satisfies

$$\varpi_p(\ell_p + j)\Gamma_{p,+} = \varpi_p(\ell_p + j + 1)\Gamma_{p,-}.$$

The stationary distribution of the finite connected chain is therefore ϖ_p, and the joint stationary distribution of independent modes is the target ϖ. The input specifies only two boundaries per mode, without writing the weights of all $\prod_p d_p$ states individually. If inputs are initially integers supplied in positional notation, obtaining their prime exponents remains part of input conversion.

The six weights in Figure 6 thus arise jointly from two local windows and the same exchange law. Delivering a sample in an actual run additionally requires bounding the evolved distribution's error relative to the target, starting from the prepared initial state.

5.1.2 Relaxation Spectrum and Joint-Distribution Error

Sampling is complete when the entire readout distribution is within the specified error of its target. To determine the waiting time, we first derive the single-mode relaxation spectrum and then combine independent-mode errors into a joint-distribution bound. Consider a reflecting chain of length $d \ge 2$ with upward and downward rates $\alpha, \delta > 0$. Its interior generator on observables is

$$(Qf)_j = \alpha(f_{j+1} - f_j) + \delta(f_{j-1} - f_j), \qquad \varpi_j \propto (\alpha/\delta)^j,$$

Terms crossing either endpoint are omitted. Set $D = \mathrm{diag}(\sqrt{\varpi})$. The similarity transform $S = DQD^{-1}$ makes $-S$ a real symmetric tridiagonal matrix, with off-diagonal entries $-\sqrt{\alpha\delta}$, interior diagonal $\alpha + \delta$, and endpoint entries α, δ. The interior eigenvalue equation becomes

$$u_{j+1} + u_{j-1} = 2\cos\theta\, u_j, \qquad \gamma = \alpha + \delta - 2\sqrt{\alpha\delta}\cos\theta.$$

The reflecting boundaries are equivalent to $u_{-1} = \sqrt{\delta/\alpha}\, u_0$ and $u_d = \sqrt{\alpha/\delta}\, u_{d-1}$. Taking

$$u_j = \sin((j+1)\theta) - \sqrt{\delta/\alpha}\sin(j\theta)$$

satisfies the left boundary; substitution at the right gives $\gamma \sin(d\theta) = 0$. The nonzero modes take $\theta_m = m\pi/d$, $m = 1, \ldots, d-1$, giving $d-1$ distinct eigenvalues. Together with the stationary zero mode, these form the complete d-dimensional spectrum. Substitution of the actual rates yields

$$\boxed{\gamma_{p,m} = \kappa_p \left[1 - \frac{2\sqrt{z_p}}{1+z_p}\cos\frac{m\pi}{d_p}\right], \quad m = 1, \ldots, d_p - 1.}$$

Write $\gamma_p = \gamma_{p,1}$. Coordinates of length $d_p = 1$ are fixed by preparation and do not enter the comparison of slowest relaxation rates; no exchange waiting time is needed if all coordinates are fixed. Otherwise, the independent occupation generator is a tensor sum of local generators, with spectrum given by sums of local eigenvalues. If the task family has uniform $z_p \le z_* < 1$ and $\kappa_p \ge \kappa_* > 0$, then

$$\boxed{\mathrm{gap}(\mathcal{L}_{\mathrm{occ}}) = \min_{p:d_p \ge 2} \gamma_p \ge \gamma_* := \kappa_* \frac{(1-\sqrt{z_*})^2}{1+z_*} > 0.}$$

For example, $\beta \ge \beta_{\min} > 0$ permits $z_* = 2^{-\beta_{\min}}$. Under these conditions, the occupation-chain gap bound does not decrease as windows grow. Translating local relaxation rates into full-sample error also requires the initial state and mode independence. Other coherence or label degrees of freedom do not thereby acquire the same gap.

Use the lower-corner initial state $|n_-\rangle$, denote the occupation distribution of mode p by $\nu_{p,t}$, and define

$$\chi^2(\nu\|\varpi) = \sum_x \frac{(\nu(x) - \varpi(x))^2}{\varpi(x)}.$$

The lower-corner stationary mass satisfies $\varpi_p(\ell_p) \ge 1 - z_*$, hence $\chi^2_p(0) = 1/\varpi_p(\ell_p) - 1 \le C_* := z_*/(1-z_*)$. Expand the nonconstant part of the relative density in the reversible eigenmodes above. Each mode amplitude decays at least as $e^{-\gamma_* t}$, giving $\chi^2_p(t) \le C_* e^{-2\gamma_* t}$. Both initial state and evolution are modewise independent, so expectations of squared relative densities multiply:

$$1 + \chi^2(\nu_t\|\varpi) = \prod_p (1 + \chi^2_p(t)) \le \exp\!\left(N_{\mathrm{mode}} C_* e^{-2\gamma_* t}\right).$$

Applying Cauchy–Schwarz, $\|\nu - \varpi\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\chi^2(\nu\|\varpi)}$, gives the full-sample error bound

$$\boxed{\|\nu_t - \varpi\|_{\mathrm{TV}} \le \frac{1}{2}\sqrt{\exp(N_{\mathrm{mode}} C_* e^{-2\gamma_* t}) - 1}.}$$

For $0 < \varepsilon \le 1/2$, running until

$$\boxed{t_\varepsilon = \frac{1}{2\gamma_*}\left[\log\frac{N_{\mathrm{mode}} C_*}{\log(1+4\varepsilon^2)}\right]_+, \qquad [x]_+ = \max(x, 0),}$$

ensures joint total variation at most ε. With fixed $z_* < 1$ and a uniform lower bound on mode clock rates, the required exchange time is $O(\gamma_*^{-1}\log(N_{\mathrm{mode}}/\varepsilon))$. Starting from the specified lower corner, this bound controls every sample coordinate jointly, rather than only one mode's mean.

Figure 7 uses $\beta = 1$ and $\kappa_{11} = \kappa_{13} = 1$ for the window of Figure 6. One may take $z_* = 1/11$, $C_* = 0.1$, and $\gamma_* \simeq 0.447229$. For error 0.01, the bound gives a guaranteed time of approximately 6.94812. Direct evolution of the finite chain may converge faster than this uniform bound.

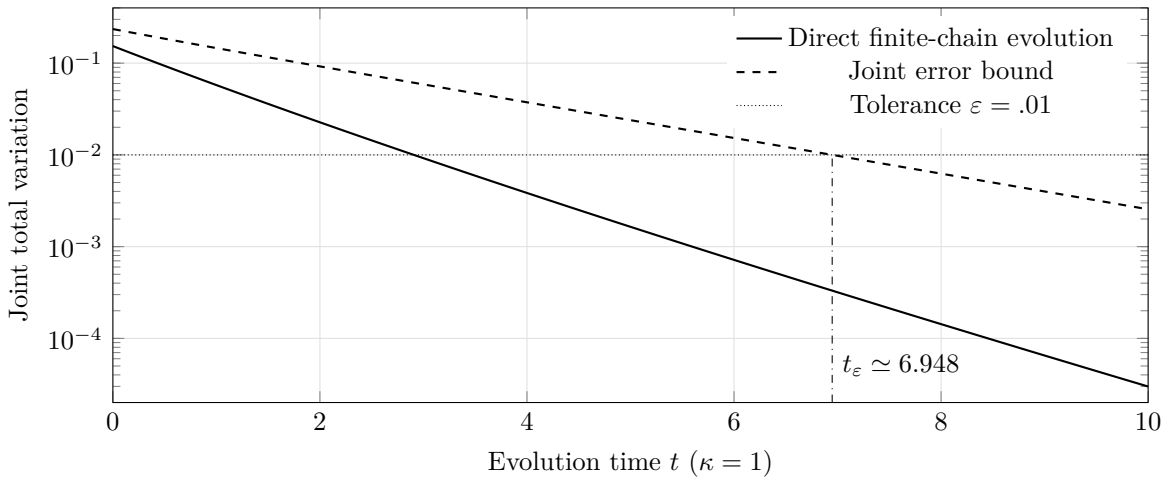


Figure 7: Joint convergence for the six-state window. The modes are independent, $\beta = \kappa_{11} = \kappa_{13} = 1$, and the initial state is $|11\rangle$. The solid curve comes from uniformization of the finite chain; the dashed curve uses the uniform gap bound in the text. Both are model calculations, not device observations. The stated time bound is a sufficient waiting time for the required tolerance.

5.1.3 Condition Updates and Sequential Sampling

A new condition $m \mid n$ is expressed by the same boundary class. Given the prime valuations of m, set $n'_- = \mathrm{lcm}(n_-, m)$. If $n'_- \nmid n_+$, the feasible set is empty and the program reports infeasibility; otherwise,

$$\{n_- \mid n \mid n_+,\ m \mid n\} = \{n'_- \mid n \mid n_+\},$$

Conditioning preserves relative weights $n^{-\beta}$, and renormalization gives

$$\boxed{\varpi_{n_-,n_+;\beta}(\,\cdot \mid m \mid n) = \varpi_{\mathrm{lcm}(n_-,m),n_+;\beta}\cdot}$$

The corresponding local update is $\ell'_p = \max(\ell_p, v_p(m))$. Multiple conditions accumulate through modewise maxima. After preparing $|n'_-\rangle$ afresh, the same exchange law and completion-time bound apply. Adding $13 \mid n$ in Figure 6 removes the row $v_{13} = 0$ and resets the lower corner.

Alternatively, drawing independent exact samples from the original distribution and rejecting those violating the new condition gives acceptance probability

$$P = \frac{Z_{n'_-,n_+;\beta}}{Z_{n_-,n_+;\beta}} = \prod_p z_p^{\ell'_p-\ell_p}\,\frac{1 - z_p^{u_p-\ell'_p+1}}{1 - z_p^{u_p-\ell_p+1}}.$$

The number of trials until first acceptance is geometric with mean $1/P$. In the two-mode example at $\beta = 1$, $P = 14/183$, requiring approximately 13.07 trials on average. The exchange-time bound for direct boundary updating contains no such rejection-trial factor; update costs comprise boundary setting, preparation of the new lower corner, and energy supply.

Record feedback may also select condition m_h at round j from history h. Each round prepares the corresponding lower corner afresh and uses a fresh environment independently prepared conditional on that history. If correlations with the old environment are retained, the conditional run law must be rechecked in the joint model. Let $K_j(h,\cdot)$ and $\widetilde{K}_j(h,\cdot)$ be the ideal and actual complete-record kernels, using the same causal control rule and satisfying, at every ideally reachable history,

$$\|K_j(h,\cdot) - \widetilde{K}_j(h,\cdot)\|_{\mathrm{TV}} \le \varepsilon_j.$$

With identical initial records, let $P_j, \widetilde{P}_j$ be the history distributions through round j. Inserting $P_{j-1}\widetilde{K}_j$ between them and using contraction under kernels gives

$$\begin{aligned}\|P_j - \widetilde{P}_j\|_{\mathrm{TV}} &\le \|P_{j-1}\widetilde{K}_j - \widetilde{P}_{j-1}\widetilde{K}_j\|_{\mathrm{TV}} + \|P_{j-1}K_j - P_{j-1}\widetilde{K}_j\|_{\mathrm{TV}} \\ &\le \|P_{j-1} - \widetilde{P}_{j-1}\|_{\mathrm{TV}} + \varepsilon_j.\end{aligned}$$

Zero-probability histories are completed by valid kernels, and infeasible conditions and timeouts remain records. Induction bounds the complete-history error over N_{run} rounds by $\min(1, \sum_j \varepsilon_j)$. If all rounds share mode-count bound $N_{\max}$ and the same z_*, γ_*, assigning error $\varepsilon/N_{\mathrm{run}}$ per round permits cumulative exchange time

$$O\left(N_{\mathrm{run}}\gamma_*^{-1}\log\frac{N_{\max}N_{\mathrm{run}}}{\varepsilon}\right).$$

Multiplicative conditional sampling therefore forms a reusable family of finite boundary programs. For the stated exponent inputs and exchange model, window compilation realizes divisibility exactly; evolution from the lower corner until t_ε yields a joint sample distribution within total variation ε of the target; new divisibility conditions compile into windows of the same class and retain the run guarantee after fresh preparation; and, for history-dependent conditions, per-round kernel errors cumulatively control the complete-history error. These results follow respectively from unique factorization, detailed balance, reversible spectral expansion, and contraction of record kernels.

The same occupation structure thus connects task representation, boundary updating, and sample formation, allowing single-run results to enter sequential programs. The cumulative exchange time assumes independent parallel modes. The full task also includes input conversion, boundary updating, preparation, reset, and readout.

5.2 Arithmetic Tasks

Sampling delivers a sample from a specified distribution; arithmetic tasks further specify values or relations to obtain from inputs. Direct operations transform inputs by known rules, whereas answers specified only by constraints require candidate generation and verification. These tasks likewise obtain inputs and deliver results through the interaction module. With external conditions fixed, we can examine how internal occupations, phases, and constraints participate.

5.2.1 Multiplication and Phase Addition

A multiplication task supplies n, m and requests nm. Write the input as occupation state $|n\rangle$ and apply the restricted multiplication K_m of Section 4 within its admissible domain, obtaining $|nm\rangle$. The output occupations satisfy

$$\boldsymbol{v}_\Lambda(nm) = \boldsymbol{v}_\Lambda(n) + \boldsymbol{v}_\Lambda(m).$$

During subsequent retention, disable actions changing these occupations and read the mode eigenvalues to obtain the product in exponent representation. If the integer in positional notation is required, perform the corresponding reconstruction. A complete instrument reports out-of-range inputs as failures; writing, occupation increments, control, and readout each impose actual time and precision requirements.

Addition modulo q uses phase representation. For given $a, c \in \mathbb{Z}/q\mathbb{Z}$, prepare $F_q|a\rangle$, apply V_c, and then perform the inverse transform and position readout:

$$F_q|a\rangle \xrightarrow{V_c} F_q|a+c\rangle \xrightarrow{F_q^{-1}} |a+c \bmod q\rangle.$$

The required coherence is between physical components. Complete dephasing in the $|k\rangle$ basis before V_c gives

$$\mathcal{E}_k(F_q|a\rangle\langle a|F_q^\dagger) = \sum_k |k\rangle\langle k|F_q|a\rangle\langle a|F_q^\dagger|k\rangle\langle k| = I/q.$$

Subsequent unitary operations preserve I/q, reducing the correct-reading probability from 1 to $1/q$. By contrast, dephasing in the final output basis after obtaining $|a+c\rangle$ does not affect that reading. Protection of phase-slope encoding therefore depends on the basis and timing of dephasing; discrimination of a classical carrier's global phase cannot replace this acceptance test.

These tasks directly use the established operator relations: multiplication changes occupations, while phase addition uses relative phases and their readout. Ideal relations determine candidate results; implementation errors and output conversion determine delivered accuracy.

5.2.2 Evaluation of Coherent Responses

Quantum-simulation constructions have already linked autocorrelation responses of logarithmic spectra to finite Dirichlet sums [8]. Here we incorporate that response into BC occupation encoding and boundary programs, giving normalization and truncation errors for actual finite support.

Multiplication and phase addition map known inputs to output labels. Function evaluation can also use a probe response formed jointly by several labels: a program prepares a weighted superposition, lets components accumulate different phases, and extracts the desired weighted sum from the return amplitude. For $s = 2\sigma > 1$, prepare on the actual finite encoding set $\mathcal{N}$

$$|\psi_{\sigma,\mathcal{N}}\rangle = Z_{\mathcal{N}}(s)^{-1/2} \sum_{n\in\mathcal{N}} n^{-\sigma}|n\rangle, \qquad Z_{\mathcal{N}}(s) = \sum_{n\in\mathcal{N}} n^{-s}.$$

On an occupation box, amplitudes admit local preparation through the product structure of Section 4. After $U(t) = e^{iHt}$, the relation $U(t)|n\rangle = n^{it}|n\rangle$ gives return amplitude

$$R_{\mathcal{N}}(t) = \langle\psi_{\sigma,\mathcal{N}}|U(t)|\psi_{\sigma,\mathcal{N}}\rangle = \frac{\sum_{n\in\mathcal{N}} n^{-s+it}}{Z_{\mathcal{N}}(s)}.$$

We retain the positive-sign evolution parameter here. If the device denotes forward free evolution by e^{-iHt}, the response uses $-t$, or realizable reverse control obtains $U(t)$. The ideal global limit gives

$$R(t) = \frac{\zeta(s-it)}{\zeta(s)}, \qquad R_\chi(t) = \langle\psi_\sigma|U_\chi U(t)|\psi_\sigma\rangle = \frac{L(s-it,\chi)}{\zeta(s)}.$$

These series converge absolutely for $s > 1$. A finite device obtains finite sums, omitting ideal probability mass

$$\eta = \frac{1}{\zeta(s)} \sum_{n\notin\mathcal{N}} n^{-s}.$$

The finite-support distribution is the ideal distribution conditioned on the retained encoding set; their total variation distance is η. Thus, for phase or character factors of magnitude at most 1, finite and ideal responses differ by at most 2η. Recovering an unnormalized function value from the normalized response also requires the normalization constant and its error.

Complex amplitudes are read through two self-adjoint quantities. Set $W_\chi(t) = U_\chi U(t)$ and define

$$A_\chi(t) = \frac{W_\chi(t) + W_\chi(t)^\dagger}{2}, \qquad B_\chi(t) = \frac{W_\chi(t) - W_\chi(t)^\dagger}{2i}, \qquad W_\chi(t)^\dagger = e^{-iHt}U_{\bar\chi}.$$

Their expectations are the real and imaginary parts of R_χ, respectively, and their norms are at most 1. Measurement requires suitable references and couplings; auxiliary-system interference likewise requires the controlled action or a valid extension. For N_{samp} independently prepared and measured samples, each component's sample-mean variance is at most $1/N_{\mathrm{samp}}$ and its root-mean-square statistical error is $O(N_{\mathrm{samp}}^{-1/2})$. If character action is implemented by filtering the coprime sector, the filtering probability must be retained; the normalized conditional response after filtering cannot replace the unconditional response above. Phase-control, probe-preparation, and truncation errors are counted separately.

With the direct instrument of Section 4 that does not resolve inversion pairs, readable components are also symmetry-constrained. At $t = 0$, $B_\chi(0) = (U_\chi - U_{\bar\chi})/(2i)$ is inversion-odd on the corresponding unit-group carrier. With a coherent reference or general t, readout scope depends jointly on H, the reference system, and the joint instrument. Averaging conjugate characters gives

$$\frac{R_\chi(t) + R_{\bar\chi}(t)}{2} = \frac{L(s - it, \chi) + L(s - it, \bar\chi)}{2\zeta(s)},$$

The result generally remains complex. Evaluation estimates the required components through reference coupling and repeated measurement, combining statistical, truncation, and normalization errors in the final function value.

5.2.3 Constraint Solving

A solving task specifies relations an answer must satisfy and outputs a candidate state satisfying them. One implementation writes finite constraints as energies, concentrates a low-temperature distribution on satisfying states, and reads and verifies candidates. For a subset-product example with binary occupations, take

$$S = \{11, 13, 17, 19, 23\}, \qquad N = 3289, \qquad v_p \in \{0, 1\},$$

and require $\prod_{p \in S} p^{v_p} = N$. Using the inputs S and $\log N$, construct

$$\boxed{H_{\mathrm{pin}}(\boldsymbol{v}) = \left(\sum_{p \in S} v_p \log p - \log N \right)^2 .}$$

Unique factorization ensures that zero-penalty states are precisely the satisfying subsets. Compilation specifies only the target logarithm and coupling coefficients; expanding the square reveals two-mode terms $2v_p v_r \log p \log r$. Hereafter H_{pin} is the implemented total effective energy, including compensation for the BC base energy. If other terms remain, the lowest-energy sector and equilibrium distribution must be determined from the actual total energy.

The program prepares binary modes and uses transitions covering the entire admissible configuration graph. At fixed temperature, an irreducible actual chain satisfying detailed balance for this total energy converges to the Gibbs distribution; an annealing schedule [20] changes its concentration. Cross terms place this example outside the independent-mode generator of Section 5.1, so its running time must be determined from the joint chain.

Let D be the finite configuration count, $g_0 > 0$ the zero-energy degeneracy, and Δ the smallest positive penalty gap. Separating zero-energy and positive-energy terms gives

$$P_\beta(H_{\mathrm{pin}} = 0) = \frac{g_0}{g_0 + \sum_{E>0} e^{-\beta E}} \geq \frac{1}{1 + \frac{D - g_0}{g_0} e^{-\beta \Delta}}.$$

This is a low-temperature bound for the equilibrium target. If the finite-time distribution is within total variation distance $\varepsilon_{\mathrm{mix}}$ of equilibrium, the lower bound on the target-event probability is reduced by at most $\varepsilon_{\mathrm{mix}}$. Physical coefficient errors must also lie below the required energy resolution. For example, a uniform total-energy deviation at most $\delta_E < \Delta/2$ leaves a separation of at least $\Delta - 2\delta_E$ between the original lowest-energy group and the rest.

Read candidates at the prescribed sampling time and verify them by exact integer multiplication. For these inputs, $11 \cdot 13 \cdot 23 = 3289$, so $(1, 1, 0, 0, 1)$ is the directly checkable unique zero-penalty occupation. This value comes from finite-constraint verification and is not device-run data. The chosen prime-set example illustrates energy compilation and readout; it is not a hardness benchmark.

Deliver an answer only after verification; otherwise continue sampling or restart under the prescribed rule. Early stopping triggered by occupation of the target sector requires an actually

measurable target-event instrument. With reliable verification, the repeated protocol delivers only correct candidates; completion time depends on per-trial hit probability, sample correlations, and restart costs.

If N is absent from the finite subset-product set, the minimum penalty is strictly positive. Failure to observe zero penalty in finite time can report only current candidates and incompletion; proving unsatisfiability requires exhaustive search or another certificate. Low hit probabilities, metastable residence, and long relaxation may produce similar finite-window records and must be distinguished through statistical tests for the actual chain.

General integer constraints can also become finite penalties within a fixed witness bound. The Davis–Putnam–Robinson–Matiyasevich result [21] establishes the Diophantine expressibility of recursively enumerable relations, but supplies no few-body coupling, precision, or mixing-time guarantee. Implementing cross-variable constraints and determining their spectral gaps remain obligations of each solving task.

5.2.4 Search and Candidate Verification

The preceding solver alters candidate probabilities through constraint energies. Candidates may instead be generated successively by explicit update rules, with each verification deciding whether to deliver or continue searching. Addition shifts on a residue carrier change integer labels, while verification still uses occupation or constraint relations. Let the integer step be $h \neq 0$ and take a prime $r \nmid h$. For nonzero integer labels $n, n+h$, they cannot both be divisible by r, hence

$$v_r(n)v_r(n+h) = 0.$$

Under natural density, $\mathbb{P}(v_r(n) = a) = (1 - r^{-1})r^{-a}$. Given $v_r(n) \geq 1$, the latter valuation must vanish. Given $v_r(n) = 0$, residue classes satisfying $v_r(n+h) = b \geq 1$ automatically exclude $r \mid n$, so

$$\mathbb{P}(v_r(n+h) = b \mid v_r(n) = 0) = r^{-b}, \qquad \mathbb{P}(v_r(n+h) = 0 \mid v_r(n) = 0) = \frac{r-2}{r-1}.$$

Geometric summation gives $\mathbb{E}v_r = 1/(r-1)$ and $\mathrm{Var}(v_r) = r/(r-1)^2$. Since the product always vanishes, the covariance is $-1/(r-1)^2$, and the one-step correlation coefficient is $-1/r$.

This describes local valuation changes under integer shifts without wraparound. Wraparound on finite residue domains and interpretation as integer candidates must be specified separately. A fixed step also preserves its corresponding congruence class and does not ensure connectivity of the full search graph. Multiple steps require checking admissible-edge connectivity first.

For example, on a finite candidate set, take symmetric proposal weights $a(x, y) = a(y, x)$ and construct

$$q(x, y) = a(x, y) \min\{1, e^{-\beta(E(y)-E(x))}\}$$

as Metropolis transitions [22], satisfying Gibbs detailed balance edge by edge. Connectivity gives a unique stationary distribution, but hitting time still depends on the full proposal graph, energy, and target set. The coefficient $-1/r$ above does not bound mixing of the full search chain. Successful verification yields a valid output; reaching the cutoff without a hit retains an incomplete record. Mixing, verification, and retries jointly determine search cost.

5.3 Detection and Control Tasks

Tasks acting directly on external signals and actions also impose explicit timing requirements. Detection decides signal presence from finite-time records; control turns available records into actions bringing external variables into a target range. Arithmetic processing supports record discrimination and action selection, while interaction determines how signals enter and results return to the outside.

5.3.1 Detection within a Finite Observation Window

Consider a weak signal transduced into counts, with rates $0 < \lambda_0 < \lambda_1$ under background-only and signal-present hypotheses. On a window where the Poisson model applies, N_T has distribution Poisson$(\lambda_i T)$. Choose observation cutoff T and $\lambda_0 < a < \lambda_1$ in advance, reporting signal presence when $N_T \geq \lceil aT \rceil$.

The Poisson moment-generating function is $\mathbb{E}_i e^{sN_T} = \exp[\lambda_i T(e^s - 1)]$. For the background upper tail take $s > 0$, and for the signal lower tail take $s < 0$. Minimizing the exponential bound at $s = \log(a/\lambda_i)$ gives

$$\mathbb{P}_0(N_T \geq \lceil aT \rceil) \leq e^{-TI_0(a)}, \qquad \mathbb{P}_1(N_T < \lceil aT \rceil) \leq e^{-TI_1(a)},$$

$$I_i(a) = a \log(a/\lambda_i) - a + \lambda_i.$$

The threshold a allocates weight between false-alarm and missed-detection exponents. The log-likelihood ratio of the two Poisson count laws is

$$\log \frac{P_1(N_T)}{P_0(N_T)} = N_T \log \frac{\lambda_1}{\lambda_0} - (\lambda_1 - \lambda_0)T.$$

Taking its expectation under the signal-present hypothesis and dividing by T gives information per unit time

$$D(\lambda_1 \| \lambda_0) = \lambda_1 \log \frac{\lambda_1}{\lambda_0} - \lambda_1 + \lambda_0.$$

These error probabilities concern a decision at the predetermined cutoff. A program stopping upon crossing a time-dependent boundary uses a different first-passage event, whose error must be recalculated. A fixed finite threshold will eventually be crossed even by persistent background counts. A detection result therefore includes both the decision and its error guarantee; one threshold crossing alone is not a correctness guarantee.

The device must support the required count range, reset, and readout within the observation window. Saturation, dead time, and non-Poisson noise change the count law and must enter the actual test. Longer observation lowers statistical error within this model, while transduction and readout errors determine attainable total accuracy.

5.3.2 Servo Control and Stable Retention

A servo selects actions from records already read, and actuators then change the environment. Let environmental variables belong to a normed space. One cycle of transduction, processing, and actuation gives

$$y_{k+1} = F(y_k) + \xi_k.$$

where F is the noiseless map of the declared control rule and ξ_k collects measurement and actuation deviations. On an actually reachable invariant operating domain, suppose $F(y_*) = y_*$, $\|F(y) - F(y_*)\| \leq \theta \|y - y_*\|$, and $0 \leq \theta < 1$. Repeated substitution gives

$$\boxed{\|y_k - y_*\| \leq \theta^k \|y_0 - y_*\| + \sum_{j=0}^{k-1} \theta^{k-1-j} \|\xi_j\|.}$$

If $\|\xi_j\| \leq \eta$ over a finite operating window, the bound becomes $\theta^k \|y_0 - y_*\| + \eta(1-\theta^k)/(1-\theta)$. If noise is bounded only probabilistically, the guarantee applies only on the corresponding event. For example, if the noise magnitude in each cycle exceeds η with probability at most δ, the first K cycles all satisfy the bound with probability at least $1 - K\delta$. With the moment bound $\mathbb{E}\|\xi_j\| \leq \bar{\eta}$, taking expectations yields the corresponding mean-error bound. Shot noise generally has no deterministic bound over infinite time; a single noise scale cannot be treated as a permanent-retention guarantee.

Closed-loop completion requires y_* to belong to the acceptance set and the sum of initial-deviation and noise terms to enter tolerance within the specified cycle count or retention window. Actual time is the number of cycles multiplied by the corresponding loop period, including transduction, processing, and actuation delays.

Stable retention may also use an attraction basin of the actual dynamics. For a stable steady state, local recovery rates come from linear response or an appropriate generator; periodic motion requires transverse-stability and phase-drift analysis. A chosen weak-noise barrier model with approximate escape rate $r_{\mathrm{esc}} \simeq \kappa_0 e^{-\beta \Delta_{\mathrm{B}}}$, where Δ_{B} is the barrier, can estimate metastable retention time. Only if the escape hazard is additionally proved at most $r_{\max}$ throughout the operating window does one obtain

$$P_{\mathrm{stay}}(T) \geq e^{-T r_{\max}} \geq 1 - T r_{\max}.$$

This separates recovery and residence into two testable requirements. A stable motion pattern or a local spectral rate alone does not determine escape time under arbitrary noise. The amount of sustained control supplied by structure depends on these recovery and retention relations and on boundary-maintenance costs.

5.4 Composite Tasks and Continuous Operation

Sensing records may not directly determine an action. A machine may need to combine records with a prior, infer likely targets, solve for admissible candidates, and finally select and execute an action. Evidence and prior jointly determine the state distribution of an inference step; inference, solving, and actuation connect sequentially through records and states. If the environment changes further, new records after actuation enter the next computation. Shared couplings among several devices also require specifying their states, timing, and admissible actions.

5.4.1 Evidence Fusion and Inference

Inference updates candidate-state weights using sensing records. Retain the reference inverse temperature $\beta_0 > 0$ of Section 4, let the prior be $P_0(\boldsymbol{v})$, and assume observations are conditionally independent given candidate $\boldsymbol{v}$, with likelihoods $L_i(\boldsymbol{v})$. First require $Z_{\mathrm{post}} = \sum_{\boldsymbol{v}} P_0(\boldsymbol{v}) \prod_i L_i(\boldsymbol{v}) > 0$; if all weights vanish, report incompatible evidence or infeasible conditions. On positive-weight support, if the actual total energy is implemented as

$$E_{\mathrm{tot}}(\boldsymbol{v}) = \beta_0^{-1} \left[-\log P_0(\boldsymbol{v}) - \sum_i \log L_i(\boldsymbol{v}) \right] + C,$$

then equilibrium at $\beta = \beta_0$ has weights

$$\boxed{P(\boldsymbol{v} \mid \mathrm{data}) = \frac{P_0(\boldsymbol{v}) \prod_i L_i(\boldsymbol{v})}{\sum_{\boldsymbol{w}} P_0(\boldsymbol{w}) \prod_i L_i(\boldsymbol{w})}.}$$

Here C is the energy zero, and any remaining base energy must already be included in E_{tot}. Correlated observations require a joint likelihood; zero weights are implemented by support restrictions or controlled penalty approximations. Changing temperature with this energy fixed raises the posterior to power β/β_0 and renormalizes it. Low temperature concentrates weight on maximum-posterior states, but obtaining them still depends on actual mixing and sampling.

Local soft evidence gives an inference class with uniform completion-time guarantees. Retain the occupation windows and lower-corner initial state of Section 5.1 and, at reference inverse temperature β_0, set

$$E_p(k) = (\log p) k + \beta_0^{-1} V_p(k), \qquad \operatorname{osc} V_p := \max V_p - \min V_p \leq R.$$

Each e^{-V_p} is a local likelihood factor. The target is $\varpi_V(n) \propto n^{-\beta_0} \prod_p e^{-V_p(v_p(n))}$, supported on the original divisibility window. In addition to writable energies, require autonomous occupation dynamics, independent modes, and edge rates

$$q_{V,p}(x,y) = \frac{\kappa_p}{1+\exp[\beta_0(E_p(y)-E_p(x))]}, \qquad |x-y|=1.$$

These reduce to Section 5.1 for $V_p = 0$ and satisfy target detailed balance. Let ϖ_0 denote the original single-mode distribution, and write $\varpi_V = w\varpi_0/Z_w$ and $w = e^{-V}$. Choose constants $m_0, M_0 > 0$ such that

$$m_0\varpi_0 \le \varpi_V \le M_0\varpi_0, \qquad M_0/m_0 = e^{\operatorname{osc} V}.$$

The reversible conductance of a neighboring edge is

$$c_V(x,y) = \kappa_p \frac{\varpi_V(x)\varpi_V(y)}{\varpi_V(x)+\varpi_V(y)} \ge m_0 c_0(x,y).$$

The inequalities follow from monotonicity in both positive variables and degree-one homogeneity of the fraction on the right. The variational expression for variance, $\operatorname{Var}_\varpi f = \min_c \sum_x \varpi(x)(f(x)-c)^2$, also gives $\operatorname{Var}_{\varpi_V} f \le M_0 \operatorname{Var}_{\varpi_0} f$. Comparing numerator and denominator in the Dirichlet quotient yields

$$\boxed{\gamma_{V,p} \ge e^{-\operatorname{osc} V_p}\gamma_{0,p}.}$$

The lower-corner mass further satisfies $\varpi_{V,p}(\ell_p) \ge e^{-R}(1-z_*)$. Thus the joint-error and completion-time derivation of Section 5.1 applies with replacements

$$\gamma_* \longmapsto e^{-R}\gamma_*, \qquad C_* \longmapsto \frac{e^R}{1-z_*} - 1.$$

Local evidence amplitude therefore has an explicit effect on running time. The evidence tables contain a total of $\sum_p d_p$ entries; their writing precision and edge responses are actual resources. Cross-mode evidence or a shared bath requires a joint generator. For bounded readings obtained after independent preparations, sample-mean root-mean-square error still decreases as $N_{\mathrm{samp}}^{-1/2}$; correlated trajectories require accounting for correlation time.

5.4.2 Sequential Programs and Module Composition

Consider identifying a composition and choosing an action, with external features mapped by a calibrated encoding to a finite subset-product model. Transduction first estimates target $\log N$ and its uncertainty; inference converts this into candidate integers or weights; the arithmetic stage solves and verifies the subset product; and the actuator chooses an action from the confirmed composition. Mapping external features to integer labels is an actual calibration condition. A signal cannot be treated as an integer product merely because it has harmonics.

In this chain, each stage's delivered result becomes the next stage's input. The identified target label is retained in a register to set preparation targets and couplings for solving; the verified composition record then selects actuation. QND readout and decoupled registers provide one implementation, while general states may pass directly through admissible quantum channels. Any addressing, storage, or conditional selection performed by an external electronic controller belongs to this actual composite process.

Connection requires an output admissible as the next input, compatible conserved-charge flows, realizable coupling, and coordinated timing and stopping rules. Parallel processes compose by tensor products when joint initial states, boundaries, environments, and dynamics are independent. Initial correlations or shared couplings require joint states and evolution. Shared constraints alone do not uniquely determine transition rates or retention times.

Figure 8 places actual operation and output testing side by side for identification, solving, and actuation. States change along the physical run. The terminal test checks whether the action meets requirements, retaining verification failure, actuation failure, and timeout outcomes. Pulling this test back through each trace adjoint computes the same final-outcome probability on the initial state.

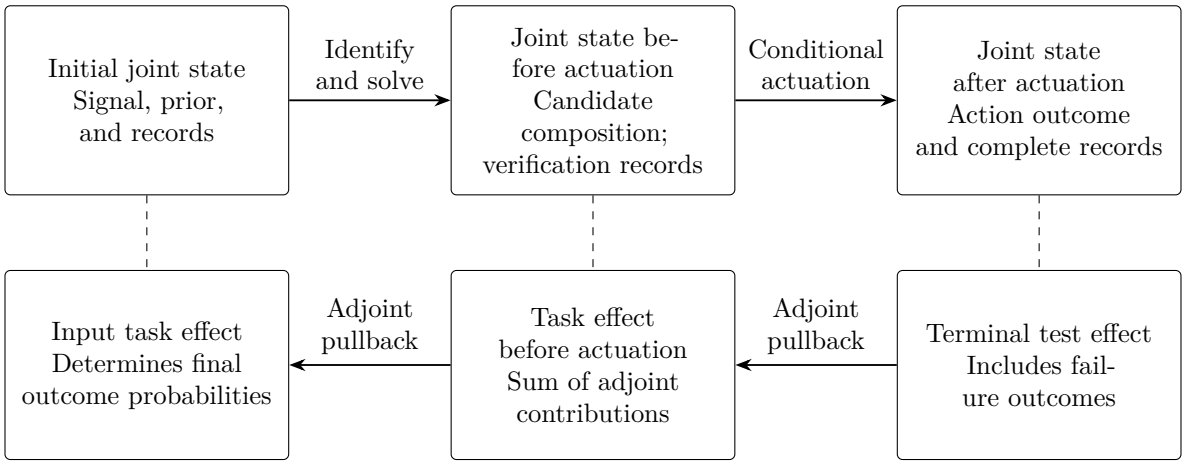


Figure 8: States and task effects in identification, solving, and actuation. The upper row shows the physical process. The lower row pulls terminal tests back through trace adjoints, yielding the same outcome probability at every stage. Feedback records select subsequent instruments. Pulling back a test does not reverse physical time.

This test does not depend on a particular three-stage division. Consider a program with a deterministic upper bound on step count and discrete records, expanded into a bounded-depth history tree. Node h retains all systems, controllers, records, and environment that will still participate, with branch instrument $\mathcal{I}_{h,r}$. Each component is completely positive and trace-nonincreasing; including failures makes their sum trace-preserving. Let ω_h be the unnormalized joint state at the node. Then

$$\omega_{hr} = \mathcal{I}_{h,r}(\omega_h), \qquad p(h) = \operatorname{Tr} \omega_h, \qquad \omega_\varnothing = \rho_{\mathrm{in}}.$$

Probabilistic preparation and its failures can also be initial branches. Actual terminal tests are effects $E_{h,y} \geq 0$ satisfying $\sum_y E_{h,y} = I$; output y includes reported success, failure, or complete history. Define recursively

$$Q_{h,y} = \begin{cases} E_{h,y}, & h \text{ is terminal,} \\ \displaystyle\sum_r \mathcal{I}^*_{h,r}(Q_{hr,y}), & h \text{ continues.} \end{cases}$$

At every nonterminal node, trace duality gives

$$\sum_r \operatorname{Tr}[Q_{hr,y}\mathcal{I}_{h,r}(\omega_h)] = \operatorname{Tr}[Q_{h,y}\omega_h].$$

Iterating to the root yields the outcome probability for the complete computation:

$$\boxed{p(y) = \operatorname{Tr}[Q_{\varnothing,y}\rho_{\mathrm{in}}].}$$

For this task, y may indicate action completion or retain the full record from identification to actuation. Unnormalized states retain branch probabilities, event-triggered stopping changes terminal positions in the tree, and feedback records determine subsequent instruments. Continuous records replace sums by instrument measures and integrals. General state postconditions require corresponding test families or state distances; reference systems and record–state correlations must remain in the joint object when relevant.

This also tests module substitution. For example, after replacing the solving stage, the actuator should receive task-equivalent outputs for identical inputs, including records and probabilities. The modules must therefore induce the same action on allowed downstream tests. Let $\mathcal{V}_h$ be the real linear span of allowed downstream task effects and I at node h, and require

$$\mathcal{I}_{h,r}^{*}(\mathcal{V}_{hr}) \subseteq \mathcal{V}_h.$$

If normalized inputs ρ, σ have equal expectations on $\mathcal{V}_h$, then, for every $A \in \mathcal{V}_{hr}$,

$$\mathrm{Tr}[A\mathcal{I}_{h,r}(\rho)] = \mathrm{Tr}[\mathcal{I}_{h,r}^{*}(A)\rho] = \mathrm{Tr}[\mathcal{I}_{h,r}^{*}(A)\sigma] = \mathrm{Tr}[A\mathcal{I}_{h,r}(\sigma)].$$

Taking $A = I$ gives equal branch probabilities. Normalizing at a common nonzero probability leaves conditional outputs equivalent for all allowed tests. Applied node by node, this equivalence is preserved throughout the program.

Thus, with a common encoding and identical record conventions, if two modules satisfy branchwise

$$\boxed{\mathcal{I}_{h,r}^{*}(A) = \widetilde{\mathcal{I}}_{h,r}^{*}(A), \qquad A \in \mathcal{V}_{hr}, \quad \text{for every } r,}$$

substitution into the task-effect recursion leaves the root effect, and hence composite task statistics, unchanged. Merely preserving the same effect space does not imply this substitution relation. If reference correlations must be preserved, the test space must include the corresponding joint tests.

Record-selected actions also organize finite sequential control. Occupation writing and reading, phase operations, Fourier transforms, record branching, and exchange stages supply component operations. For example, use L addressable prime modes for a finite-alphabet tape of length at most L, with additional head-position and finite-control states. A simulation step reads the current mode, updates the finite control from the symbol read, changes occupation to write a symbol, and moves the head. For a fixed alphabet, the number of local occupation changes is bounded; addressing mode i still has device-dependent cost.

For fixed tape length and step count, this gives a finite-resource sequential implementation, including registers, controller, and addressing system. Increasing tape length requires a uniformly constructed machine family and renewed accounting for register, control, and addressing growth. When processing units are coupled, local charges of shared modes need not remain conserved, and retention times must be recalculated from the actual joint network. Module connection is thus determined by executable state transfer and a common output-test relation.

5.4.3 Stagewise Errors and Actuation Outcomes

Returning to identification, solving, and actuation, correct connection must be followed by an error guarantee for the entire delivered chain. If identification yields a continuous estimate, its conversion into a discrete target must account for misclassification near candidate boundaries. Multiplication verification confirms only the target received; it cannot itself correct an earlier misidentification as another integer. A final-action guarantee must therefore propagate stage errors through the same process to its endpoint.

Let complete-output errors at the stages be $\varepsilon_1, \varepsilon_2, \varepsilon_3$, measured in total variation or trace distance, uniformly over downstream reachable inputs and any reference systems that must be retained. Inserting an intermediate process with one ideal and one actual stage, then applying the triangle inequality and channel contraction, gives

$$\boxed{\varepsilon_{\mathrm{tot}} \leq \varepsilon_1 + \varepsilon_2 + \varepsilon_3.}$$

Section 8.4.4 gives the general multistage form and reference-system requirements. If the last two stages have Lipschitz gains L_2, L_3 in the chosen metric, the same recursion yields

$$\varepsilon_{\mathrm{tot}} \leq L_3 L_2 \varepsilon_1 + L_3 \varepsilon_2 + \varepsilon_3.$$

An appropriate Wasserstein distance, for example, can use this form once kernel gains are proved. Fidelity error has no general triangle inequality and cannot simply be added in the same way. Final acceptance depends on whether the terminal action reaches the target. Stage failures and cutoffs must remain in the output; reporting only success-conditioned error is insufficient. Identification accuracy, solving reliability, and actuation deviations jointly determine complete-task delivery quality.

5.4.4 Continuous Tracking

After an action, the environment may already have changed, making the preceding identification stale. Feeding new post-actuation records back into transduction and inference turns the chain into a tracking loop. Each cycle updates candidates, solves for an action, and acts externally to maintain accuracy within a prescribed operating window. The preceding posterior must first be predicted forward to the current time under an environmental-change model before incorporating new evidence. Direct reuse is an approximation requiring a static or sufficiently slowly varying setting. Retained environmental memory and intercycle correlations remain part of the joint run.

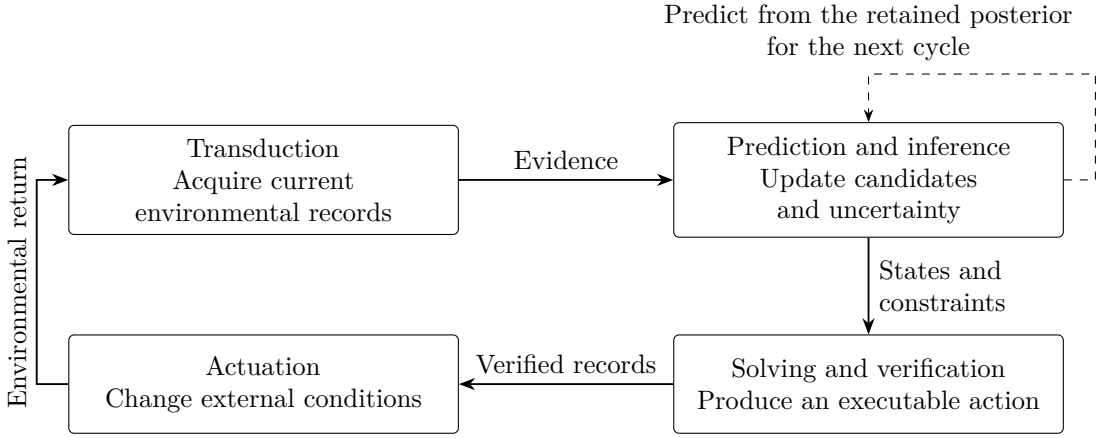


Figure 9: A serial tracking loop. Solid arrows show physical and record transfer within the current cycle; the dashed arrow carries the statistical state into the next prediction. All four durations contribute to information delay. Accuracy also depends on actual closed-loop stability, noise, and target variation.

The durations of the four serial stages give

$$T_{\text{loop}} = T_{\text{tr}} + T_{\text{inf}} + T_{\text{sol}} + T_{\text{act}}.$$

This describes a serial loop; pipelined or overlapping execution requires separate calculations of throughput period and single-item information delay. Environmental correlation bandwidth B_{env} must be coordinated with transduction bandwidth B_{tr}, effective closed-loop recovery rate κ_{loop}, and processing delay. In common time units, a scale requirement is

$$B_{\text{env}} \lesssim \min\{B_{\text{tr}}, \kappa_{\text{loop}}, T_{\text{loop}}^{-1}\}.$$

This is a necessary temporal-resolution check, not by itself a guarantee of stability or tracking accuracy. Feedback gain, phase delay, noise, and environmental drift must also enter the actual control relation.

For example, take $y_{k+1} = F_k(y_k) + \xi_k$, with target at cycle k satisfying $F_k(y_{*,k}) = y_{*,k}$. Suppose trajectories remain throughout the operating window in a domain with the same contraction bound $\theta < 1$, and target drift per cycle is at most vT_{loop}. On the noise event $\|\xi_k\| \le \eta$, set $e_k = \|y_k - y_{*,k}\|$. Then

$$e_{k+1} \le \theta e_k + \eta + vT_{\text{loop}}, \qquad e_k \le \theta^k e_0 + \frac{\eta + vT_{\text{loop}}}{1-\theta}(1-\theta^k).$$

Processing time therefore enters tracking error directly. Maintaining tolerance in each rolling window requires jointly bounding noise, recovery gain, and loop latency, with the corresponding probability stated. Long relaxation or repeated restarts in the solving stage increase information age and change the usable range of the entire loop.

Sensors and actuators sharing an environment require analysis of their joint closed loop; independent-loop bounds cannot be applied directly to a coupled system without analysis. Completion of continuous computation means sustained reporting or control within the prescribed operating window, rather than necessarily reaching a stopped state.

5.4.5 Simulation of Finite Dynamics

A segment of machine evolution may itself be an output, reproducing specified observations or dynamical statistics of another finite system. A direct example is the occupation birth–death process on a given window: encode target level labels as occupations and match reflecting boundaries and edge rates, and the generator of Section 5.1 realizes the target chain term by term. Sampling requires a near-stationary sample; dynamical simulation requires reproduction of target evolution at specified times. With matching initial states and readouts, occupation statistics agree at any given time; time-series statistics additionally require matching monitoring protocols.

More generally, let $\mathcal{L}$ be the encoded target generator and $\widetilde{\mathcal{L}}$ the actual generator. On a common finite carrier, if both generate quantum channels and $\|\widetilde{\mathcal{L}} - \mathcal{L}\|_\diamond \le \eta$, their semigroup difference is

$$e^{t\widetilde{\mathcal{L}}} - e^{t\mathcal{L}} = \int_0^t e^{(t-s)\widetilde{\mathcal{L}}}(\widetilde{\mathcal{L}} - \mathcal{L})e^{s\mathcal{L}}\,\mathrm{d}s.$$

A channel has diamond norm 1, so the norm above is at most $t\eta$. With state distance defined by half the trace norm, the generator deviation contributes output error at most $t\eta/2$, in addition to encoding, initial-state, and readout errors. Leakage outcomes must first be included in the common carrier; evolution restricted to a subspace that is not invariant under the dynamics cannot simply be treated as a trace-preserving quantum channel.

This makes target matching a testable simulation condition. Available couplings, spectra, and controls determine the realizable range. Writing every target rate individually can also implement controlled simulation, but its writing resources and structural origin require separate assessment. Equilibrium observations, spectral responses, and ground-state candidates can use the processes above under their respective preparation and measurement protocols.

Multiplicative conditional sampling provides a complete example from local windows to a joint distribution and completion time; spectral comparison extends the guarantee to inference with local soft evidence. Task-effect recursion tests composite outputs, error propagation connects stage accuracies, and closed-loop relations connect running time to tracking capability. These examples show how the same components can deliver values and samples, or retain states and records to meet behavioral requirements continuously within specified time windows.

6 Characteristics of the Computer

The task examples exhibit several common characteristics: factor composition participates directly in operations, local boundaries support program reuse, conserved records and task-required coherence can be maintained separately, and states and records can continue into subsequent computations. These characteristics determine suitable data, operating modes, and resource allocations.

6.1 Information and Addressing

BC encoding distinguishes components through prime modes and represents quantities by their occupations. Internal data composition can therefore be operated on directly: selecting a mode

selects a factor component, and changing occupations changes the corresponding integer structure. This representation jointly determines addressing, encoding, and information capacity.

6.1.1 Address Structure

Occupation addresses have a factorizable multicomponent structure. For finite mode set Λ and cutoffs L_p, the occupation basis vector for an integer label is

$$\boxed{|n\rangle = \bigotimes_{p\in\Lambda} |v_p(n)\rangle, \qquad n = \prod_{p\in\Lambda} p^{v_p(n)}, \qquad 0 \le v_p(n) \le L_p.}$$

Unique factorization assigns distinct integer labels to distinct occupation vectors. The mode address p identifies a component, and its local eigenvalue $v_p(n)$ specifies the component's data. Multiplication, divisibility tests, and window updates can then be organized along these components. The global integer label thus has an explicit correspondence with local composition.

This address structure suits tasks repeatedly accessing factor information. Once occupation encoding is supplied, condition $m \mid n$ can be checked through $v_p(m) \le v_p(n)$ mode by mode, without reconstructing a positional integer and extracting the same factorization again. An addressing device still selects physical modes, with access time depending on spatial, frequency, or control implementation. General superpositions also contain amplitudes and correlations between labels, carried by the joint state space.

6.1.2 Data Encoding

Occupation encoding directly accepts factor lists, symbol strings, and structural data with fixed labels. For a known factor list, exponents are the occupations to write. For a finite-alphabet string $s = (s_1, \ldots, s_k)$, fix a position-to-mode correspondence and write $v_{p_i} = s_i$. With a fixed alphabet and implemented modewise writing, the latter takes a number of local write calls proportional to string length.

Structural data may express inclusion through divisibility. For example, with fixed vertex labels and a shared edge-to-prime dictionary, encode graph edge set E as square-free label n_E. Then

$$n_E = \prod_{e\in E} p_e, \qquad E_1 \subseteq E_2 \iff n_{E_1} \mid n_{E_2}.$$

The same correspondence gives

$$n_{E_1\cap E_2} = \gcd(n_{E_1}, n_{E_2}), \qquad n_{E_1\cup E_2} = \operatorname{lcm}(n_{E_1}, n_{E_2}).$$

Data relations thereby become established arithmetic relations, allowing later programs to reuse occupation checks and condition updates. This concerns edge-set relations under common labels; vertex relabeling and subgraph isomorphism require additional programs.

If only the positional representation of an ordinary integer n is supplied, obtaining its full occupation vector includes factorization: a factor list gives occupations, and occupations give the complete factor list. Direct-input benefits therefore primarily concern data already containing composition information, and symbolic or structural data assignable modewise. Required input conversions are counted for each task.

6.1.3 Information Capacity

The logarithmic spectrum directly relates the number of available labels to the energy window. Let $E \ge 0$ be a hard energy ceiling in the adopted units and $\mathcal{N}_{\Lambda,L}$ the finite occupation-label set. The number of admissible labels is

$$d_{\Lambda,L}(E) = \#\{n \in \mathcal{N}_{\Lambda,L} : \log n \le E\}.$$

Injectivity of integer labels gives $d_{\Lambda,L}(E) \le \lfloor e^E \rfloor$. Encoding with mutually orthogonal occupation basis vectors therefore bounds the noiseless classical information distinguishable in one use by

$$\boxed{C_{\Lambda,L}(E) = \log_2 d_{\Lambda,L}(E) \le \log_2 \lfloor e^E \rfloor \le E/\log 2.}$$

The first bound is attained when the label set includes $1, \ldots, \lfloor e^E \rfloor$, allowing direct conversion between address range and required energy. A single mode restricted to occupations $0, 1$ provides one bit if the energy window includes both labels; combined modes enlarge the address set.

Nonuniform data can preferentially use low-energy addresses. Suppose symbol probabilities are ordered as $P_1 \ge P_2 \ge \cdots \ge P_m$ and addresses $1, \ldots, m$ are available. Swapping an inverted pairing lowers mean $\log n$ cost, so the optimal assignment writes symbol i to address i. Since $iP_i \le 1$,

$$\boxed{A_{\min} := \sum_i P_i \log i \le \sum_i P_i \log(1/P_i) = H(P).}$$

Here $H(P)$ is Shannon entropy in natural-log units [23]. This gives an explicit encoding that assigns frequent data to low-energy addresses. If available addresses are sparse, $a_1 < a_2 < \cdots$, the optimal ordering remains valid, but cost becomes $\sum_i P_i \log a_i$ and is no longer guaranteed below $H(P)$.

Capacity also requires sufficient physical degrees of freedom. The complete occupation box has dimension $\prod_p (L_p + 1)$. Encoding it in m units of local dimension at most d requires, by dimension comparison, $m \log d \ge \sum_p \log(L_p + 1)$. The capacity relations describe both encoding organization and the scale needed to carry it.

6.2 Programming and Control

BC programs select executable processes through boundaries and control conditions. In a divisibility-window task, the program specifies admissible occupations, and the machine produces results through established exchange relations. Changed inputs permit local condition updates. Acquired records can select the next boundary and operations, supporting reuse and feedback.

6.2.1 Boundary Programming

Divisibility conditions have a directly compilable local form. Window $n_- \mid n \mid n_+$ is equivalent to $\ell_p \le v_p(n) \le u_p$ on each mode. Adding lower-bound condition $d \mid n$ and upper-bound condition $n \mid m$ gives the update

$$\boxed{n'_- = \operatorname{lcm}(n_-, d), \qquad n'_+ = \gcd(n_+, m).}$$

Prime by prime, this takes the maximum lower exponent and minimum upper exponent. If $n'_- \mid n'_+$, the local intervals form a new window; otherwise the conditions are incompatible and emptiness can be reported before exchange starts. Combining conditions and checking feasibility use the same coordinates.

This follows from the distributive structure of a finite divisor lattice. Coordinatewise,

$$\min(x, \max(y, z)) = \max(\min(x, y), \min(x, z)),$$

Thus gcd and lcm are meet and join in the divisibility order. For condition events $E_m = \{n : m \mid n\}$, $E_m \cap E_s = E_{\operatorname{lcm}(m,s)}$. The divisibility projections of Section 4 test these events, while local boundaries select admissible ranges. If all nonzero cutoffs are one, the divisor lattice additionally has coordinatewise Boolean complements; general multivalued occupations retain distributivity.

Programs can therefore express a class of compositional conditions directly. After conditions are written, preparation, exchange rates, evolution time, and readout jointly perform the run. Boundaries specify admissible states; the chosen dynamics determine how they form results.

6.2.2 Program Reuse

Window programs reuse internal exchange rules and error guarantees after condition updates. For the window distribution of Section 5 with relative weights $n^{-\beta}$, adding $d \mid n$ with a nonempty new window gives

$$\boxed{\varpi_{n_-,n_+;\beta}(\,\cdot \mid d \mid n) = \varpi_{\mathrm{lcm}(n_-,d),n_+;\beta\cdot}}$$

Section 5.1.3 proves this conditioning relation. Updating the relevant lower bounds and preparing the new lower-corner state preserves the internal exchange law, bounds on local spectral gaps and on the stationary probabilities of the initial states, the target-distribution form, and the finite-time guarantee.

This reuse avoids retries in rejection sampling from the original distribution. Independent exact original-distribution samples followed by filtering require $1/P$ draws on average at acceptance probability $P > 0$. Directly setting the conditional window removes that factor from the exchange-time bound (Figure 10). Boundary setting, lower-corner preparation, and energy supply remain update costs.

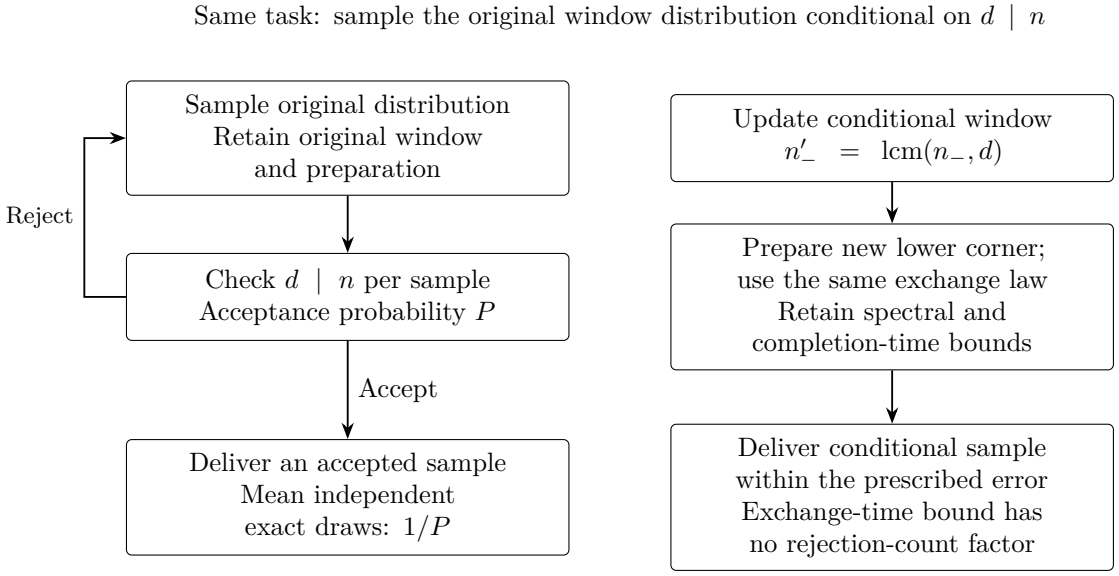


Figure 10: Conditional updating and program reuse. The left protocol draws independent exact samples from the original distribution and rejects inadmissible ones. The right writes the condition into the boundary and directly runs the window program, reusing the exchange law and time guarantee of Section 5. The comparison concerns these two protocols; classical direct conditional sampling can also exploit probability factorization.

6.2.3 Feedback Control

Retained records directly select subsequent programs, making boundary programming adaptive. Let h be the acquired history, from which the control rule selects condition b_h. With branch instrument $\mathcal{I}_{h,r}^{b_h}$ and new record r, unnormalized states satisfy

$$\omega_{hr} = \mathcal{I}_{h,r}^{b_h}(\omega_h), \qquad p(hr) = \mathrm{Tr}\,\omega_{hr}.$$

State updates retain branch probabilities, and record updates determine the next control. Conditional sampling can revise a window from the preceding result, verification can choose acceptance

or retry, and tracking can adjust actuation from new observations. Branches call the same implemented operations, enabling fixed components to perform different run sequences.

The record-kernel bound of Section 5.1.3 further ensures that single-round errors at most ε_j on every ideally reachable history give complete-history error at most $\sum_{j=1}^{N} \varepsilon_j$ over N rounds under the same causal policy. Feedback accuracy thus composes along records, retaining controllers and environmental correlations that continue to participate.

6.3 Storage Characteristics

The BC computer can retain occupation eigenvalues as records for later programs. Retention disables actions changing selected occupations, readout measures their eigenvalues, and rewriting reopens the corresponding exchanges. Distinct conditions for holding, reading, and updating allow records to be reused across computational stages.

6.3.1 Storage by Conservation

The conservation structure of holding dynamics can preserve occupation records. Let P_m be spectral projections of N_p, with holding-stage GKSL generator $\mathcal{G}$, Hamiltonian H_{hold}, and jump operators L_α. If every P_m commutes with H_{hold} and L_α, then

$$\mathcal{G}^*(P_m) = i[H_{\text{hold}}, P_m] + \sum_\alpha \left(L_\alpha^\dagger P_m L_\alpha - \tfrac{1}{2}\{L_\alpha^\dagger L_\alpha, P_m\} \right) = 0.$$

The Hamiltonian term vanishes, and the three terms within each dissipator cancel. Differentiating the holding evolution gives

$$\boxed{\operatorname{Tr}[P_m e^{t\mathcal{G}}(\rho)] = \operatorname{Tr}(P_m \rho) \qquad \text{for all } m,\ t \geq 0.}$$

This preserves the entire eigenvalue distribution and hence occupation expectations. Once a definite label is written, holding cannot transfer it to another label. In this ideal model, no periodic refresh is needed to compensate occupation loss during retention.

Such storage suits program parameters, condition labels, and acquired discrete results. Retaining interlabel coherence additionally requires preservation of the corresponding off-diagonal relations; tasks storing only eigenvalues can determine their requirements from these conservation conditions.

6.3.2 Repeated Readout

Conserved labels support repeated reading without changing an already read result. With a Lüders instrument of spectral projections, the state after first reading m is

$$\rho_m = \frac{P_m \rho P_m}{\operatorname{Tr}(P_m \rho)}.$$

This state lies entirely in the support of P_m. If subsequent holding evolution continues to fix P_m, then

$$\boxed{\Pr(m \text{ read again} \mid m \text{ previously read}) = \operatorname{Tr}[P_m e^{t\mathcal{G}}(\rho_m)] = 1.}$$

The same record can therefore be used successively for verification, conditional branching, and actuation control. Preparing another register from the read label also delivers these orthogonal records to different components. QND measurement preserves the repeatability of the selected observable's outcome [12].

The repeatedly deliverable objects here are occupation labels; the first Lüders measurement removes coherence across labels. For occupation programs, this record repeatability suffices to connect processing stages.

6.3.3 Retention and Reset

Residual leakage quantitatively determines retention time. If the actual holding generator is $\widetilde{\mathcal{G}}$ and a label projection satisfies $\|\widetilde{\mathcal{G}}^*(P_m)\|_\infty \le \epsilon_m$, its probability changes at absolute rate at most ϵ_m at every time. Integration gives

$$\left|\mathrm{Tr}[P_m e^{t\widetilde{\mathcal{G}}}(\rho)] - \mathrm{Tr}(P_m\rho)\right| \le t\epsilon_m.$$

Starting from label m, this bounds the probability of still reading m at time t below by $1 - t\epsilon_m$ under ideal spectral readout. Residual occupation-changing actions thus translate directly into storage-time and terminal-label error specifications; instrument misclassification is added separately. In the symmetric unit-group exchange model of Section 4.1.3, character eigenmodes satisfy $\langle a_\chi(t)\rangle = e^{-d_\chi t}\langle a_\chi(0)\rangle$, giving decay scale d_χ^{-1}. Record lifetime follows its actual dynamical model.

Reset actively changes a record for the next task. Refresh requirements during retention and rewriting costs can be arranged separately: residual leakage determines the former, while information erased and actual energy changes determine the latter. With an initially Gibbs bath at temperature T, initially uncorrelated with the record, no usable side information, restored auxiliary resources, and coupling off at both endpoints, entropy balance gives [24]

$$Q_{\mathrm{env}} \ge k_B T(S_{\mathrm{in}} - S_{\mathrm{out}}).$$

Here S uses natural logarithms. Combining with the first law, $W_{\mathrm{on}} = \Delta E + Q_{\mathrm{env}}$, gives $W_{\mathrm{on}} \ge \Delta F$, where $F = E - k_B T S$. Complete reset of energy-degenerate records has $S_{\mathrm{out}} = 0$; a uniform bit gives a lower bound $k_B T \log 2$ on environmental heat absorption. Physical energy units must be restored when applying these relations. A computing cycle can therefore incur rewriting costs when updating while retaining records during intervals satisfying conservation conditions.

6.4 Operational Characteristics

BC operations use occupation changes, phase responses, and constrained exchange. Arithmetic relations specify input changes, spectra specify phase accumulation, and boundaries and exchange jointly specify statistical formation. For decomposable tasks, these actions can also proceed in parallel across modes.

6.4.1 Arithmetic Operations

Multiplication can be implemented directly as an increment of the occupation vector. Within the valid domain of Section 4, restricted multiplication K_m satisfies

$$K_m|n\rangle = |mn\rangle, \qquad \boldsymbol{v}(mn) = \boldsymbol{v}(m) + \boldsymbol{v}(n).$$

Written factor composition therefore participates directly in the operation. Outputs retain the same occupation encoding and can undergo further multiplication, divisibility tests, or storage. The inverse implements division on the corresponding divisibility domain; repeated operations require intermediate states to remain admissible.

This action is compatible with the logarithmic spectrum. Using $H|n\rangle = (\log n)|n\rangle$ and subtracting on each basis vector gives

$$\boxed{[H, K_m] = (\log m)K_m.}$$

Outside the valid domain, K_m vanishes and the relation still holds. A multiplier thus determines both occupation and energy increments. If finitely many integer coefficients satisfy $\sum_p a_p \log p = 0$, exponentiation and unique factorization imply all $a_p = 0$. Distinct prime compositions admit no such nontrivial exact integer frequency relation. Near-resonances within finite linewidths are treated according to actual resolution.

Phase representation supplies another arithmetic action. The finite Fourier transform and modulation of Section 4 satisfy

$$F_q^{-1} V_c F_q |a\rangle = |a + c \bmod q\rangle.$$

Multiplication and modular addition therefore have explicit ideal operator implementations, yielding discrete labels in specified encodings. The arithmetic relations themselves introduce no rounding; actual accuracy depends jointly on valid domains, encoding conversion, control, and readout.

6.4.2 Function Evaluation

The same logarithmic spectrum organizes a family of arithmetic-function responses. For finite occupation box $\mathcal{N}$ and $s = 2\sigma > 1$, the probe of Section 5 is $|\psi_{\sigma,\mathcal{N}}\rangle = Z_{\mathcal{N}}(s)^{-1/2} \sum_{n\in\mathcal{N}} n^{-\sigma} |n\rangle$, where $Z_{\mathcal{N}}(s) = \sum_{n\in\mathcal{N}} n^{-s}$. With the positive-sign evolution parameter $U(t) = e^{iHt}$, its return amplitude is

$$\boxed{R_{\mathcal{N}}(t) = \langle \psi_{\sigma,\mathcal{N}} | U(t) | \psi_{\sigma,\mathcal{N}} \rangle = \frac{\sum_{n\in\mathcal{N}} n^{-s+it}}{Z_{\mathcal{N}}(s)}.}$$

Occupation-box weights factorize locally, allowing modewise probe preparation. With preparation rules and spectrum fixed, changing the evolution duration changes evaluation parameter t; realizable character actions give additional weighted responses. If the device uses e^{-iHt} for forward evolution, the corresponding parameter is $-t$.

Complex responses are obtained from self-adjoint real and imaginary components. Section 5.2.2 shows that, under a unit norm bound, N_{samp} independent repetitions give single-component root-mean-square error at most $N_{\text{samp}}^{-1/2}$; omitted ideal mass η contributes response truncation error at most 2η. Unnormalized function values additionally require the appropriate normalization constants.

This characteristic assigns a family of finite arithmetic sums to one spectrum and readout process. Actual instruments determine coherence requirements: expectations of the diagonal evolution operator here depend only on occupation probabilities, so the same occupation distribution gives the same self-adjoint-component expectations. Coherence needed by reference or auxiliary systems must be maintained during readout. Reusable response structure and coherence resources are determined separately by the evaluation protocol.

6.4.3 Statistical Formation

Local boundaries and exchange relations can jointly form target distributions. For window $\ell_p \le v_p \le u_p$ and inverse temperature $\beta > 0$, Section 5.1.1 derives by detailed balance of independent nearest-neighbor exchange

$$\boxed{\varpi(n) = Z^{-1} n^{-\beta} \mathbf{1}_{\{n_-|n|n_+\}}.}$$

By supplying only two boundaries per mode and the required run parameters, a program organizes weights over $\prod_p (u_p - \ell_p + 1)$ states, reducing the need to enumerate target weights state by state. Classical factorizable distributions also permit this parameter compression. The BC implementation additionally maps these parameters to mode boundaries and physical exchanges, allowing direct participation in sample formation.

Local evidence enters similarly. Implementing $E_p(k) = (\log p)k + \beta_0^{-1} V_p(k)$ at reference inverse temperature β_0 adds factor $e^{-V_p(k)}$ to equilibrium weights. For the specified edge rates, Section 5 proves local gap bound $\gamma_{V,p} \ge e^{-R} \gamma_{0,p}$ when $\operatorname{osc} V_p \le R$. Local evidence updates thus preserve both product statistics and calculable run guarantees. Cross-mode evidence is handled by the corresponding joint coupling.

6.4.4 Parallel Operation

Independent modes can form a joint result simultaneously. From local spectra and their tensor sum, Section 5.1.2 establishes the occupation-generator gap

$$\boxed{\operatorname{gap}(\mathcal{L}_{\mathrm{occ}}) = \min_{p:d_p\geq 2} \gamma_p.}$$

Here $d_p = u_p - \ell_p + 1$; coordinates with only one allowed occupation level do not enter the minimum, and no exchange waiting is required if every coordinate is fixed. Independent composition produces no positive decay rate below the slowest local mode. If $z_p = p^{-\beta} \leq z_* < 1$ and per-mode clock rates satisfy $\kappa_p \geq \kappa_* > 0$, a uniform lower bound is $\gamma_* = \kappa_*(1 - \sqrt{z_*})^2/(1 + z_*)$. With each mode independently prepared at its window lower endpoint and with $C_* = z_*/(1 - z_*)$, the joint-error bound gives a sufficient time to meet total variation tolerance $0 < \varepsilon \leq 1/2$:

$$\boxed{t_\varepsilon = \frac{1}{2\gamma_*}\left[\log \frac{N_{\mathrm{mode}} C_*}{\log(1 + 4\varepsilon^2)}\right]_+.}$$

Here $[x]_+ = \max(x, 0)$, and N_{mode} is the mode count. At fixed mode clock rates and z_*, exchange time grows only logarithmically with mode count, and window lengths do not enter this uniform bound. This is a parallel guarantee from the specified initial state to the complete joint sample.

The benefit requires simultaneous provision of mode clock rates. If total clock-rate budget κ_{tot} is fixed and shared equally, $\kappa_p = \kappa_{\mathrm{tot}}/N_{\mathrm{mode}}$, the same guarantee becomes $O((N_{\mathrm{mode}}/\kappa_{\mathrm{tot}}) \log(N_{\mathrm{mode}}/\varepsilon))$. Input writing, lower-corner preparation, and output access also take time. The result explicitly relates parallel resources to exchange-stage completion time.

6.5 Interaction Characteristics

The interaction module lets external inputs change computational conditions directly and lets internal results act on other systems. Transduction, state transfer, record feedback, and arithmetic use the same task description, enabling continuous computation in a changing environment.

6.5.1 Input Transduction

Calibrated couplings bring distinguishable external features into occupations, phases, and boundary conditions. Section 4 denotes this relation by $\mathcal{T}_\Lambda : s_{\mathrm{phys}}(t) \mapsto b(t)$. Inputs change event rates in count detection, phase accumulation in phase response, and candidate weights in conditional inference. Each input enters subsequent processing through the variables it actually affects.

Such transduction connects measurement continuously to inference. Under the energy-writing relation of Section 4.2.2, if actual total energy includes the BC base energy and represents prior P_0, conditionally independent positive likelihoods L_i give weights $P_0 \prod_i L_i$ at the same reference inverse temperature β_0. Correlated observations use a joint likelihood. Inputs update realizable evidence terms without first delivering a complete posterior table. Evidence computation, writing, and calibration, together with transduction bandwidth and noise, determine the available information.

6.5.2 Result Delivery

The BC computer can deliver records, quantum states, or actions. Occupation readout supplies discrete labels, response measurements supply statistics, state channels pass intermediate results to compatible components, and actuators convert confirmed records into external actions. Delivery form follows the next task's actual requirements.

A common encoding lets adjacent tasks directly reuse intermediate results. Product occupations can enter divisibility tests, conditional-sample occupations can select feedback branches, and verified labels can select actions. This avoids reconstructing positional integers and re-extracting the same

factor lists at handoffs. Compatible quantum channels can also pass intermediate states directly, retaining coherence and correlations needed downstream.

Delivery has explicit composition conditions: outputs must be admissible inputs to the next stage, with compatible record conventions, timing, and conserved-charge flows. The effect recursion of Section 5 gives complete-result probability $p(y) = \mathrm{Tr}(Q_y \rho_{\mathrm{in}})$ and tests module substitution against the same downstream observations. Output forms may differ, while the complete task retains a common final acceptance requirement.

6.5.3 Continuous Interaction

Computation can receive new inputs while retaining states and records. Post-actuation external conditions reenter through transduction, existing records select new control branches, and internal states continue updating under the specified dynamics. One-shot tasks can thereby compose into sustained detection, servo control, and tracking.

Continuous-operation performance is directly tied to processing delay. In the tracking model of Section 5.4.4, trajectories remain in one contraction domain throughout the operating window; the noiseless maps share contraction factor $\theta < 1$, target drift per unit time is at most v, and loop period is T_{loop}. Let e_k denote tracking error at cycle k. On the event that noise in every cycle is at most η, the recursion gives

$$\boxed{e_k \le \theta^k e_0 + \frac{\eta + v T_{\mathrm{loop}}}{1-\theta}(1-\theta^k).}$$

The first term decays with cycle count; the second contains noise and information-delay contributions. With operating domain, noise bound, and recovery coefficient fixed, reducing the total period can lower the drift contribution to the tracking-error bound. The guarantee's probability is determined by the noise event over that operating window.

Continuous interaction thus makes computational speed directly relevant to external tracking. State retention preserves useful information across cycles, while record feedback lets new evidence promptly change subsequent actions. Previous results still require time propagation under the environmental model, with necessary intercycle correlations retained.

6.6 Reliability

Reliability can be configured around information actually used by a task. Occupation programs require preservation of labels and transition statistics, phase-based modular addition requires preservation of intermediate relative phases, and candidate solving can use independent verification to ensure correct delivery. These operating modes admit distinct protection and acceptance conditions.

6.6.1 Coherence Requirements

Required coherence can be determined from the complete subsequent readout. Fix the proposed dephasing time and take the joint state ρ then as input, retaining prior records together with any controllers and environment that will still participate. Let $\{P_\ell\}$ be complete orthogonal label projections on the dephased factor, extended identically on the rest, and define $\mathcal{E}_P(\rho) = \sum_\ell P_\ell \rho P_\ell$. Let Ψ be the subsequent channel through task delivery, O a terminal self-adjoint observable, and $O_{\mathrm{eff}} = \Psi^*(O)$ its effective observable pulled back to that time. For all input states,

$$\boxed{\begin{aligned} &\mathrm{Tr}[O\Psi(\mathcal{E}_P(\rho))] = \mathrm{Tr}[O\Psi(\rho)] \quad \text{for all } \rho \\ &\iff \quad [O_{\mathrm{eff}}, P_\ell] = 0 \quad \text{for all } \ell. \end{aligned}}$$

The difference between the two readings is $\mathrm{Tr}\,\rho[\mathcal{E}_P^*(O_{\mathrm{eff}}) - O_{\mathrm{eff}}]$. It vanishes for all density operators exactly when the self-adjoint operator in brackets vanishes. Fixedness $\sum_\ell P_\ell O_{\mathrm{eff}} P_\ell = O_{\mathrm{eff}}$

then gives $P_m O_{\text{eff}} = P_m O_{\text{eff}} P_m = O_{\text{eff}} P_m$. Conversely, commutation and completeness of the projections recover fixedness, proving the necessary and sufficient condition.

For multistep monitoring and feedback, future history h from the same time has branch map $\mathcal{J}_h$; feedback unfolds from prior records retained in the joint input and newly acquired records. Report y has input effect $Q_y = \sum_{h \mapsto y} \mathcal{J}_h^*(I)$. Applying the criterion to every Q_y determines whether dephasing preserves the entire report distribution. Full-history delivery requires checking each history effect; quantum-state or reference-correlation delivery requires the corresponding joint tests. On a restricted reachable input set, commutation remains sufficient, while necessity depends on that set's linear span.

Protection can thus follow the program. Occupation tasks may use only label probabilities; complete dephasing in the physical component basis at the $F_q|a\rangle$ stage of phase-based modular addition gives I/q, preserved by later unitaries, reducing correct-readout probability from one to $1/q$. Figure 11 compares the two task classes. Protection can focus on coherences affecting results and expand with allowed downstream controls.

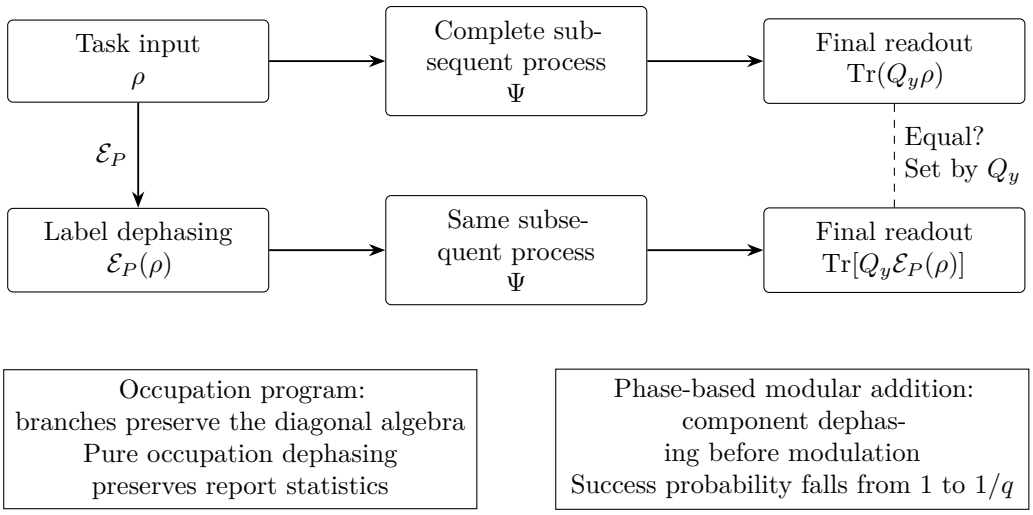


Figure 11: The task determines required coherence. Both paths use identical subsequent operations, recording rules, and terminal tests, differing only by label dephasing at the specified point. Equal probabilities for all inputs are equivalent to commutation of the task effect with every label projection. The examples concern occupation programs and phase-based modular addition. Quantum-state delivery additionally requires joint tests.

6.6.2 Structural Protection

Occupation-program protection follows from a common structure: terminal report effects lie in the occupation-diagonal algebra, and the adjoint of every evolution stage and instrument branch preserves it. Finite multiplication, divisibility checks, occupation QND readout, occupation exchange, and record-based control can all be organized this way. Denote this algebra by

$$\mathcal{D}_{\text{occ}} = \left\{ \sum_{\boldsymbol{v}} f(\boldsymbol{v}) P_{\boldsymbol{v}} \right\}.$$

For finite multiplication, use the valid domain D_m of Section 4. Pulling a diagonal readout back to the input gives

$$K_m^\dagger f(N) K_m = \sum_{\boldsymbol{v} \in D_m} f(\boldsymbol{v} + \boldsymbol{v}(m)) P_{\boldsymbol{v}} \in \mathcal{D}_{\text{occ}}.$$

A failure branch uses the complementary valid-domain projection or terminates and reports failure, still preserving diagonal tests. Spectral projections for divisibility checks and occupation

QND readout share this property. Selecting these operations from classical records preserves $\mathcal{D}_{\mathrm{occ}}$ branchwise under the adjoints.

A diagonal Hamiltonian and occupation jumps $L_{wv} = \sqrt{\Gamma_{wv}}|w\rangle\langle v|$ give

$$\mathcal{L}^*(f) = \sum_v \sum_w \Gamma_{wv}[f(w) - f(v)]P_v \in \mathcal{D}_{\mathrm{occ}}.$$

The adjoint of pure occupation dephasing $\mathcal{E}_{\mathrm{occ}}$ is the identity on this algebra. Adding dephasing at strength $\eta \geq 0$ leaves the generator's restriction to this invariant subspace unchanged. Uniqueness of finite-dimensional evolution therefore yields

$$\boxed{e^{t[\mathcal{L}+\eta(\mathcal{E}_{\mathrm{occ}}-\mathrm{id})]^*}(A) = e^{t\mathcal{L}^*}(A), \qquad A \in \mathcal{D}_{\mathrm{occ}}.}$$

Compositions of evolution stages and instrument branches continue to preserve this algebra, so added pure dephasing does not change report statistics for the entire occupation-program class. Actual transition rates, controls, and instruments retain their prescribed forms here; noise that changes them requires separate control of output deviations.

This is protection at the program level: multiplicative conditional sampling, occupation arithmetic, and their feedback use diagonal statistics, and pure interlabel dephasing preserves their report distributions. Phase operations require protection determined by the complete-readout criterion above. The same computer may use different retention requirements for these task classes.

6.6.3 Result Verification

Independently verifiable tasks connect candidate generation to correct delivery. A verifier rechecks the required integer relation for product candidates or every acceptance condition for constraint candidates. With error-free acceptance, the program delivers only verified results, and randomness in candidate generation affects the number of runs.

Suppose that, conditional on any history without prior acceptance, a round produces and accepts an answer with probability at least $p_* > 0$ and has conditional expected duration at most T_*. Let N be the first accepted round. Conditional probabilities give

$$\Pr(N > m) \leq (1 - p_*)^m, \qquad \mathbb{E}N = \sum_{m \geq 0} \Pr(N > m) \leq p_*^{-1}.$$

Let T_j be the duration of round j. Conditional expectation of the nonnegative sum yields

$$\boxed{\mathbb{E}T_{\mathrm{total}} = \sum_{j \geq 1} \mathbb{E}[\mathbf{1}_{\{N \geq j\}} T_j] \leq T_* \sum_{j \geq 1} \Pr(N \geq j) \leq T_*/p_*.}$$

Thus, under error-free acceptance and these uniform conditions, every delivered answer meets the task requirements and total expected time has an explicit bound. If false acceptance is possible, its probability over the entire stopping process requires separate control. Without a candidate by a finite cutoff, the program reports incompletion.

Statistical results gain precision through repeated readout. Independent identically distributed samples with finite variance σ_f^2 have sample-mean variance $\sigma_f^2/N_{\mathrm{samp}}$. Independent decisions with error at most $1/2 - a$, $a > 0$, have majority-vote error at most $e^{-2N_{\mathrm{samp}}a^2}$. Correlated samples require accounting for correlation time, and shared systematic biases are controlled separately. Candidate verification ensures that delivered answers satisfy relations; statistical repetition supplies the declared confidence. They provide appropriate reliability mechanisms for different outputs.

6.7 Overall Advantages

The machine's overall advantages arise from these characteristics working together within one task: one representation continues through operations, condition updates, sample formation, and result handoffs.

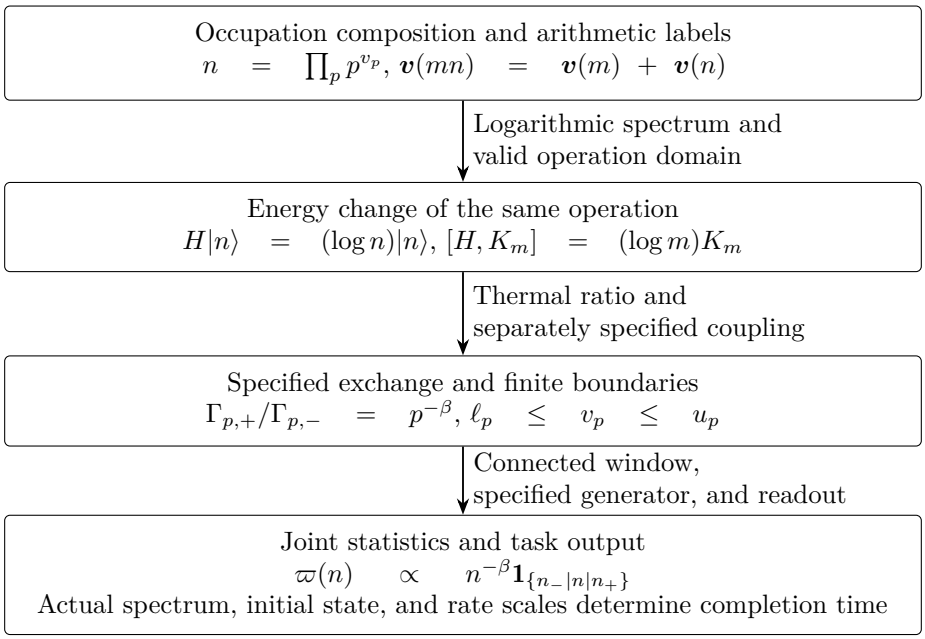
6.7.1 Structural Advantages

The principal structural advantage is that common arithmetic relations connect data, operations, energy, and statistics. Occupation vectors express factor composition and multiplication increments; the logarithmic spectrum converts the same multiplier into an energy difference; specified thermal exchange connects that difference to statistical weights. In the window task, this is expressed by

$$\boldsymbol{v}(mn) = \boldsymbol{v}(m) + \boldsymbol{v}(n) \quad \Longrightarrow \quad E(mn) - E(n) = \log m,$$

$$\frac{\Gamma_{p,+}}{\Gamma_{p,-}} = p^{-\beta} \quad \Longrightarrow \quad \varpi(n) \propto n^{-\beta} \mathbf{1}_{\{n_-|n|n_+\}}.$$

The first line uses occupation encoding and the logarithmic spectrum; the second uses the specified independent nearest-neighbor exchange and window boundaries. Figure 12 connects these relations into a complete process, with actual mode clock rates and initial states additionally supplying finite-time results.



The boundary program selects windows, preparation, control, and observation;
Readout records and return coupling organize subsequent tasks.

Figure 12: The shared arithmetic structure in the window task. Arithmetic labels organize both occupation changes and energy differences. Exchange ratios and target statistics additionally require the stated physical conditions. Absolute rates, channels, and controls depend on implementation. Finite-time delivery uses the guarantees established in Section 5.

This common structure lets condition changes retain compilation and run guarantees while reducing representation conversions between adjacent steps. Conserved storage, protection of occupation statistics, and phase retention are configured by task, aligning the scope of reuse with protection requirements.

6.7.2 Task Suitability

These advantages are clearest for tasks compatible with arithmetic composition and local conditions. Their supporting structures and established benefits are summarized below.

Task	Computational benefit supported by the structure
Factor and labeled structural data	Factor composition enters occupations directly; multiplication, divisibility tests, and handoffs retain the same representation
Multiplicative conditional sampling	Local windows form a joint distribution; condition updates preserve programs and time guarantees, avoiding the retry factor of rejection from the original distribution
Local-evidence inference	Local energy terms update weights; bounded evidence amplitudes retain quantitative spectral-gap comparisons
Independent-window sampling	With simultaneous mode clock rates, the exchange stage reaches a joint-error guarantee in time logarithmic in mode count
Occupation programs	With transition laws and instruments fixed, pure occupation dephasing leaves complete report statistics unchanged; protection follows the information used by the task
Arithmetic-response evaluation	A common probe construction and spectrum organize finite function responses through duration and weight controls
Continuous detection and control	Records and states connect subsequent processes; compatible connections reduce intermediate conversions, and loop delay determines tracking accuracy

Benefits can combine: local evidence may be updated, samples formed in parallel, verified results retained, and control selected from those records. Cross-mode constraints require joint dynamics, and new phase operations require corresponding protection. Actual tasks determine suitability.

6.7.3 Comparison of Paradigms

Universal quantum simulation [25], dissipative computation [6], and programmable Rydberg arrays [26] also perform tasks through controlled dynamics, stationary states, or feedback. Bosonic occupation representations of factor multiplicities [9] and logarithmic-spectrum schemes seeking unknown factors [10,11] are more directly related, as discussed in the introduction.

The advance here is concrete: finite boundary programs and their run guarantees. Divisibility windows admit uniform compilation, specified exchange forms the conditional distribution, condition updates reuse proved time bounds, and complete records and delivered states continue under the same task tests. The window tasks of Section 5 accept supplied exponent representations and directly perform conditional sampling. Seeking unknown factors from ordinary integers has different input and solution requirements. These distinctions place comparisons at the level of complete tasks, beyond encoding, spectra, or individual mechanisms.

Computing approach	Emphasis in task organization
Classical digital computation	Symbol encodings, instructions, and control states organize operations; structured algorithms may also exploit factorization and direct conditional sampling
Gate-based quantum computation	Encoded-state processing is compiled into gates and measurements, with intermediate information protected according to the encoding and noise model
Programmable quantum simulation	Tunable couplings, geometry, drives, and measurements realize target processes, potentially combined with feedback and classical optimization
BC state-space computing	A common arithmetic structure connects occupation data, boundary conditions, spectra, exchange, and readout, with reusable relations within supported task families

These approaches may overlap in capability and connect through encoding and control. The BC proposal is distinguished by the consistent use of these arithmetic relations across the machine and by the associated task guarantees. Comparisons of total time, energy, and scale require identical tasks and input–output specifications with complete resource budgets, to establish the engineering consequences of the structural advantages.

Verification can also accompany output. For tasks requiring mathematical proof, the machine may retain the operations and relations supporting a result, in addition to reporting values or samples, and organize them into independently checkable certificates.

7 Certificate Readout

The computer can organize relations supporting a conclusion into independently verifiable certificates. BC arithmetic actions and the logarithmic spectrum provide correlation expressions, and positivity combines correlations at different times into inequalities. Starting with a finite three-time combination, we construct the Mertens inequality, then use analytic steps to obtain boundary nonvanishing, the prime number theorem, and prime-counting asymptotics in arithmetic progressions of fixed modulus.

7.1 Certificate Output

7.1.1 Output Objects

A certificate consists of a target statement, finite expressions, and verification steps. Expressions specify operations, coefficients, parameters, and their combinations; verification steps supply algebraic reductions, positivity relations, and necessary error bounds. A parameterized statement additionally specifies parameter ranges and the derivation from finite instances to the general conclusion. Recipients can then check each equality and inequality and determine the applicable inputs.

For example, if the output contains a finite operator expression z and an identity for target quantity Q,

$$Q = \varphi(z^* z),$$

positivity of φ gives $Q \geq 0$. The proof object is the square structure on the right and its exact correspondence with Q. Actual readouts can test correlations at specified parameters, while the general inequality follows from the algebraic derivation throughout the declared parameter range. Both use the relations recorded explicitly in the certificate.

In the preceding computer construction, the arithmetic module performs finite operations and forms the associated states; the interaction module supplies parameter input, correlation readout, and record retention. The certificate describes how these operations support the target statement and can be checked by an independent classical program. If actual preparation or measurement supplies only intervals, those intervals must enter the final estimate, with parameters and error records retained.

7.1.2 Verification Relations

We use two internal verification classes. The first comprises algebraic relations of arithmetic actions and time evolution, including composition, adjoints, and phase changes of multiplication actions. The second comprises state positivity and invariance, including $\varphi(a^* a) \geq 0$ and $\varphi(\sigma_t(a)) = \varphi(a)$. They support expression rewriting and inequality formation, respectively.

Targets involving infinite series or asymptotic distributions additionally require convergence, analytic continuation, and Tauberian theorems. These steps enter where used: finite positivity gives finite inequalities, convergence permits limits, and boundary properties then yield counting results. A complete certificate records both finite algebraic grounds and analytic theorems.

Verified identities and inequalities can feed subsequent certificates. Figure 13 shows the verification chain for the construction below.

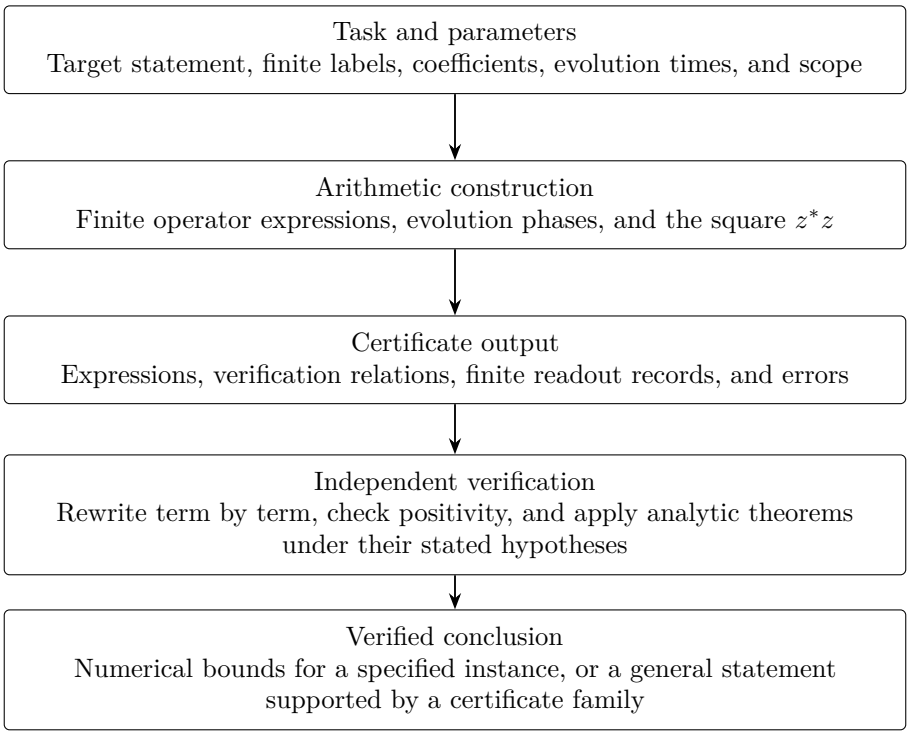


Figure 13: Certificate construction and verification. The arithmetic module supplies finite relations; the interaction module handles parameter input, correlation readout, and records. General statements follow from parameterized certificate derivations and their analytic hypotheses. A finite run corresponds to a specified instance.

7.2 Certificate Construction

7.2.1 Arithmetic Correlations

In the ideal BC representation, multiplication and the logarithmic spectrum satisfy [3]

$$\mu_m|n\rangle = |mn\rangle, \qquad \mu_m^*\mu_m = I, \qquad H|n\rangle = (\log n)|n\rangle.$$

For the associated time evolution $\sigma_t(a) = e^{itH}ae^{-itH}$,

$$\sigma_t(\mu_m) = m^{it}\mu_m.$$

Let φ be a normalized positive linear functional, hence a state, invariant under this evolution. Distinct multiplication labels are orthogonal in this state. Indeed,

$$\varphi(\mu_m^*\mu_n) = (n/m)^{it}\varphi(\mu_m^*\mu_n) \quad \text{for all } t \in \mathbb{R},$$

Thus the value vanishes for $m \neq n$; for $m = n$, isometry and normalization give one. Hence

$$\boxed{\varphi(\mu_m^*\mu_n) = \delta_{mn}.}$$

Take a finite set $S \subset \mathbb{N}$ and finite expression $x = \sum_{n\in S} c_n\mu_n$. Its correlation function is

$$C_x(t) := \varphi(x^*\sigma_{-t}(x)) = \sum_{n\in S} |c_n|^2 n^{-it}.$$

Arithmetic labels thus enter phases, with squared coefficient magnitudes as correlation weights. The following argument uses only positivity, time invariance, and these arithmetic relations. Thermal equilibrium states can supply such invariant states, but no additional KMS boundary identity is needed here.

7.2.2 A Positive Combination

Combine three time translates of the same expression with coefficients $1, 2, 1$:

$$z = x + 2\sigma_{-t}(x) + \sigma_{-2t}(x).$$

Write $x_j = \sigma_{-jt}(x)$ and $j = 0, 1, 2$. Expanding the square gives

$$\begin{aligned} z^*z = {} & x_0^*x_0 + 4x_1^*x_1 + x_2^*x_2 \\ & + 2(x_0^*x_1 + x_1^*x_0) + 2(x_1^*x_2 + x_2^*x_1) \\ & + (x_0^*x_2 + x_2^*x_0). \end{aligned}$$

Invariance turns the three diagonal terms into $C_x(0)$, $4C_x(0)$, and $C_x(0)$. Both adjacent-time correlations equal $C_x(t)$, and the first-to-last correlation equals $C_x(2t)$. Therefore

$$\boxed{\frac{1}{2}\varphi(z^*z) = 3C_x(0) + 4\,\mathrm{Re}\,C_x(t) + \mathrm{Re}\,C_x(2t) \geq 0.}$$

Coefficients $3, 4, 1$ arise from this square expansion, collecting equal-time, adjacent-time, and first-to-last contributions. The input-combination coefficients are $1, 2, 1$, whereas the output-inequality coefficients are $3, 4, 1$; they play distinct roles.

Positivity can also be seen label by label. Setting $\theta = t \log n$ gives

$$\frac{1}{2}|1 + 2e^{-i\theta} + e^{-2i\theta}|^2 = 3 + 4\cos\theta + \cos 2\theta = 2(1 + \cos\theta)^2 \geq 0.$$

The finite certificate collects these labelwise squares into a correlation inequality. Checking finite coefficients, phases, and the square expansion verifies an instance; the same derivation applies to every finite S and every real t.

7.2.3 Finite Realization

This finite certificate connects exactly to the truncated operations of Section 4. Choose a finite occupation box W containing 1 and all labels in S, let P_W be its projection, and write

$$K_m = P_W \mu_m P_W, \qquad H_W = H|_W, \qquad \sigma_t^W(a) = e^{itH_W} a e^{-itH_W}.$$

Truncated multiplication may fail on out-of-range inputs, but for the specified reference input $|1\rangle$, $K_m|1\rangle = |m\rangle$. Since P_W commutes with H,

$$\sigma_t^W(K_m) = m^{it}K_m, \qquad \langle 1|K_m^*K_n|1\rangle = \delta_{mn} \quad (m, n \in S).$$

Hence $X_W = \sum_{m\in S} c_m K_m$ satisfies

$$\langle 1|(\sigma_u^W X_W)^* \sigma_v^W X_W|1\rangle = \sum_{m\in S} |c_m|^2 m^{i(v-u)}.$$

This agrees term by term with the corresponding finite two-time correlation in the ideal representation. Every finite certificate instance therefore has this finite-model realization, preserving the three-time positivity relation unchanged. The result concerns the specified input and finite correlations; truncated operations on the full box still obey the valid-domain conditions of Section 4.

Physical realization additionally requires input preparation, control, and correlation readout. An operator linear combination does not automatically define a deterministic executable channel. Device proposals must supply couplings, auxiliaries, success conditions, and measurement errors. The certificate specifies what finite correlations these implementations must reproduce and what relation their final results must satisfy.

7.3 Arithmetic Proofs

7.3.1 Prime-Power Weights

Connecting positivity certificates to prime distribution requires weights representing the Euler product. For real $\sigma > 1$ and finite cutoff N, define

$$x_N = \sum_{p^k \le N} \frac{p^{-k\sigma/2}}{\sqrt{k}}\, \mu_{p^k}, \qquad D_N(s) = \sum_{p^k \le N} \frac{p^{-ks}}{k},$$

where p runs over primes and $k \ge 1$. Inputs are finite prime-power labels and parameters N, σ, t; outputs retain labels, weights, and the square decomposition as an independently verifiable correlation certificate. Distinct prime powers have distinct integer labels, so orthogonality gives

$$\varphi(x_N^* x_N) = D_N(\sigma), \qquad \varphi(x_N^* \sigma_{-t}(x_N)) = D_N(\sigma + it).$$

Substituting x_N into the same positive combination yields, for every N,

$$3D_N(\sigma) + 4\,\mathrm{Re}\, D_N(\sigma + it) + \mathrm{Re}\, D_N(\sigma + 2it) \ge 0.$$

For $\mathrm{Re}\, s > 1$, the logarithmic Euler series converges absolutely and

$$\log \zeta(s) = \sum_{p,k \ge 1} \frac{p^{-ks}}{k}.$$

The logarithm is defined by this convergent series, with real part $\log|\zeta(s)|$. Taking scalar limits $N \to \infty$ term by term gives the Mertens inequality

$$\boxed{3 \log \zeta(\sigma) + 4 \log|\zeta(\sigma + it)| + \log|\zeta(\sigma + 2it)| \ge 0.}$$

Only convergence of three scalar correlations is required. Each x_N remains a finite expression; the proof assumes no norm-convergent infinite operator sum. Explicit convergence thus turns verified finite positivity into an analytic-function inequality.

7.3.2 Boundary Nonvanishing

To examine the line $\mathrm{Re}\, s = 1$, first use partial summation:

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{u\} u^{-s-1}\, du.$$

The integral converges uniformly on compact subsets of $\mathrm{Re}\, s > 0$, defining a holomorphic function there. Thus ζ extends to this half-plane, with only a simple pole at $s = 1$ of residue 1.

Let $t_0 \ne 0$ and suppose $\zeta(1 + it_0) = 0$, with order of vanishing $m \ge 1$. Let $\ell \ge 0$ be the order of vanishing of ζ at $1 + 2it_0$, taking $\ell = 0$ if nonzero. Set $\sigma \downarrow 1$. The left side of the Mertens inequality becomes

$$(3 - 4m - \ell) \log \frac{1}{\sigma - 1} + O(1).$$

Since $3 - 4m - \ell < 0$, this tends to negative infinity, contradicting nonnegativity. Therefore

$$\boxed{\zeta(1 + it) \ne 0 \qquad (t \ne 0).}$$

The key is the relation between square-expansion coefficients and zero and pole orders. Finite positivity supplies $3, 4, 1$, and analytic continuation makes boundary orders meaningful; together they prove nonvanishing.

7.3.3 Prime Counting

Boundary nonvanishing then determines average prime counts. Use the von Mangoldt function $\Lambda_{\mathrm{VM}}(n)$ of Section 4, equal to $\log p$ for $n = p^k$ and zero otherwise. For $\operatorname{Re} s > 1$,

$$F(s) := -\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 1} \frac{\Lambda_{\mathrm{VM}}(n)}{n^s}.$$

The coefficients $\Lambda_{\mathrm{VM}}(n)$ are nonnegative. Boundary nonvanishing and the simple pole at $s = 1$ imply that $F(s) - 1/(s-1)$ extends holomorphically to a neighborhood of every point of $\operatorname{Re} s = 1$.

We apply the Wiener–Ikehara theorem [27]: if a Dirichlet series $F(s) = \sum a_n n^{-s}$ with nonnegative coefficients converges for $\operatorname{Re} s > 1$ and $F(s) - A/(s-1)$ extends continuously to the boundary, where $A \geq 0$, then $\sum_{n \leq x} a_n = Ax + o(x)$. The local holomorphic extension here satisfies this condition. Hence

$$\psi(x) := \sum_{n \leq x} \Lambda_{\mathrm{VM}}(n) \sim x.$$

To pass to primes themselves, set $\vartheta(x) = \sum_{p \leq x} \log p$. Higher prime powers arise only from $p \leq \sqrt{x}$, with each such prime contributing at most $\log x$, so

$$0 \leq \psi(x) - \vartheta(x) \leq \sqrt{x} \log x = o(x).$$

Thus $\vartheta(x) \sim x$. Partial summation gives

$$\pi(x) = \frac{\vartheta(x)}{\log x} + \int_2^x \frac{\vartheta(u)}{u \log^2 u}\, du.$$

Using $\vartheta(u) = O(u)$ and splitting the integral at $\sqrt{x}$ bounds the integral term by $O(x/\log^2 x)$. Finally,

$$\boxed{\pi(x) \sim \frac{x}{\log x}.}$$

This proof connects finite squares, the logarithmic Euler series, boundary nonvanishing, and prime counting. The prime number theorem is classical; the contribution here is its explicit construction in the certificate language, with the analytic theorems connecting inequalities to counting retained clearly.

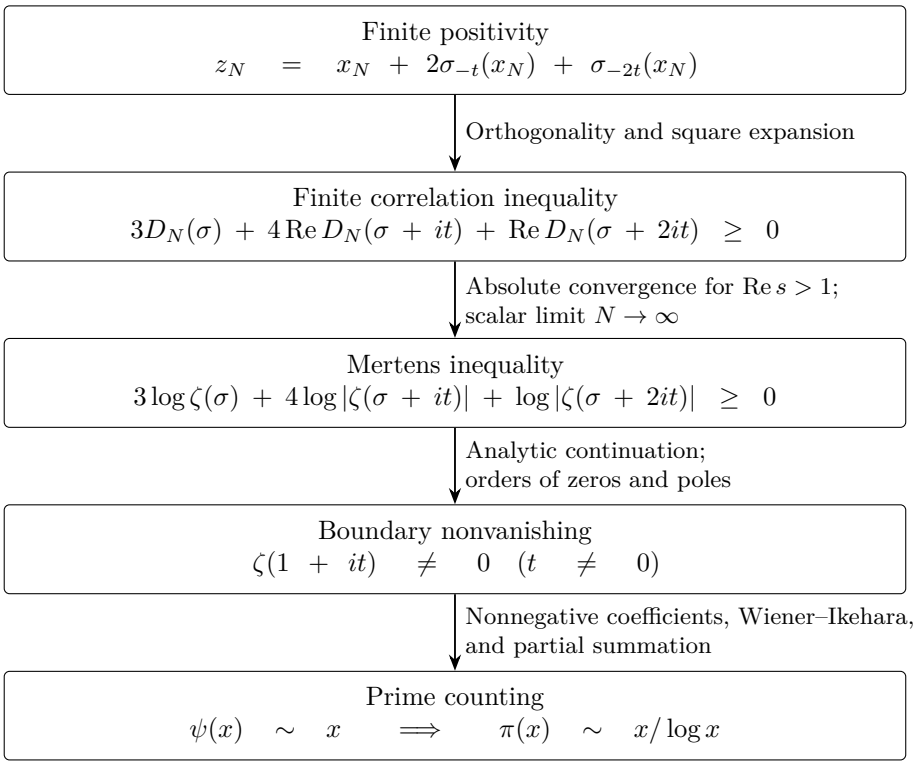


Figure 14: The complete chain from finite certificates to the prime number theorem. Arrows state the justification at each step. The character version reuses the square template. A positive-coefficient argument for real nonprincipal characters at $s = 1$ completes boundary nonvanishing, yielding the arithmetic-progression asymptotic for each fixed modulus.

7.3.4 Arithmetic Progressions

The same construction retains residue-class information. Fix modulus $q \geq 2$, let χ_0 be the principal character modulo q and χ any Dirichlet character modulo q, and define

$$x_{\chi,N} = \sum_{p^k \leq N} \frac{p^{-k\sigma/2}\chi(p)^k}{\sqrt{k}}\,\mu_{p^k}.$$

Use the combination

$$z_{\chi,N} = x_{\chi_0,N} + 2\sigma_{-t}(x_{\chi,N}) + \sigma_{-2t}(x_{\chi^2,N}).$$

For $p \nmid q$, set $u = \chi(p)^k p^{-ikt}$. Then $|u| = 1$, and this label has coefficient $p^{-k\sigma/2}(1+2u+u^2)/\sqrt{k}$ in the combination. For $p \mid q$, all three terms vanish. The square expansion therefore still gives exactly $3, 4, 1$. Taking the absolutely convergent logarithmic Euler-series limit yields

$$\begin{aligned} 0 \leq\ & 3\log L(\sigma,\chi_0) + 4\log|L(\sigma+it,\chi)| \\ & + \log|L(\sigma+2it,\chi^2)|. \end{aligned}$$

The principal-character function satisfies

$$L(s,\chi_0) = \zeta(s)\prod_{p\mid q}(1-p^{-s}),$$

and hence has a simple pole at $s = 1$. A nonprincipal character has zero period sum and bounded partial sums, so partial summation extends its L function holomorphically to $\mathrm{Re}\,s > 0$.

If $t \neq 0$ and $L(1+it,\chi)$ vanishes, comparing the orders of zeros and poles in the three terms again yields a term tending to negative infinity, a contradiction. If $t = 0$ and χ is nonreal, χ^2 remains nonprincipal and the same comparison applies. Only real nonprincipal characters at $s = 1$ remain.

For such χ, set $a(n) = (1 * \chi)(n) = \sum_{d|n} \chi(d)$. This is multiplicative and satisfies, on prime powers,

$$a(p^r) = \begin{cases} r+1, & \chi(p) = 1, \\ 1 \text{ or } 0, & \chi(p) = -1,\ r \text{ even or odd, respectively}, \\ 1, & p \mid q. \end{cases}$$

Hence $a(n) \geq 0$ and $a(m^2) \geq 1$. If $L(1,\chi) = 0$, then $G(s) = \zeta(s)L(s,\chi)$ is holomorphic on $\operatorname{Re} s > 0$, while $G(s) = \sum a(n)n^{-s}$ holds for $\operatorname{Re} s > 1$. Its Taylor disk about $s = 2$ contains $s = 1/2$. Differentiating by absolute convergence and interchanging sums by nonnegativity gives

$$\begin{aligned} G(1/2) &= \sum_{k\geq 0} \sum_{n\geq 1} a(n) n^{-2} \frac{((3/2)\log n)^k}{k!} \\ &= \sum_{n\geq 1} a(n) n^{-1/2} \geq \sum_{m\geq 1} \frac{1}{m} = \infty, \end{aligned}$$

contradicting holomorphy. This explicitly expands Landau's positive-coefficient argument in the present case [28], completing $L(1,\chi) \neq 0$.

All nonprincipal-character L functions are therefore nonzero on $\operatorname{Re} s = 1$; the principal-character function is also nonzero there apart from its pole at $s = 1$. Fix $(a,q) = 1$. Character orthogonality gives

$$\begin{aligned} F_{q,a}(s) &:= \sum_{\substack{n\geq 1 \\ n\equiv a \pmod q}} \frac{\Lambda_{\mathrm{VM}}(n)}{n^s} \\ &= \frac{1}{\varphi(q)} \sum_{\chi \pmod q} \overline{\chi(a)} \left(-\frac{L'(s,\chi)}{L(s,\chi)} \right). \end{aligned}$$

Here $\varphi(q)$ is Euler's totient function. This series has nonnegative coefficients, and $F_{q,a}(s) - 1/(\varphi(q)(s-1))$ is holomorphic locally at every boundary point. Applying Wiener–Ikehara again, removing higher prime powers, and partially summing as above yields

$$\boxed{\pi(x;q,a) \sim \frac{x}{\varphi(q)\log x} \qquad (q \text{ fixed},\ (a,q) = 1).}$$

Characters thus connect the same certificate template to residue-class counting. This proof applies to each fixed modulus; moduli growing with x require separate uniform quantitative estimates.

7.4 Scope of Proof

7.4.1 Relational Information

The proof uses time-evolution phases of arithmetic labels, orthogonality between labels, and positivity across several times. By retaining these relations, finite correlations form function inequalities constraining boundary zeros. This demonstrates a specific capability: organizing expressible state-space relations into independently verifiable mathematical grounds.

To compare conclusions supported by different retained information, fix a certificate system using only divisibility counts. Given finite integer range $1 \leq n \leq x$, divisor set $\mathcal{D}$ containing 1, main terms b_d, and termwise errors $r_d \geq 0$, its admissible data are

$$w(n) \geq 0, \qquad |S_d(w) - b_d| \leq r_d, \qquad S_d(w) = \sum_{\substack{n\leq x \\ d|n}} w(n).$$

This Type I information gives total weight in each divisibility class. If finite coefficients α_d satisfy pointwise domination $\sum_{d\in\mathcal{D}} \alpha_d \mathbf{1}_{d|n} \geq \Phi(n)$, then

$$\sum_{n\leq x} w(n)\Phi(n) \leq \sum_d \alpha_d b_d + \sum_d |\alpha_d| r_d.$$

Reverse pointwise domination gives lower bound $\sum_d \alpha_d b_d - \sum_d |\alpha_d| r_d$. Errors are weighted by actual certificate coefficients; an unweighted error sum cannot replace them.

Optimal certificate bounds equal the optimal values of the nonnegative-weight model under the specified constraints. This follows from finite-dimensional linear-programming duality: assign nonnegative multipliers to both sides of each error band and eliminate multipliers for positivity constraints, obtaining pointwise domination and weighted error above. When the constraint set is nonempty, the finite total-mass bound for $d = 1$ makes it bounded, so finite linear-programming strong duality applies. Its proof scope can therefore be studied by finding distinct weight models satisfying the same constraints.

7.4.2 The Parity Limitation

The sieve parity principle shows how information organization limits resolution of prime weights [29]. We establish the result for the specified system using three explicit models. Fix $\varepsilon \in (0, 1/2)$, let $D = x^{1-\varepsilon}$, and let $\mathcal{D}$ be the square-free integers at most D. The Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ distinguishes parity of the total prime-factor count. Construct

$$w_0(n) = \mathbf{1}_{(\sqrt{x},x]}(n), \qquad w_\pm(n) = (1 \pm \lambda(n))\mathbf{1}_{(\sqrt{x},x]}(n).$$

All three are nonnegative. Writing $L_\lambda(y) = \sum_{n\leq y} \lambda(n)$, complete multiplicativity gives

$$\begin{aligned} S_d(w_\pm) = \frac{x - \sqrt{x}}{d} + e_d \\ \pm \lambda(d)\left[L_\lambda(x/d) - L_\lambda(\sqrt{x}/d)\right], \qquad |e_d| \leq 1. \end{aligned}$$

To place the models within common error bands, use the classical quantitative cancellation estimate $|L_\lambda(y)| \leq Cy\exp(-c\sqrt{\log y})$ $(y \geq 2)$. Its analytic input is the analogous bound for Möbius partial sums $M(y) = \sum_{n\leq y} \mu(n)$; see [30]. Logarithmic prefactors are absorbed by reducing the exponential constant. Since $\lambda(n) = \sum_{a^2|n} \mu(n/a^2)$,

$$L_\lambda(y) = \sum_{a\leq\sqrt{y}} M(y/a^2).$$

apply the quantitative bound for $a \leq y^{1/4}$ and the trivial bound elsewhere, obtaining $O(ye^{-c'\sqrt{\log y}})$ and $O(y^{3/4})$, respectively, and hence the required Liouville estimate. This is additional analytic input for these countermodels; the preceding certificate proof was completed independently.

Since $x/d \geq x^\varepsilon$, and the lower-end partial sum is bounded by its term count, for sufficiently large x all models admit

$$\begin{aligned} b_d &= \frac{x - \sqrt{x}}{d}, \\ r_d &= 1 + \frac{Cx}{d}e^{-c\sqrt{\varepsilon\log x}} + \frac{\sqrt{x}}{d}\mathbf{1}_{\{d\leq\sqrt{x}\}}. \end{aligned}$$

Summing these errors gives

$$\begin{aligned} R := \sum_{d\in\mathcal{D}} r_d &\leq D + O\Big(x\log x\, e^{-c\sqrt{\varepsilon\log x}} + \sqrt{x}\log x\Big) \\ &\ll_\varepsilon xe^{-c_\varepsilon\sqrt{\log x}}. \end{aligned}$$

Thus, even with divisor range reaching $x^{1-\varepsilon}$, all three models remain feasible under the specified termwise error budgets.

Now take $P(z) = \prod_{p \le z} p$ and target $\Phi_z(n) = \mathbf{1}_{\{\gcd(n,P(z))=1\}}$. Within $(\sqrt{x}, x]$, integers satisfying $\Phi_{\sqrt{x}}(n) = 1$ are exactly primes: two prime factors both exceeding $\sqrt{x}$ would already have product exceeding x. Set $Q = \pi(x) - \pi(\sqrt{x})$. Then

$$\boxed{\sum_n w_+(n)\Phi_{\sqrt{x}}(n) = 0, \qquad \sum_n w_0(n)\Phi_{\sqrt{x}}(n) = Q, \qquad \sum_n w_-(n)\Phi_{\sqrt{x}}(n) = 2Q.}$$

Every valid lower bound using only these Type I constraints must therefore be nonpositive, and every valid upper bound at least $2Q$. Nonnegativity supplies a zero lower bound, so the optimal provable lower bound is exactly zero. These three models do not determine whether the optimal upper bound equals $2Q$.

This identifies the resolution of the specified information: three models satisfy compatible divisibility-count constraints while retaining different parity and prime weights. The preceding certificate uses intertime correlations and analytic relations; these countermodels identify the limits of conclusions from the stated Type I bands alone.

7.4.3 Relations between Certificates

The BC system also contains divisibility information. In the Gibbs state on the full integer spectrum, for $\beta > 1$, divisibility projection $E_d = \mu_d \mu_d^*$ satisfies

$$\varphi_\beta(E_d) = \frac{\sum_{d|n} n^{-\beta}}{\zeta(\beta)} = d^{-\beta}.$$

Divisibility weights are thus a family of readouts of the full system. For the divisibility projection $E_d^W = P_W E_d P_W$ on a finite occupation box, the expectation is

$$\varphi_{\beta,W}(E_d^W) = \frac{\sum_{\substack{n \in W \\ d|n}} n^{-\beta}}{\sum_{n \in W} n^{-\beta}},$$

This retains the actual window conditions. Once divisibility expectations are identified with readouts, using finite-window positivity still requires retaining its support conditions.

A simple example illustrates this. On $1 \le n \le 8$,

$$f(n) = 1 - \mathbf{1}_{2|n} - \mathbf{1}_{3|n} - \mathbf{1}_{5|n} - \mathbf{1}_{7|n} + \mathbf{1}_{6|n} \ge 0.$$

Direct replacement of each divisibility indicator by d^{-1} changes the same coefficient combination into

$$1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{5} - \frac{1}{7} + \frac{1}{6} = -\frac{1}{105}.$$

The negative value persists for $\beta > 1$ sufficiently close to 1, where the full-spectrum Gibbs state is well defined. The reason is that positivity of f holds only on the original finite window; for example, $f(30) = -1$. Direct full-spectrum replacement with unchanged coefficients therefore fails to preserve this finite-window certificate's positivity.

Comparing certificate systems requires preserving objects, parameters, windows, and positivity together. We have established concrete certificate families, the stated Type I limitation, and one failed direct translation. Complete translations that preserve positivity still require formal constructions. Any strict ordering of the systems by deductive strength likewise requires a proof.

Extensions to three-prime sums and average distributions over moduli require organizing convolution, Cauchy–Schwarz, and Parseval into estimates uniform in a main parameter, with bilinear,

modulus, and arc analyses supplied. Automatic certificate construction and search efficiency require their own algorithms and resource analyses.

Certificate readout thus connects finite operations to checkable mathematical proofs. A three-time square supplies positivity, and prime-power weights with analytic steps support general counting results. A device must implement the finite correlation measurements and deliver records with parameters and errors. These testable run objects define the engineering requirements of the certificate function.

8 Engineering Implementation

Engineering brings occupation, exchange, phase, and interaction relations into one device. Occupation increments must match prescribed labels and energy changes, boundary controls must physically suppress out-of-range channels, and readout must satisfy task acceptance together with preparation and execution errors. The two-mode window of Section 5 supplies a concrete prototype for testing finite encoding, conditional sampling, and program updates before extending arithmetic, certificate, and interaction functions.

8.1 Implementation Requirements

8.1.1 Finite Carrier

The physical carrier first determines which states the machine can retain. For a finite prime set Λ, let N_p be the occupation observable of mode p, restricted by $0 \le N_p \le L_p$. With integer labels $n = \prod_p p^{v_p(n)}$, a declared encoding relates device and model spaces. The required spectrum is

$$\boxed{H_{\mathrm{dev}} = E_* \sum_{p\in\Lambda} (\log p) N_p, \qquad \Delta_p = E_* \log p, \qquad \omega_p = \Delta_p/\hbar.}$$

Here $E_* > 0$ is a physical energy scale, Δ_p the adjacent-occupation energy difference, and ω_p the corresponding angular frequency, connecting the dimensionless logarithmic spectrum to actual frequencies. Modes may be vibrational degrees of freedom, cavity modes, or encoded finite levels. The chosen carrier requires calibration of occupations, energy gaps, and allowed actions.

For exchange-environment temperature T, the dimensionless inverse temperature is

$$\beta = \frac{E_*}{k_{\mathrm{B}} T}, \qquad e^{-\Delta_p/(k_{\mathrm{B}}T)} = p^{-\beta}.$$

Write actual evolution time as τ. Phase evolution uses dimensionless parameter $t = E_*\tau/\hbar$, whereas exchange completion depends on transition rates per unit physical time. Energy calibration and exchange rate scales therefore require separate measurements; frequency alone does not determine relaxation speed.

A program selects windows $\ell_p \le N_p \le u_p$ within the configured carrier. Preparation must reach the required lower corner, holding must suppress channels declared closed, and readout must resolve task-required occupations or phases. Actual auxiliaries, out-of-range states, and failure records are included in the device description.

8.1.2 Preserving the Relations

The same devices must preserve the correspondence between arithmetic actions and physical evolution. For admissible-window projection P_W, restricted multiplication $K_m = P_W \mu_m P_W$ must satisfy

$$[H_{\mathrm{dev}}, K_m] = E_* (\log m) K_m, \qquad K_m^\dagger K_m = P_{D_m},$$

Here D_m is its valid domain: inputs whose products remain admissible. Energy differences specify exchange energy, while the valid domain specifies deliverable outputs. Window changes update the domain, and actual instruments retain out-of-range outcomes.

Sampling additionally requires the correct exchange ratio on every open paired edge. If both directions share one paired matrix element and an environment at temperature T supplies energy, the target relation is

$$\frac{\Gamma_{p,+}}{\Gamma_{p,-}} = p^{-\beta}.$$

Section 5 gives an explicit guarantee for flat levelwise rates. Paired nonflat exchange can retain the same target distribution, with time guarantees from the gap comparison in Section 8.2. Actual rates, autonomy of occupation statistics, independent mode clock rates, and thermal ratios are calibrated separately. Phase-dependent tasks must additionally preserve coherences actually entering readout, for times determined by their control and measurement processes.

Figure 15 places these requirements in one finite ancilla-collision device. The arithmetic module carries occupations and exchange; the interaction module receives conditions, controls boundaries, reads states, and retains results. Operating energy, environmental refresh, and the controller each have explicit roles.

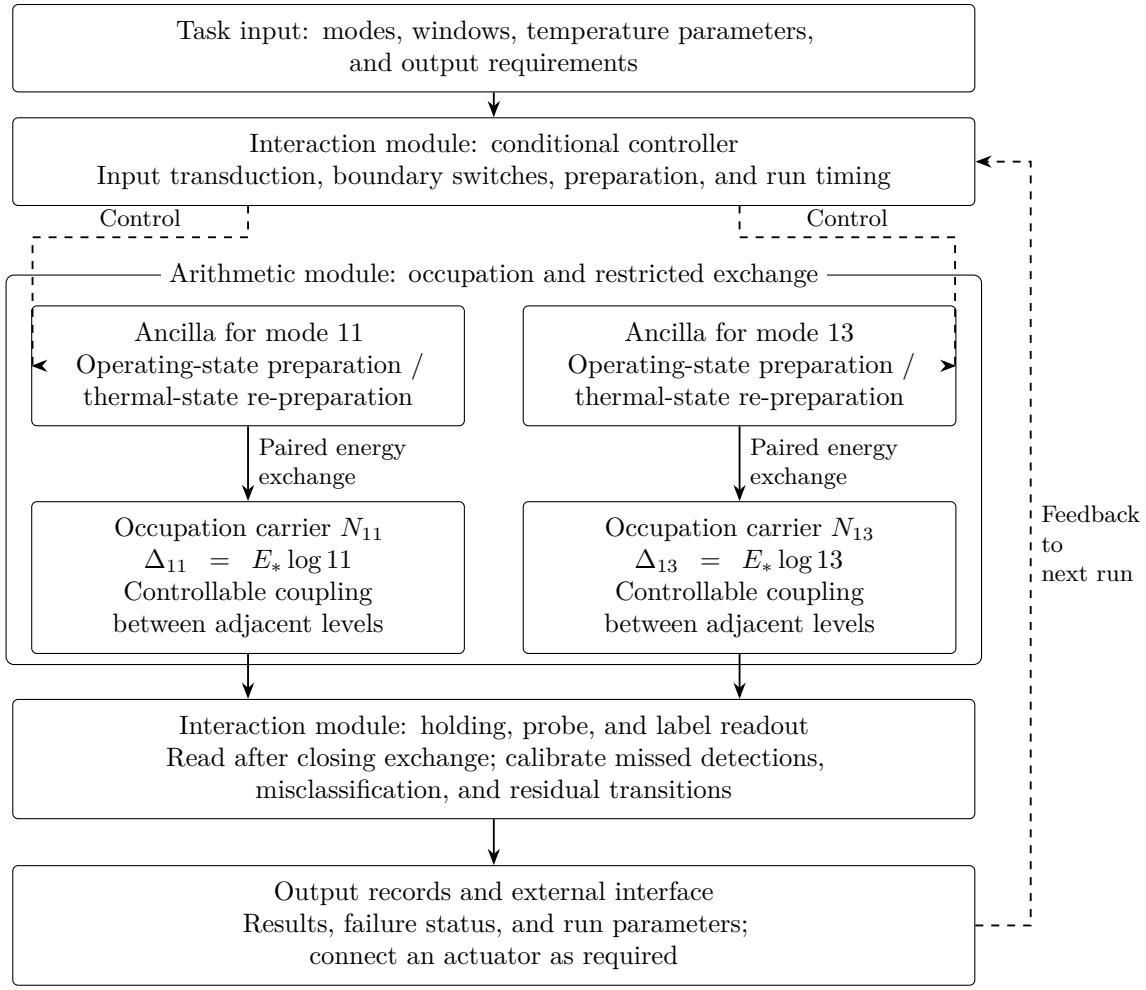


Figure 15: An ancilla-collision realization of the finite machine. Solid arrows denote input, physical exchange, or readout transfer; dashed arrows denote control and feedback, as labeled. Both modules jointly perform the task, with ancilla preparation, refresh, and control included in machine resources. A continuous reservoir may connect through its actual exchange channels; a phase carrier may be added for later tasks.

8.2 Device Realizations

8.2.1 Occupations and Multiplication

Occupation registers store prime exponents in distinguishable finite levels. Trapped-ion vibrational modes may use phonon numbers, and superconducting cavities photon numbers; coupling to auxiliary two-level systems supports preparation, control, and readout. Device selection follows the required logarithmic energy gaps. Spectral calibration covers the working occupation range to preserve the correspondence between labels, occupation changes, and energy changes.

For a chosen local window, denote the flat upward shift by

$$A_p = \sum_{j=\ell_p}^{u_p-1} |j+1\rangle\langle j|.$$

It has identical matrix elements at every internal level and stops at the upper endpoint. An ordinary oscillator creation operator instead satisfies $a^\dagger|j\rangle = \sqrt{j+1}|j+1\rangle$. In oscillator number-state encoding, the required shift is

$$A_p = P_W a^\dagger (N_p+1)^{-1/2} P_W.$$

Equally spaced spectra and flat levelwise actions are therefore distinct requirements. Bare linear-drive matrix elements increase with occupation; a flat shift requires number-selective control, nonlinear coupling, or suitable pulses. Equally spaced transitions share one frequency, so levelwise compensation needs a selection mechanism beyond frequency alone.

An explicit multiplication design uses an auxiliary two-level system with the same gap Δ_p. Denote its lower and upper states by $|g\rangle, |e\rangle$, write $\sigma_- = |g\rangle\langle e|$, and take

$$H_0 = \Delta_p\big(N_p \otimes I + I \otimes |e\rangle\langle e|\big),$$
$$H_{\text{int}} = \hbar g_p\big(A_p \otimes \sigma_- + A_p^\dagger \otimes \sigma_+\big).$$

Since $[N_p, A_p] = A_p$ and $[|e\rangle\langle e|, \sigma_-] = -\sigma_-$, the energy changes cancel and $[H_0, H_{\text{int}}] = 0$. In the interaction picture, each pair $|j,e\rangle$ and $|j+1,g\rangle$ exchanges independently. With $\theta_p = g_p\tau_{\text{int}}$,

$$|j,e\rangle \longmapsto \cos\theta_p|j,e\rangle - i\sin\theta_p|j+1,g\rangle, \qquad j < u_p.$$

At $\theta_p = \pi/2$, the ancilla releases one energy quantum, occupation rises by one, and the integer label is multiplied by p. The upper-end input $|u_p, e\rangle$ does not transfer and can be retained as a failure branch through ancilla readout. Composite multipliers use restricted operations along prime directions, with energy and ancilla-reset costs counted per actual call.

Trapped-ion experiments have demonstrated nearly deterministic phonon addition and subtraction using an auxiliary two-level system and pulse control while preserving coherence of the tested states [31], supporting normalized shifts. Superconducting-cavity number-state preparation has experimental foundations [32], and a recent preprint reports finite-dimensional gates in truncated photon-number spaces using Jaynes–Cummings interactions and ancilla rotations [33]. These supply concrete occupation-control methods. For multiplication, control must implement the shift on its valid domain and the upper-end failure branch. Complete-transfer pulses and continuous partial exchanges require calibration of their respective actions and timing. Drive energy and ancilla reset are included in operation costs.

Occupation indices may also be encoded in other finite levels or modes. Coherent control of adjacent Wannier–Stark states has been demonstrated in tilted optical lattices [34], and boundaries formed through auxiliary rings have been demonstrated in synthetic frequency dimensions [35]. These support adjacent actions and finite-range control, respectively. Their use here requires calibrating encoded gaps, exchange channels, and readout within the same working states, so mode indices carry the prescribed occupation meaning.

8.2.2 Exchange and Boundaries

The auxiliary two-level exchange structure can also implement thermalization. Independently prepare each ancilla in the thermal mixture

$$\tau_{B,p} = p_g|g\rangle\langle g| + p_e|e\rangle\langle e|, \qquad p_e = \frac{z_p}{1+z_p}, \quad p_g = \frac{1}{1+z_p}, \quad z_p = p^{-\beta},$$

and remove and reprepare it after use, giving a repeated-interaction collision model [36]. A complete collision channel is

$$\Phi_p(\rho) = \mathrm{Tr}_B\left[U_p(\rho \otimes \tau_{B,p})U_p^\dagger\right], \qquad U_p = e^{-i\theta_p(A_p\otimes\sigma_- + A_p^\dagger\otimes\sigma_+)}.$$

Upward occupation transfer along an internal edge requires an initially excited ancilla; downward transfer requires its ground state. The occupation-transition probabilities for a single collision are therefore

$$a_p = p_e \sin^2\theta_p, \qquad b_p = p_g \sin^2\theta_p, \qquad \frac{a_p}{b_p} = z_p.$$

Unpaired endpoint states do not exchange beyond the boundary. Diagonal inputs remain diagonal, and the output occupation distribution depends only on the input occupation distribution, so the occupation block is a reflecting birth–death chain. For $\sin\theta_p \neq 0$ and $0 < T < \infty$, internal edges are connected and the unique stationary weights follow z_p^j.

If an event model calls the complete collision channel on each mode through an independent Poisson clock of rate r_p, then

$$\mathcal{L}_p = r_p(\Phi_p - I), \qquad \Gamma_{p,+} = \frac{\kappa_p z_p}{1+z_p}, \quad \Gamma_{p,-} = \frac{\kappa_p}{1+z_p}, \quad \boxed{\kappa_p = r_p \sin^2\theta_p.}$$

The occupation block then matches the continuous-time model of Section 5. For fixed-interval collisions, use Φ_p^M directly to calculate the result after M collisions. Actual collision duration, ancilla refresh, and dead time determine which clock model applies; event rates must be calibrated from the complete operating process.

Flat levelwise rates give the explicit model above. Sampling also permits different strengths on different edges: each paired forward and reverse transition shares its edge factor, leaving target weights determined by their ratio and convergence speed by edge strength. For one mode, suppress the mode index, set $z = p^{-\beta}$, and use on window $\ell \le j \le u$

$$q_{j,j+1} = \frac{\kappa z}{1+z}c_j, \qquad q_{j+1,j} = \frac{\kappa}{1+z}c_j, \qquad c_j > 0 \quad (\ell \le j < u).$$

Here $\kappa > 0$ is the reference rate and c_j the dimensionless edge strength; channels outside the window are closed. Normalized geometric weights satisfy edgewise

$$\pi_j = \frac{z^{j-\ell}}{\sum_{r=\ell}^{u} z^{r-\ell}}, \qquad \pi_j q_{j,j+1} = \pi_{j+1} q_{j+1,j}.$$

Thus π remains the unique stationary distribution when internal edges are connected. Let Q_c be this occupation generator and Q_1 the flat generator with $c_j = 1$ on the same window. Their equilibrium measures coincide, and their Dirichlet forms satisfy

$$\begin{aligned}
\mathcal{E}_c(f,f) &= -\langle f, Q_c f\rangle_\pi \\
&= \sum_{j=\ell}^{u-1} \pi_j \frac{\kappa z}{1+z} c_j |f(j+1) - f(j)|^2 \\
&\ge c_{\min}\mathcal{E}_1(f,f), \qquad c_{\min} = \min_{\ell \le j < u} c_j.
\end{aligned}$$

Taking variational infima with the same denominator $\mathrm{Var}_\pi(f)$ and substituting the flat-chain gap yields, for $d = u - \ell + 1 \geq 2$,

$$\boxed{\gamma_c \geq c_{\min}\gamma_1, \qquad \gamma_1 = \kappa\left[1 - \frac{2\sqrt{z}}{1+z}\cos\frac{\pi}{d}\right].}$$

For independent modes starting from a product state, write $C_p = \chi^2(\nu_{p,0}\|\pi_p)$. Local reversible-chain $L^2(\pi_p)$ contraction and the product-distribution χ^2 identity give

$$\begin{aligned}\chi^2(\nu_\tau\|\pi) &= \prod_p \left[1 + \chi^2(\nu_{p,\tau}\|\pi_p)\right] - 1 \\ &\leq \exp\left(\sum_p C_p e^{-2\gamma_{p,c}\tau}\right) - 1.\end{aligned}$$

Taking $\underline{\gamma} = \min_p(c_{p,\min}\gamma_{p,1})$ and $C_\Sigma = \sum_p C_p$, Cauchy–Schwarz yields

$$\|\nu_\tau - \pi\|_{\mathrm{TV}} \leq \frac{1}{2}\sqrt{\exp\left(C_\Sigma e^{-2\underline{\gamma}\tau}\right) - 1}.$$

The minimum ranges over nontrivial windows. A single-state window requires no mixing and contributes $C_p = 0$; if all windows are singletons there is no exchange-mixing error. Measured edge strengths thus enter finite-time guarantees. The flat model retains its exact spectrum as a reference, whereas a nonflat implementation uses its own generator or this bound to determine completion time.

A continuous reservoir provides another realization. Trapped-ion experiments have demonstrated vibrational-mode environments with tunable temperature and dissipation rate and studied local-temperature effects in a two-mode system [37]. For bosonic jumps actually restricted to a window, write $P = \sum_{j=\ell}^{u} |j\rangle\langle j|$ and engineer, within that space,

$$\begin{aligned}\mathcal{L}_B(\rho) &= \Gamma_B(\bar{n}+1)\mathcal{D}[PaP](\rho) + \Gamma_B\bar{n}\mathcal{D}[Pa^\dagger P](\rho), \\ \mathcal{D}[L](\rho) &= L\rho L^\dagger - \tfrac{1}{2}\{L^\dagger L, \rho\}.\end{aligned}$$

The occupation-edge strengths are $c_j = j + 1$. Choosing $\bar{n} = (p^\beta - 1)^{-1}$ $(\beta > 0)$ gives forward-to-reverse ratio $p^{-\beta}$ and reference rate $\kappa = \Gamma_B(2\bar{n} + 1)$. Ordinary bosonic matrix elements can therefore serve sampling, while controlled normalized shifts perform coherent multiplication. Actual heating, cooling, and reset establish the engineered environment, with drive energy and control timing included in operation.

Program boundaries enter dynamics by selecting open exchange channels. In the model, the summation range of A_p and jump-operator projections express this selection. In a device, number-selective control or level isolation must suppress the corresponding out-of-range channels. For a window containing only occupations 1 and 2, retain their bidirectional exchange while suppressing $1 \to 0$ and $2 \to 3$. Boundary controls remain active during exchange; measured residual leakage enters the error budget.

Superconducting-cavity experiments restricting driven dynamics by forbidding a selected photon number [38] provide a basis for finite-range control. Application to these windows requires combining boundary control with actual dissipation channels and recalibrating internal gaps and paired rates. Changing a boundary changes admissible states and exchange channels, letting the same carrier execute a new conditional task.

8.2.3 Phases and Readout

Phase operations require orthogonal components with independently controllable relative phases. For a finite modulus q, encode $|a\rangle$ in the single-excitation subspace of q physical modes, implement

V_c through mode phase shifts, and F_q through a multiport interferometer. Any finite unitary decomposes into two-mode transformations and phase shifts corresponding to tunable beam splitters and phase shifters [39]. The arithmetic relation is

$$F_q^{-1} V_c F_q |a\rangle = |a + c \bmod q\rangle.$$

Programmable spatial optical circuits have experimentally controlled finite-mode unitaries [40], providing a basis for these phase units. Phase shifters set relative phases, interferometers change basis, and mode-resolved detection reads outputs. Tests include relative phases, loss, and multi-excitation leakage; changing only a common vector phase does not implement modular addition. Conversion between different frequencies additionally requires modulation or pumping, whose drive is included in the operation.

For a truncated number-state oscillator, phase rotation may implement V_c, while a separately compiled controlled operation implements the Fourier transform. A Fourier gate on a three-level number-state subspace of a superconducting cavity has been reported experimentally [33], motivating occupation and phase tasks on one carrier. The actual dimension, gate error, and conversion timing are calibrated for the implementation.

Occupation readout may use a probe coupling commuting with occupation. For example, the dispersive interaction between an ancilla qubit and mode p is

$$H_{\mathrm{read}} = \frac{\hbar \chi_p}{2} N_p \otimes \sigma_z, \qquad [H_{\mathrm{read}}, N_p] = 0.$$

Probe phase changes with occupation without causing occupation transitions in the ideal model. Number-resolved dispersive response has been demonstrated in superconducting circuits [41], as have preparation and tomography of finite photon-number superpositions [32]. These support candidate components; assembling this device must preserve the selected encoding, exchange law, and boundaries.

Disabling occupation-changing exchange during readout separates sampling and holding. Residual transitions during finite holding, probe-induced dephasing, and readout misclassification contribute separately. Occupation-sample delivery protects labels; coherent-state or phase-response delivery additionally protects the coherence used by readout.

8.2.4 Interaction Connections

The interaction module turns these devices into a machine that receives tasks and delivers results. The input writes supplied prime exponents, temperature parameters, and run conditions into a controller. It sets boundaries, schedules preparation and exchange, and records readouts with the current parameters. Sensor inputs retain their calibrated transduction relations and uncertainties.

The conditional controller is part of the actual device. Electronic circuits, programmable logic, or suitable direct couplings can perform addressing, timing, recording, and feedback. The chosen implementation's bandwidth, computation, randomness sources, and power belong to machine resources. Transition probabilities formed by thermal ancillas and joint exchange are checked through their actual physical process.

Components require consistent state correspondences. Passing occupation labels to a residue carrier uses the encoding of Section 4 and transforms required operations as well. Reading an integer and processing it classically retains conversion and processing costs. For quantum-state transfer between modules, encoding, transmission, and subsequent measurement jointly determine delivery error.

Input, actuation, and recording also require coordinated timing. A sample identifier links the current boundary, preparation-complete signal, exchange start and stop, and output state. Infeasibility, preparation failure, and timeout use distinct outcome labels. After an actuator changes the external environment in response to confirmed records, new inputs enter the next run. Energy exchange, control, and records thereby form a closed machine process.

8.3 Prototype Assembly

8.3.1 Two-Mode Prototype

The six-state task needs only two occupation directions to test finite encoding, boundaries, exchange, and readout together. A prototype may use two distinguishable trapped-ion vibrational modes or superconducting-cavity modes, connecting occupation carriers, auxiliary control, exchange environments, and readout. Selection depends on achievable energy gaps, boundary control, and joint operation within one device.

We use flat levelwise rates as a reference implementation to specify the prototype spectrum and completion time. Nonflat implementations satisfying paired detailed balance in Section 8.2 retain the target distribution, with timing from their actual generator or gap comparison. Take $\Lambda = \{11, 13\}$, configure a carrier covering $0, 1, 2$, and open during sampling

$$1 \le N_{11} \le 2, \qquad 0 \le N_{13} \le 2, \qquad \beta = 1.$$

The corresponding integers satisfy $11 \mid n \mid 11^2 13^2$. Prepare occupations 1 and 0, disable additional phase drives inside the windows, and connect independent exchange. The mode gaps are $E_* \log 11$ and $E_* \log 13$; a common temperature $k_{\mathrm{B}} T = E_*$ gives different but specified thermal ratios.

For the flat reference implementation, suppose both measured mode clock rates are $\kappa_0 > 0$. Require

$$(\Gamma_{11,+}, \Gamma_{11,-}) = \kappa_0 (1/12, 11/12), \qquad (\Gamma_{13,+}, \Gamma_{13,-}) = \kappa_0 (1/14, 13/14).$$

In the collision scheme this means choosing $r_p \sin^2 \theta_p = \kappa_0$. Both mode clock rates and ancilla-refresh capabilities must be supplied simultaneously. If only alternating calls are available, recalibrate against the event time each mode actually receives.

The joint equilibrium distribution and its updated conditional distribution are listed below. Fractions follow exactly from the finite partition functions.

Integer label n	Occupation (N_{11}, N_{13})	Six-state target probability	After adding $13 \mid n$
11	$(1, 0)$	1859/2196	0
121	$(2, 0)$	169/2196	0
143	$(1, 1)$	143/2196	143/168
1573	$(2, 1)$	13/2196	13/168
1859	$(1, 2)$	11/2196	11/168
20449	$(2, 2)$	1/2196	1/168

These probabilities arise from two windows and paired exchange. The device writes boundaries, mode clock rates, and temperature conditions; six-state target weights serve independent verification.

8.3.2 Run Timing

A run first checks that the input window is nonempty, disables sampling exchange, prepares $|11\rangle$, and confirms the boundaries. It then enables independent exchange, stops at the specified time, holds occupations, reads both labels, reconstructs the integer, and delivers the result. Samples and failures use one counting convention.

Express the exchange stage in dimensionless time $s = \kappa_0 \tau$. The nonzero relaxation rates of the six-state occupation generator are

$$\frac{\gamma}{\kappa_0} \in \left\{ 1 - \frac{\sqrt{13}}{14},\ 1,\ 1 + \frac{\sqrt{13}}{14},\ 2 - \frac{\sqrt{13}}{14},\ 2 + \frac{\sqrt{13}}{14} \right\}.$$

The joint gap is approximately $0.742461\kappa_0$. Using the uniform bound $\gamma_* \simeq 0.447229\kappa_0$ and $C_* = 0.1$ of Section 5.1.2, the specified lower-corner state reaches total variation tolerance 0.01 by sufficient time $\tau_{0.01} \simeq 6.948124/\kappa_0$. At this time, the ideal generator's numerical total variation is approximately 3.31615×10^{-4}. Device deviations enter through the calibration and error composition of Section 8.4.

A complete cycle includes

$$T_{\mathrm{cycle}} = T_{\mathrm{set}} + T_{\mathrm{prep}} + T_{\mathrm{run}} + T_{\mathrm{read}} + T_{\mathrm{reset}}.$$

This applies to serial execution. Pipelining requires reporting sample throughput period and individual input-to-output latency separately. Normalized exchange time converts to physical time only after measuring κ_0.

8.3.3 Condition Updating

Adding condition $13 \mid n$ to the sampling program changes the lower bound of mode 13 from 0 to 1 while retaining upper bound 2; mode 11 is unchanged. The controller closes the edge joining 0 and 1, prepares the new lower corner $|143\rangle$, and reopens retained internal exchange.

The updated admissible set is

$$143 \mid n \mid 20449, \qquad P_{\mathrm{new}}(n) = P_{\mathrm{old}}(n \mid 13 \mid n).$$

giving the four-state target in the table. In the flat reference implementation, both modes now have two admissible occupations and the joint relaxation spectrum is $\{0, \kappa_0, \kappa_0, 2\kappa_0\}$. The same device, thermal ratios, and internal exchange law remain in use; changes concern boundaries and preparation. Paired weighted implementations retain the same conditional weights, with updated spectra determined by retained edge strengths.

The prototype realizes Section 5.1.3 through actual edge closure and fresh preparation. Update duration and energy are recorded separately. Both output distributions, leakage across closed edges, and record consistency jointly determine acceptance (Figure 16).

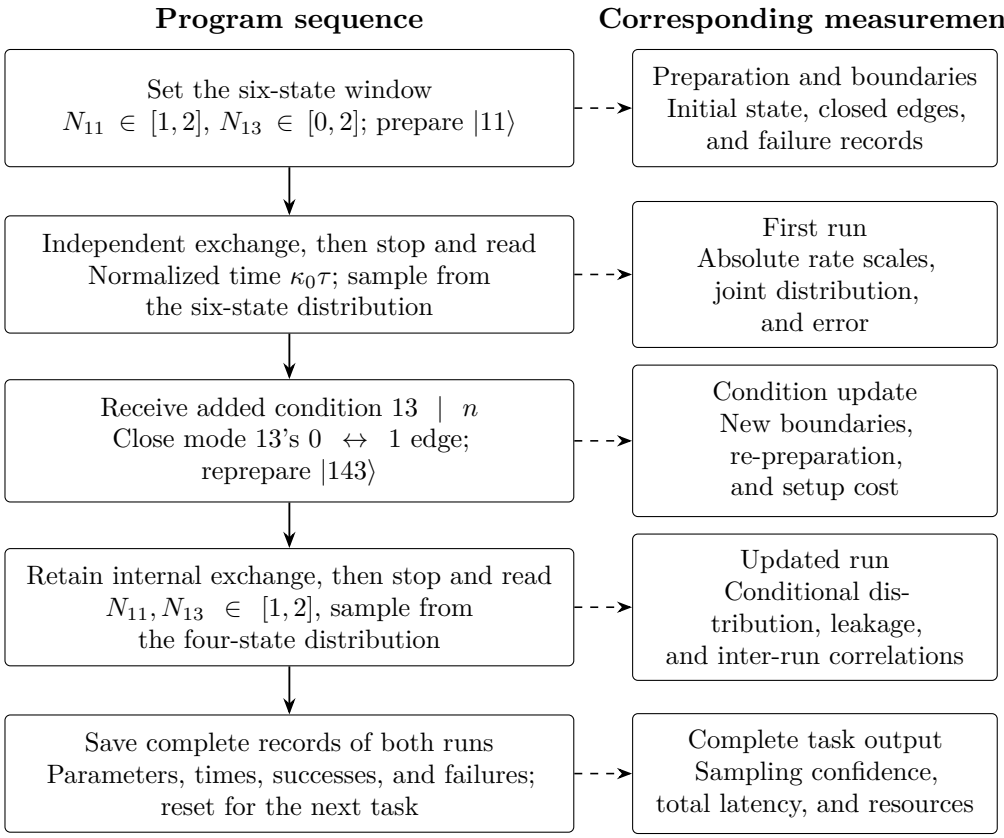


Figure 16: Two rounds of conditional sampling and experimental verification for flat reference exchange. The left column gives execution order; the right lists measurements retained at each stage. The second run reuses the carrier and internal exchange while updating boundaries and lower-corner preparation. Both rounds' statistics and resources come from complete records.

8.4 Experimental Validation

8.4.1 Component Calibration

Calibration begins with directly measurable quantities. Measure levelwise gaps and occupation readout, then calibrate restricted raising and reverse actions on known inputs. Within the valid domain, check occupation increments, ancilla energy changes, and success probabilities; at boundaries, check suppressed channels and out-of-range records.

For the flat coherent-exchange design of Section 8.2, calibrate matrix elements level by level. If a paired edge has matrix element m_j, ideal resonant two-state exchange satisfies $P_{j\to j+1}(\tau) = \sin^2(g_p|m_j|\tau)$, and its first complete transfer obeys

$$\tau_j^{\max} = \frac{\pi}{2g_p|m_j|}, \qquad \Omega_j := \frac{\pi}{2\tau_j^{\max}} = g_p|m_j|.$$

The flat design gives $\Omega_j/\Omega_{\ell_p} = 1$, while bare oscillator linear coupling gives $\sqrt{(j+1)/(\ell_p+1)}$. Loss alters maximum transfer, so fits should report the model and residual deviations together. Dissipative transition rates depend on squared matrix-element magnitudes and require separate measurement from Ω_j defined here. Coherent multiplication gates are tested on occupation inputs and their superpositions for complete-operation amplitudes, levelwise phases, and failure branches. Composite-pulse implementations require calibration of their complete channels.

Sampling calibration uses different initial occupation states to measure short-time transitions in both directions along every internal edge and the full time-dependent occupation probabilities $P_j(\tau)$. The flat reference checks constant rates edgewise; paired weighted implementations check

target ratios and actual strengths, deriving spectra or bounds from the reconstructed generator. Equilibrium adjacent ratios test thermal weights; short-time transfers and full occupation curves jointly test rates and finite-time predictions.

Readout uses known labels to calibrate a confusion matrix, retaining missed detections and invalid results. Phase units additionally test $F_q^{-1}V_cF_q$ and necessary coherence properties. These measurements specify components and feed the machine error budget.

8.4.2 Mechanism Tests

Testing exchange mechanisms requires varying physical conditions while recording controls. With coupling pulses and event rules fixed, temperature changes should move target adjacent ratios according to $e^{-\Delta_p/(k_{\mathrm{B}}T)}$. Matched system and ancilla gap changes should likewise produce the corresponding response. Retuning, ancilla preparation, and other control actions during scans belong to the setup and must accompany results.

The thermal ancilla mixture is an important input in the collision construction. A coherent rotation preparing a superposition of $|g\rangle$ and $|e\rangle$ generally retains bath coherence and cannot directly replace $\tau_{B,p}$. Thermal contact, classical preparation with known probabilities, or dephasing with discarded records can realize the mixture. The refreshed ancilla's declared independence from the current input must then be checked.

Control experiments may retain the static spectrum while changing exchange channels, or vary levelwise couplings at fixed temperature. The first tests admissible processes; the second distinguishes equilibrium weights from relaxation speed. Controller inputs, randomness sources, and event-by-event actions also belong in the comparison. Agreement with one ratio alone does not exclude alternative mechanisms.

8.4.3 Task Acceptance

Acceptance after assembly concerns complete task output. The six-state prototype must reproduce all output probabilities and finite-time convergence under common preparation, boundaries, and clock rates. The four-state update must retain both its new domain and correct conditional weights. Records must distinguish settings, successful outputs, preparation failures, and timeouts across the two runs.

Mathematical predictions, numerical checks, and hardware measurements have different roles. The table compares them for the flat reference prototype; numerical entries are checked against the present finite-model calculation and Section 5. Paired weighted implementations retain target probabilities but update finite-time references using measured rates.

Object	Mathematical and numerical flat-model result	Required hardware measurements
Local exchange	Upward-to-downward ratios 1/11 and 1/13; constant rates satisfy detailed balance	Forward and reverse transitions on every internal edge, absolute rate scales, and boundary leakage
Six-state equilibrium	Product of two local weights gives the six listed fractions	All output and failure probabilities, and intermode correlations
Finite time	Gap $0.742461\kappa_0$; sufficient time $6.948124/\kappa_0$	Occupation curves from several initial states and joint error at the target time
Condition update	Four-state target $(143, 13, 11, 1)/168$ in retained-label order	Updated preparation, closed edges, and complete output
Sequential delivery	Each input corresponds to settings, timing, and a report	Record consistency, end-to-end latency, reset, and inter-run correlations

A larger window from the preceding numerical study supplies a later scaling comparison. For $\Lambda = \{11, 13, 17\}$, $\beta = 0.65$, lower bounds $(1, 0, 1)$, upper bounds $(4, 3, 3)$, and mode clock rates $(1, 1.2, 0.9)$,

the 48-state flat model has occupation gap approximately 0.464049. Target total variation 0.05 has sufficient time approximately 10.067638, at which numerical total variation is approximately 0.00104041. These data check the specified finite model; measured rates determine physical time.

The project's trial-run summary archived on July 29, 2026 also records collision-gate circuits executed on the general-purpose quantum processor ibm_fez. That experiment used different parameters: $q = 7$, $\beta = 1$, four system levels, a ground-state initial condition, five collisions, and angle $\theta = 0.70$. The reported lowest adjacent-level ratio was approximately 0.560, or 0.410 with dynamical decoupling and randomization, both substantially different from the corresponding ideal finite-step value 0.12525. The protocol thus reached actual gate-circuit execution but did not meet quantitative acceptance for this metric. It is neither a hardware result for the 11/13 prototype nor validation of direct energy exchange or temperature scans. The summary and job identifiers are in [42]; quantitative attribution still requires raw counts, calibration, and controls.

Statistical acceptance also requires sample counts and confidence levels. Suppose each independent fresh preparation yields one report, with d_{out} complete report classes including declared failures. Let P be the true delivered distribution and $\widehat{P}$ the empirical distribution from M independent runs. Hoeffding bounds for each class and a union bound [43] give

$$\Pr\left(\max_y |\widehat{P}(y) - P(y)| > a\right) \leq 2d_{\mathrm{out}} e^{-2Ma^2}.$$

Thus, with confidence at least $1 - \alpha$,

$$\|P - P_{\mathrm{target}}\|_{\mathrm{TV}} \leq \|\widehat{P} - P_{\mathrm{target}}\|_{\mathrm{TV}} + \frac{d_{\mathrm{out}}}{2}\sqrt{\frac{\log(2d_{\mathrm{out}}/\alpha)}{2M}}.$$

Only when the right side lies within the predefined tolerance does this sampling specification pass acceptance. The bound is conservative but explicit about samples and confidence; continuously correlated trajectories require separate correlation control.

8.4.4 Error Composition

Moving from component specifications to machine guarantees requires controlling states and records after each component. Use a common encoding to place model and device in comparable finite spaces, and let $\mathcal{M}_j$ and $\widetilde{\mathcal{M}}_j$ be the ideal and actual complete channels for stage j. Inputs and outputs retain records, controllers, and correlated environment still participating downstream, with failures and leakage treated as outcomes.

Suppose a uniform channel-error bound holds over the declared inputs and reference correlations:

$$\frac{1}{2}\|\widetilde{\mathcal{M}}_j - \mathcal{M}_j\|_\diamond \leq \delta_j,$$

Let Δ_j be the joint-state distance after stage j. Inserting the actual channel acting on the ideal input between actual and ideal outputs, the triangle inequality and trace-distance contraction give

$$\begin{aligned}\Delta_j &\leq \frac{1}{2}\|\widetilde{\mathcal{M}}_j(\widetilde{\rho}_{j-1}) - \widetilde{\mathcal{M}}_j(\rho_{j-1})\|_1 + \delta_j \\ &\leq \Delta_{j-1} + \delta_j.\end{aligned}$$

Applying this stagewise to a program with a deterministic stage-count bound, and including initial preparation and terminal readout, yields the complete classical-report bound

$$\boxed{\|P_{\mathrm{dev}} - P_{\mathrm{target}}\|_{\mathrm{TV}} \leq \varepsilon_{\mathrm{model}} + \delta_{\mathrm{prep}} + \sum_j \delta_j + \delta_{\mathrm{read}}.}$$

Here $\varepsilon_{\mathrm{model}}$ is the ideal run's error relative to the task target, such as finite-time sampling error in Section 5. Readout error already included in a stage channel is not added again at the endpoint.

Random call counts can be truncated at a declared cutoff, with exceedance probability included in the report; a bound directly on a complete finite-time channel may instead treat that segment as one stage.

If only classical samples are delivered and downstream processing uses only them, classical-kernel bounds suffice within that statistical scope. Quantum-state delivery or retained reference correlations require the joint-channel specification above. A finite test set does not automatically establish a full diamond-norm bound; its scope follows from established model identification and calibration. Component measurements, task tests, and theoretical errors thereby meet within one output specification.

8.5 Scaling

8.5.1 Functional Extensions

The two-mode prototype first implements occupation programs. Adding a phase carrier enables modular addition and encoding conversion; ancilla-controlled evolution enables finite-correlation readout; sensing and actuation interfaces enable continuous interaction tasks. Each extension retains condition input, timing, and recording while establishing component and task acceptance for new operations.

The finite autocorrelation certificates of Section 7 provide direct measurement targets. For a finitely supported coefficient family c_n, choose a finite window W containing every label for which $c_n \neq 0$, write $Z_c = \sum_n |c_n|^2 > 0$, and prepare state ρ_c with occupation probabilities $|c_n|^2/Z_c$. It may be the corresponding pure state or a mixed state with the same occupation distribution. For finite spectrum $H_W = \sum_{n\in W} (\log n)|n\rangle\langle n|$,

$$\chi_c(t) := \mathrm{Tr}(\rho_c e^{-itH_W}), \qquad Z_c\chi_c(t) = \sum_n |c_n|^2 n^{-it} = C_x(t).$$

If the device implements ancilla-qubit-controlled evolution

$$U_{\mathrm{ctrl}} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes e^{-itH_W},$$

and initializes the ancilla in $|+\rangle$, its reduced output state is

$$\rho_{\mathrm{a}} = \frac{1}{2}\begin{pmatrix} 1 & \chi_c(t)^* \\ \chi_c(t) & 1 \end{pmatrix}.$$

The expectation values of the ancilla Pauli observables X and Y are therefore $\mathrm{Re}\,\chi_c(t)$ and $\mathrm{Im}\,\chi_c(t)$, respectively. For $t \geq 0$, physical duration is $\tau = \hbar t/E_*$, over which controlled evolution and ancilla coherence must be maintained. Negative-parameter correlations follow from $\chi_c(-t) = \overline{\chi_c(t)}$.

Correlations at settings t and $2t$ test the finite quantity of Section 7:

$$Q_x(t) = Z_c\big[3 + 4\,\mathrm{Re}\,\chi_c(t) + \mathrm{Re}\,\chi_c(2t)\big].$$

The character-weighted certificates of Section 7.3.4 additionally require cross-correlations carrying the character phases. Prepare a state on coprime labels with the finite prime-power weights of the principal character, and combine the character-phase action of Section 5.2.2 with the time evolution used here. Assign arbitrary unit-modulus phases to the other window labels to extend the phase action to a unitary. With controlled access to the combined operation, the same ancilla readout measures its response and that of its square, yielding the two required character correlations. If preparation uses coprime-sector filtering, its success probability is retained in the complete output.

Outputs retain coefficients, parameters, and measurement intervals. The device tests specified finite instances and actual errors; the general inequality still follows from the square decomposition and parameterized proof of Section 7. Interlabel coherence is not necessarily required in the occupation carrier here, whereas coherence used by auxiliary control and readout must be retained, consistent with task-dependent protection in Section 6.

8.5.2 Modes and Precision

More modes and occupation levels enlarge the label range while increasing addressing, connection, and readout demands. Physical frequencies, spatial locations, or control channels provide mode addresses. When frequency bands or control lines are shared, the worst neighboring-frequency separation, linewidth, and crosstalk determine simultaneously selectable operations.

Distinct integers can have very close energies. If admissible labels are exactly $\{1, \ldots, N\}$, the smallest total-energy difference is

$$\Delta E_{\min} = E_* \log \frac{N}{N-1} \qquad (N \geq 2).$$

Direct discrimination by total energy needs the corresponding resolution. Modewise readout instead uses local occupation instruments, with scaling costs from mode addressing, local levels, and joint reports. The selected discrimination method determines the measurement protocol's time and precision.

For independent geometric distributions over a finite mode set, with original occupations starting at zero and only $0, \ldots, L_p$ retained, the discarded single-mode probability is $p^{-\beta(L_p+1)}$. Joint truncation satisfies

$$P_{\text{tail}} \leq \sum_{p \in \Lambda} p^{-\beta(L_p+1)}.$$

Register depth must satisfy both equilibrium-tail requirements and maximum occupations of inputs and intermediate operations.

With the lower-corner preparation and mode independence of Section 5, local initial states satisfy $C_p = \pi_p(\ell_p)^{-1} - 1 \leq z_p/(1-z_p)$. If $z_p \leq z_* < 1$, $\kappa_p \geq \kappa_- > 0$, and open paired weighted edges have uniform lower bound $c_{p,j} \geq c_- > 0$, Section 8.2 gives

$$\underline{\gamma} \geq c_- \kappa_- \frac{(1-\sqrt{z_*})^2}{1+z_*}, \qquad C_\Sigma \leq N_{\text{mode}} \frac{z_*}{1-z_*}.$$

Substitution into the joint-distribution bound retains sufficient exchange time scaling as $\log(N_{\text{mode}}/\varepsilon)$. If edge strengths or mode clock rates decline with scale, use actual spectral rates; fixed total exchange budgets are treated on the same basis. Partitioned connections may reduce local crosstalk, while cross-partition constraints, encoding conversions, and aggregated readouts are budgeted from actual connections.

8.5.3 Performance Evaluation

Performance comparisons use complete tasks. For conditional sampling, first match admissible domains, input representations, target distributions, output errors, and confidence, then compare time, energy, and device scale from input availability to delivery. Supplied prime exponents and ordinary integers are different input conditions; independent samples and correlated trajectories also have different delivery specifications.

Sampler comparisons must also match reachable sets. For example, multiplication and division by primes only in a finite set Λ preserve the outside-prime factor $c_\Lambda(n) = \prod_{p \notin \Lambda} p^{v_p(n)}$, partitioning state space into disconnected classes. Mixing times and sampling costs describe the same task only after both implementations use the same reachable class and its conditional target distribution.

For factor-encoded window tasks, classical programs can directly exploit local probability factorization and should serve as comparison baselines for physical exchange. Condition-update frequency, sample demand, input–output connections, and actual parallel resources determine costs. Phase or joint-state tasks likewise require reference implementations matched to their encoding and delivery specifications.

As functions, modes, and connections grow, machine benefits come from actual reductions in conversion, retries, control, or waiting, while costs come from added carriers, clocks, ancilla refresh, and readout. Measuring both under common records and task specifications determines the structural advantages realized in a device.

The two-mode six-state prototype places encoding, exchange, and readout in concrete components and schedules; the four-state update additionally tests program reuse. Calibration, mechanism tests, and complete-output acceptance connect, while error composition links theoretical guarantees to actual delivery. Adding phase processing, finite certificate correlations, and interaction interfaces can then expand the machine's capabilities; the engineering benefits can be assessed under the same task specifications.

9 Conclusion

The BC state-space computer connects data, operations, and result formation through a common arithmetic structure. Mode occupations carry integer factor composition, multiplication directly changes those occupations, and the logarithmic spectrum assigns a definite energy increment to the same multiplier. Within the valid domain of finite operations, occupation encoding and covariance relations jointly express this connection. Encoded composition information can therefore continue through addressing, operations, and handoffs. The same spectrum organizes phase responses and, under specified thermal exchange, enters statistical output weights. Data relations thus become part of the actual processing dynamics.

$$\boxed{\begin{aligned} n &= \prod_p p^{v_p(n)}, \\ [H, K_m] &= (\log m) K_m. \end{aligned}}$$

Boundary programming lets task conditions participate directly in organizing the process. Divisibility constraints compile into modewise occupation intervals, and their combination and updating become interval adjustments. In the established independent-window model, updating boundaries and preparing the new lower corner allow the same components to reuse their internal exchange law and finite-time guarantees. Components thus serve changing task conditions. Specified independent exchange also lets modes form joint statistics simultaneously. Preparation, evolution, and readout jointly determine completion, with the required distribution produced by a constrained physical process. Data representation, program arrangement, and output verification thereby correspond within one arithmetic structure (Figure 17).

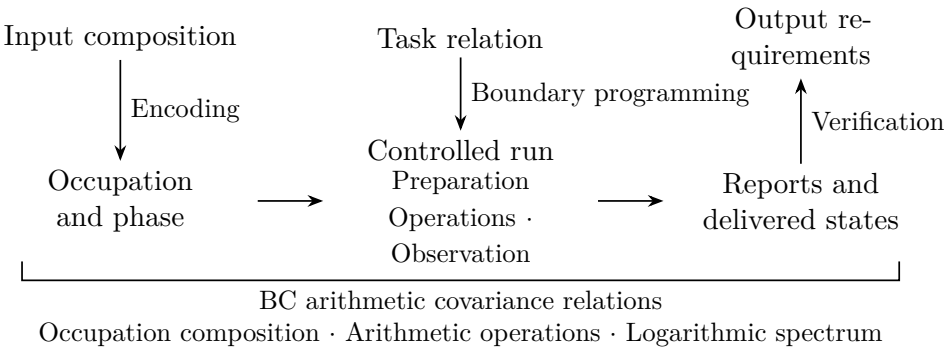


Figure 17: A common arithmetic structure connects data, programs, and outputs. The upper row represents task specifications and the lower row physical operation. Vertical arrows denote encoding, boundary programming, and verification. Occupations, operations, and the spectrum connect these stages; in the window model, exchange realizes these relations as task statistics.

The arithmetic and interaction modules jointly turn these capabilities into complete computation. External conditions enter preparation and control through transduction, internal processing produces reports, delivered states, or environmental responses, and states, records, and correlations still participating continue into the next run (Figure 18). Computation thereby extends from one-shot evaluation to continuous detection and feedback. Holding-stage conservation preserves occupation records, with nondemolition readout supporting repeated reading; the complete subsequent readout determines which coherences need protection. For occupation programs with fixed transition laws and instruments, added pure occupation dephasing leaves report statistics unchanged, while phase tasks protect the coherences they use. Program reuse, record continuation, and task-dependent retention together define characteristics of the complete machine.

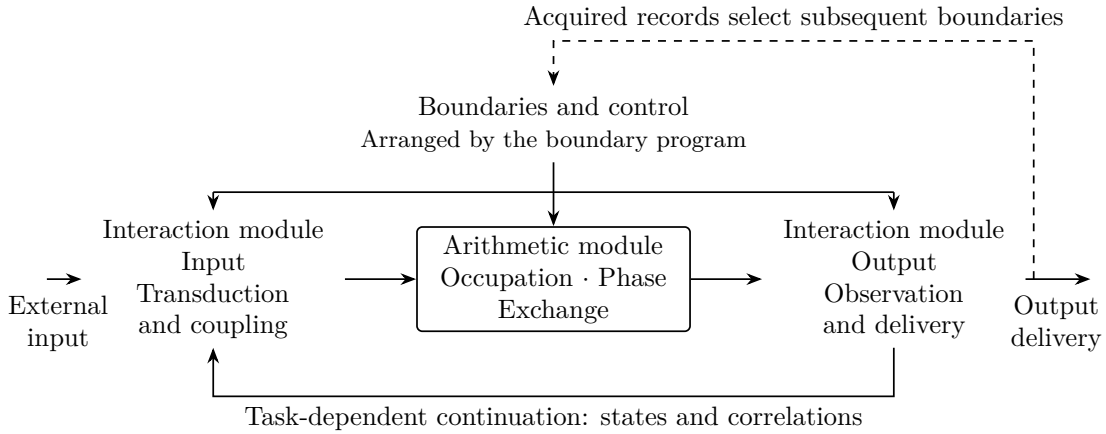


Figure 18: A boundary program organizes complete computation that can continue into subsequent runs. Input and output functions of the interaction module flank the arithmetic module. Boundaries and control arrange preparation, actions, and observation. Acquired records select later boundaries along the dashed path; states and correlations still participating continue along the lower solid path. Together these paths retain records, states, and correlations of the joint output. Delivery may also terminate the task.

These results support our proposal for a next-generation paradigm of state-space computing based on BC arithmetic covariance: data composition participates directly in operations, task relations enter programmable boundaries, actual evolution forms results, and states and records organize subsequent computation. Constructing positive combinations of finitely many correlations also yields independently verifiable certificates, with explicit analytic steps connecting them to general arithmetic conclusions. The two-mode finite prototype design translates core relations into components, timing, and acceptance requirements. Established finite-task compilation, completion guarantees, and statistical protection provide concrete grounds for this paradigm. State space thus serves both as an information carrier and as the organizing basis of computational relations, connecting data, programs, and interaction within one physical run.

A Closure of Finite Phase Operations

The translations, phases, and Fourier operations of Section 4 close on the same finite carrier. Take an odd prime q and $K \geq 1$, and set $d = q^K$, $R = \mathbb{Z}/d\mathbb{Z}$, and $\omega = e^{2\pi i/d}$. In the exponents below, $1/2$ denotes the multiplicative inverse of 2 in R. Define

$$X|x\rangle = |x+1\rangle, \qquad Z|x\rangle = \omega^x|x\rangle, \qquad D(a,b) = \omega^{ab/2}X^aZ^b.$$

Since $ZX = \omega XZ$, direct multiplication gives

$$D(a,b)D(a',b') = \omega^{(ba'-ab')/2}D(a+a',b+b').$$

This is the finite Heisenberg action with its central phase. Define the Fourier transform, quadratic phase, and multiplication by units:

$$F|x\rangle = \frac{1}{\sqrt{d}}\sum_{y\in R}\omega^{xy}|y\rangle, \qquad C|x\rangle = \omega^{x^2/2}|x\rangle, \qquad M_u|x\rangle = |ux\rangle \quad (u \in R^\times).$$

Evaluation on basis vectors gives $FXF^\dagger = Z$, $FZF^\dagger = X^{-1}$, and $CXC^\dagger = \omega^{1/2}XZ$. Hence

$$\begin{aligned} FD(a,b)F^\dagger &= D(-b,a),\\ CD(a,b)C^\dagger &= D(a,b+a),\\ M_uD(a,b)M_u^\dagger &= D(ua,u^{-1}b). \end{aligned}$$

These actions correspond to matrices $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $L_1 = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, and $\mathrm{diag}(u, u^{-1})$ on R^2. Integer powers of C give arbitrary L_r; conjugation by J gives arbitrary upper triangular shears $U_r = \left(\begin{smallmatrix} 1 & r \\ 0 & 1 \end{smallmatrix}\right)$.

These matrices generate $\mathrm{SL}_2(R)$. Indeed, the first column of a determinant-1 matrix contains a unit, since otherwise both entries are divisible by q. After left multiplication by J if necessary, the upper-left entry a may be assumed to be a unit. Then

$$\begin{pmatrix} a & b \\ c & e \end{pmatrix} = L_{c/a}\begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}U_{b/a}, \qquad e = (1+bc)/a.$$

Every symplectic transformation is therefore realized by these finite unitaries. The spectrum of Z is simple, and X cyclically permutes its eigenvectors, so an operator commuting with both X, Z must be scalar. Two unitaries implementing the same $D(a,b)$ conjugation therefore differ only by a global phase. This gives the finite Weil projective representation needed here; its global phases do not alter the conjugation relations.

Closure scope is set by the generators. For example, at $q > 3$, the cubic phase $C_3|x\rangle = \omega^{x^3}|x\rangle$ conjugates X into a shift with factor ω^{3x^2+3x+1}. The resulting phase is generally no longer linear in x, so the conjugated shift is not a Heisenberg displacement. The cubic-phase operator therefore lies outside the normalizer of this Heisenberg action. General position-dependent phase operations therefore enlarge the operation class.

B Decay of Full-Ring Labels

Consider the finite ring $\mathbb{Z}/q^K\mathbb{Z}$ of Section 4, with odd prime q and $K \geq 2$. Assign rate $\kappa/(q+1)$ to each non-loop undirected edge $\{x, qx\}$. Its continuous-time generator is

$$(Lf)(x) = \frac{\kappa}{q+1} \sum_{y:\{x,y\} \text{ is an edge}} [f(y) - f(x)].$$

Every undirected edge is counted once, and the uniform distribution satisfies detailed balance. The carrier is the entire finite ring; this exchange model differs from the independent occupation thermal chain of Section 5.

First take a primitive multiplicative character χ modulo q^K, extended by zero on nonunits. A unit x has only neighbor qx, hence $(L\chi)(x) = -\kappa\chi(x)/(q+1)$. For a nonunit of valuation 1, its unit children form a fiber of reduction modulo q^{K-1}. Primitivity makes the sum of χ over that fiber zero; other nonunits have no children with nonzero character values. Thus on the entire ring,

$$L\chi = -\frac{\kappa}{q+1}\chi.$$

Slower label decay arises because nonzero branches exchange through 0. The Teichmüller unit group T has order $q-1$, and its multiplicative action commutes with L. In any nontrivial T character sector, $f(0) = 0$. Restricting a sector function to one fixed nonzero Teichmüller-label branch determines the other branches by the character and gives a finite-tree operator absorbing at 0. Each internal point has one parent and q children; outermost points are unit leaves.

The decay operator $-L$ of this connected Dirichlet tree has a strictly positive lowest eigenvector, unique up to normalization. Tree automorphisms preserve this eigenvector and act transitively on each valuation level, so it depends only on valuation level $j = v_q(x)$. The slowest rate is therefore the lowest eigenvalue of the radial absorbing-chain decay operator. Using κ as the rate unit and writing its dimensionless generator as Q gives

$$\begin{aligned}
(Qh)_0 &= a(h_1 - h_0), \\
(Qh)_j &= a(h_{j+1} - h_j) + b(h_{j-1} - h_j), \quad 1 \leq j < K, \\
a &= \frac{1}{q+1}, \qquad b = \frac{q}{q+1}, \qquad h_K = 0.
\end{aligned}$$

Solve $-Qh = 1$ for mean absorption time. With $d_j = h_j - h_{j+1}$, one has $ad_0 = 1$ and $ad_j - bd_{j-1} = 1$. Recursion and summation yield

$$\begin{aligned}
d_j &= \frac{q+1}{q-1}(q^{j+1} - 1), \\
h_j &= \frac{q+1}{(q-1)^2}(q^{K+1} - q^{j+1}) - \frac{q+1}{q-1}(K-j).
\end{aligned}$$

Let $\mu = \gamma_{\text{tag}}/\kappa$ be the lowest eigenvalue of $-Q$ and set $\pi_j = q^{-j}/\sum_{r=0}^{K-1} q^{-r}$. The operator Q is self-adjoint under these reversible weights. Matrix $G = (-Q)^{-1}$ is nonnegative and $G1 = h$. Applying $\varphi = \mu G\varphi$ at the maximum component of the positive lowest eigenvector gives $\mu \geq 1/h_0$. Conversely, using trial vector h in the Rayleigh quotient and the quadratic-mean inequality gives

$$\boxed{\frac{1}{h_0} \leq \mu \leq \frac{\langle h, 1\rangle_\pi}{\langle h, h\rangle_\pi} \leq \frac{1}{\mathbb{E}_\pi h}.}$$

For fixed q, summing the explicit formulas shows that both h_0 and $\mathbb{E}_\pi h$ equal $(q+1)q^{K+1}/(q-1)^2$ times $1 + O(Kq^{-K})$. The two bounds imply

$$\gamma_{\text{tag}} = \kappa\frac{(q-1)^2}{(q+1)q^{K+1}}\big(1 + O(Kq^{-K})\big) = \Theta(q^{-K})\kappa.$$

This identifies the slowest decay in nontrivial Teichmüller-label sectors; primitive multiplicative characters instead obey rate $\kappa/(q+1)$ above. Different observable components on the same carrier can have different decay times. Label lifetime cannot be interpreted directly as the retention time of an arbitrary quantum state.

C Conditional Error Bound for Approximate Evolution

We give the conditional result used in the mean-field estimate of Section 4. Let $e(t) = \|A(t) - A^{\mathrm{MF}}(t)\|$ be the first-moment error and $C(t)$ a connected-correlation control quantity defined in the chosen model. Both are nonnegative and absolutely continuous on $[0,T]$, with $e(0) = C(0) = 0$. Suppose the following inequalities have been derived from the actual generator and hold almost everywhere:

$$C' \le A_0 C + B_0, \qquad e' \le A_0 e + D_0 C,$$

where $A_0 = 2M\|f\|g$, $B_0 = c_1 M^3 \|f\| g$, $D_0 = \|f\| g$, $g, c_1 \ge 0$, $M \ge 1$, and f is a fixed coupling tensor. The constants must be valid throughout the declared window.

An integrating factor in the first inequality gives

$$C(t) \le B_0 \int_0^t e^{A_0(t-s)}\, ds \le B_0 t e^{A_0 t}.$$

Substitution into the second yields

$$\begin{aligned} e(t) &\le D_0 \int_0^t e^{A_0(t-s)} C(s)\, ds \\ &\le D_0 B_0 e^{A_0 t} \int_0^t s\, ds = \frac{D_0 B_0}{2} t^2 e^{A_0 t}. \end{aligned}$$

Consequently,

$$\boxed{e(t) \le \frac{c_1}{2} M^3 \|f\|^2 (gt)^2 e^{2M\|f\| gt}.}$$

The integral estimate also covers $g = 0$. Applications must establish C and both growth bounds from the full generator. If dissipation still produces connected correlations at $g = 0$, the first inequality must include that source and be reintegrated accordingly.

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